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The present work was submitted to Chair of Process Systems Engineering

Datengestütztes Experimentdesign für die CO₂-Mineralisierung mit Stahlwerksschlacke

Data-Driven Design of Experiment for Carbon Mineralization using Steel Slag

Bachelorarbeit

Bachelor thesis

presented by

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Abstract

CO₂ mineralisation represents a promising approach for permanent carbon sequestration while valorising industrial residues as supplementary cementitious materials. However, the optimization of direct aqueous carbonation processes has been constrained by complex multi-parameter interactions, material-specific behaviour, and the limitations of traditional experimental design methodologies. This study addresses these challenges by implementing Bayesian Optimization for the first time in mineral carbonation research to systematically optimize ferrochrome slag carbonation. The process was examined using high-pressure autoclave experiments at 100 bar across a four-dimensional design space encompassing temperature (100–200 °C), reaction time (30–180 min), and calcium-based additives (CaO and CaCl₂, 0–2 wt% total Calcium). A total of 63 experiments were conducted using a batched Bayesian Optimization framework with Gaussian Process surrogate modelling. CO₂ uptake was quantified via Loss On Ignition (LOI) analysis, while **real-time feedback** based on dry mass gain enabled adaptive experiment selection throughout the campaign. The optimization yielded a predicted maximum carbonation reaction yield of 53.2% (269 g CO₂/kg slag) at 157 °C, 180 min, with 2 wt% Ca addition as CaO, representing a 76% improvement **over baseline conditions**. Response surface analysis suggests that reaction time and temperature may act as the primary drivers of carbonation yield, whereas calcium-based additives likely exert secondary enhancing effects via distinct physicochemical mechanisms. It is hypothesized that CaO promotes sustained conversion by facilitating heterogeneous nucleation and buffering local pH conditions, while CaCl₂ accelerates early-stage kinetics through ionic strength modulation and rapid Ca²⁺ **release**. The overall process behavior appears to be governed by a shift towards mass-transport limitation under optimal conditions, as indicated by kinetic modeling results and supported by scale-up observations and sedimentation patterns in the stirred reactor system. In retrospective benchmarking, Bayesian Optimization achieved a 5.7% relative improvement over the best-performing classical Design-of-Experiments strategy, with greater consistency and model confidence. The framework achieved robust model performance ($R^2 = 0.90$) despite experimental noise and provided mechanistic insights into additive-slag interactions. This study establishes Bayesian Optimization as an efficient tool for complex chemical system optimization and delivers the first systematic characterisation of calcium-based enhancement strategies in ferrochrome slag carbonation. The methodology presents a transferable framework for accelerated process development in mineral carbonation technologies.

Task Description

The utilization of carbon mineralization technology, employing industrial by-products such as steel slag, has emerged as a promising approach for reducing CO₂ emissions in the cement and construction industries. In this process, CO₂ reacts with calcium and magnesium oxides contained in the slag as silicate minerals to form stable carbonates that permanently lock the carbon as a crystalline solid. As a by-product, amorphous silica co-precipitates, providing a reactive (pozzolanic) component for conventional cement chemistry. Concurrently, the resulting material can be employed as a pozzolanic cement substitute, thereby contributing to both decarbonization and material circularity.

However, the underlying reaction is complex. It is contingent on a multitude of process parameters and is subject to non-linear kinetics as well as transport-limited phenomena. Furthermore, the high variability in slag composition poses a significant challenge in terms of the process optimization. Consequently, there is a necessity for data-driven experimentation strategies that can efficiently analyze and optimize these reaction systems, given the time-consuming and resource-intensive nature of conventional experimental approaches. The objective of this bachelor's thesis is to employ data-driven experimentation based on Bayesian optimization to characterize and improve the carbonation of a metallurgical slag from ferrochrome production in an autoclave-based process.

The work is structured into several consecutive phases. Initially, a systematic literature review was conducted to define the relevant design space, including temperature, pressure, reaction time, and additive usage. This was followed by a screening phase in which various chemical additives were experimentally evaluated to assess their individual and synergistic effects on carbonation performance under standardized conditions. Building on these insights, a Latin Hypercube Sampling strategy was employed to systematically vary key process parameters, namely temperature, reaction time, and additive dosage, in order to efficiently explore the multidimensional design space. The resulting dataset served as the foundation for training Gaussian Process surrogate models, which were subsequently used within a batched Bayesian Optimization framework. This iterative process of model-based experiment selection and continuous model updating enabled both global exploration and local exploitation of the response surface. The objective of this thesis is threefold: (i) to identify optimal reaction conditions for maximizing CO₂ uptake of the given ferrochrome slag, (ii) to establish a transferable, data-driven tool for efficient optimization of carbon mineralization across diverse feedstocks, and (iii) to derive mechanistic hypotheses on process-enhancing strategies based on systematic experimental and modeling insights.

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1 Introduction

Direct aqueous mineral carbonation is a promising strategy for Carbon Capture, Utilization, and Storage (CCUS), enabling permanent CO₂ removal via formation of stable carbonate minerals [1]. In this process, CO₂ dissolves in water and reacts with alkaline solids, typically magnesium or calcium oxides, to form thermodynamically stable carbonates [2]. While both systems are viable, research has mainly focused on magnesium **due to higher solubility and faster kinetics** [3], whereas calcium-rich systems remain underexplored despite their abundance and potential [4].

Industrial residues such as ferrochrome slag are particularly promising. With a theoretical CO₂ uptake of 510 g/kg, they offer a scalable route for carbon storage while valorizing metallurgical waste [5]. Estimates suggest that calcium-rich residue carbonation could sequester up to 8.5 billion tonnes of CO₂ annually by 2100 [6, 7]. However, experimental yields vary widely, from **10 % to 50 %**, even under comparable conditions [8]. This inconsistency is linked to complex interactions between kinetics, mass transport, and chemical modifiers [9, 10], limiting the predictive power of conventional mechanistic models, especially outside narrow regimes or in presence of catalytic additives [11]. Various enhancement strategies have been explored, including the addition of bicarbonates, chlorides, or seeding agents. For instance, NaHCO₃ increased yields from 15 % to 65 % [12], NaCl improved carbonation by 30 % [3], and MgCO₃ seeding yielded 38 % relative gains [13]. Yet, mechanistic insights into additive effects in calcium-rich systems remain limited, due to dynamic phase assemblages and evolving pH buffering, ionic strength, and surface passivation [14]. These complexities exceed the scope of classical kinetic and mechanistic models, making experimental design heavily empirical [15]. In practice, optimization still relies on static Design-of-Experiments (DoE) or trial-and-error [16], which lack the ability to handle uncertainty or dynamically target informative regions [17]. This slows hypothesis generation and renders the system a “black box,” for which deterministic strategies have limited utility. To address these limitations, flexible and data-efficient methods are needed. Bayesian Optimization (BO) provides such a framework by coupling Gaussian Process (GP) regression with uncertainty-aware acquisition functions [18]. BO has proven effective in chemical engineering contexts like fermentation and pharmaceutical formulation, reducing experimental needs by up to 33 % [19, 20]. Its

strengths, robustness to noise, capacity to model non-linearities, and suitability for low-data regimes, make it especially apt for calcium-rich slag carbonation.

1.1 Objectives and Scope of the Study

The objective of this study is to optimize direct aqueous autoclave carbonation of ferrochrome slag using a data-driven experimental framework. The scope includes batch experiments at 100 bar with as-received slag, varying temperature, reaction time, and additive composition (CaO , CaCl_2). CO_2 uptake is quantified via Loss-on-Ignition (LOI) and approximated by a surrogate model based on post-reaction mass gain. Experiments are conducted in parallel batches of three and guided iteratively by single-objective Bayesian Optimization without relying on mechanistic assumptions. No pretreatment, downstream processing, or material characterization is performed. All process parameters are constrained to reflect industrial feasibility and product compliance for use as supplementary cementitious material. Beyond performance optimization, the study aims to generate insight into additive–reactivity interactions and establish a transferable strategy for data-efficient experimentation in carbonation systems.

1.2 Structure of the Thesis

The structure of the study is as follows. Chapter 2 outlines the scientific and technical foundations of direct aqueous carbonation, including the geochemical principles of mineral carbonation, the reactivity of calcium-rich slags, and process intensification strategies. It also discusses the limitations of conventional modeling approaches and introduces Bayesian Optimization as a probabilistic design strategy for complex, uncertain systems. Chapter 3 presents the experimental methodology, including materials, reactor configuration, CO_2 uptake quantification via Loss-on-Ignition, and the implementation of the Bayesian Optimization framework. Chapter 4 reports and interprets the experimental and GP model results, with a focus on additive effects, response surface analysis, and mechanistic insights derived from the surrogate model. It further includes independent validation and retrospective benchmarking against classical DoE strategies. Finally, Chapter 5 summarizes the main findings and discusses the broader applicability of the developed framework for data-efficient experimentation and adaptive process development.

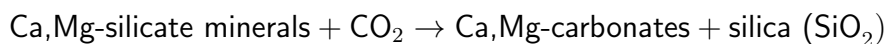
2 Scientific and Technical Background

This chapter outlines the scientific and technical foundations of aqueous mineral carbonation for CO₂ sequestration and provides the conceptual framework underlying the experimental optimization strategy. Section 2.1 introduces the basic geochemical principles of mineral carbonation and discusses engineered approaches for kinetic enhancement, with a particular focus on aqueous-phase reactions. Section 2.2 addresses the potential of steel slags as industrial feedstocks, highlighting their compositional variability and implications for process design. Section 2.3 presents established physical and chemical intensification strategies, including additive-based interventions, while Section 2.4 identifies key mechanistic, material, and methodological challenges that complicate efficient process development. Finally, Section 2.5 motivates the use of Bayesian Optimization as an adaptive design framework tailored to the complexity, uncertainty, and resource constraints of real-world carbonation systems.

2.1 Fundamentals of Mineral Carbonation for CO₂ Sequestration

Carbon mineralization harnesses natural geochemical processes to permanently store CO₂ through reactions between carbon dioxide and alkaline earth metal-bearing minerals, forming stable carbonate compounds [21]. The concept was initially proposed by Seifritz in 1990 [22] and further developed by Lackner et al. in 1995 [2]. While natural carbonation has functioned as a geological carbon sink and buffer throughout Earth's history, its inherently slow kinetics necessitates engineered acceleration for meaningful climate mitigation [23]. Given that thermodynamic equilibrium favors carbonate formation under atmospheric conditions, these processes provide a robust and permanent pathway for CO₂ sequestration [1].

The overall transformation can be summarized as:



Mineral carbonation has gained increasing interest not only due to its potential as a carbon capture and storage mechanism, but also through its dual valorization opportunity. The process simultaneously produces stable carbonates for permanent CO₂ sequestration and amorphous silica with high pozzolanic activity, making it valuable as a supplementary cementitious material (SCM) in concrete applications [24] [25]. This dual benefit transforms CO₂ sequestration from a cost center into a value-generating process, creating CO₂-negative construction materials with the potential of superior performance characteristics compared to conventional cement systems [10]. The technology's appeal is further enhanced by the abundance of suitable feedstock materials, particularly industrial residues such as steel slags, which combine high reactive oxide content and are commonly repurposed in bulk applications with limited value addition. This convergence of carbon sequestration, resource upgrading, and high-value product generation positions mineral carbonation as a key technology for sustainable industrial decarbonization strategies [26].

2.1.1 Engineered Approaches for Enhancing Carbonation Kinetics

A wide range of engineered acceleration strategies have been developed to leverage the thermodynamically advantageous equilibrium and overcome the inherent kinetic limitations. These approaches can broadly be classified into *in situ* and *ex situ* methods [27].

In-situ carbonation entails the injection of CO₂ into underground formations rich in reactive silicates, such as basalts or peridotites, where mineralization proceeds over extended timescales (months to years) [28]. While offering long-term storage stability, these approaches are limited by geological variability, restricted process controllability, and challenges in monitoring and verification [29]. Furthermore, they do not allow for material valorization or recovery of reaction products.

Ex-situ carbonation, in contrast, takes place in engineered, above-ground systems that allow precise control over reaction conditions, including feedstock properties, phase compositions, and process environment [30]. This setup enables accelerated reaction rates, continuous monitoring, and the possibility of targeted product recovery. Ex-situ processes are particularly attractive for industrial symbiosis, enabling the production of value-added materials such as SCMs from carbonated residues [24].

Aqueous Carbonation as Preferred Ex-situ Approach

Among ex-situ strategies, **aqueous carbonation** has emerged as the most technically and economically promising approach [3]. Aqueous carbonation refers to the reaction of dissolved CO₂ with alkaline earth metals in the liquid phase, resulting in the precipitation of solid carbonates. In this method, fine and thereby activated solid feedstock is dispersed into solution to form a slurry, creating a reactive medium that enhances mass transfer, dissolution kinetics, and precipitation control [9].

2.1.2 Mechanism of Aqueous Carbonation

The aqueous carbonation process proceeds through several interconnected physicochemical steps, each with distinct kinetic constraints [11]:

CO₂ dissolution and hydration: CO₂ dissolves in the aqueous phase and forms carbonic acid, which subsequently dissociates into bicarbonate and carbonate ions:



Silicate dissolution: The acidic conditions promote dissolution of alkaline minerals in the feedstock, releasing divalent metal ions into solution. For calcium-rich slags, this involves calcium silicate phases such as larnite and merwinite:



Carbonate precipitation: When sufficient concentrations of metal cations and carbonate ions are available, thermodynamically favorable precipitation occurs:



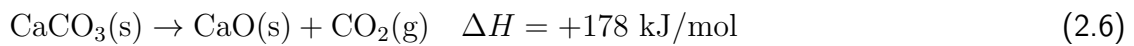
Simultaneously, dissolved silicic acid polymerizes to form amorphous silica with high pozzolanic reactivity, valuable for cement applications. The overall carbonation of calcium silicate is exothermic and yields calcium carbonate and amorphous silica, as summarized in Table 2.1.

Table 2.1: Thermodynamic parameters of selected carbonation reactions (25 °C, 1 bar) [6, 9]. **Portlandite** (Ca(OH)₂), **Brucite** (Mg(OH)₂), **Larnite** (Ca₂SiO₄), **Alite** (Ca₃SiO₅), **Merwinite** (Ca₃Mg(SiO₄)₂)

Carbonation Reaction	ΔH° (kJ/mol)	ΔG° (kJ/mol)
$\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$	−68	−73
$\text{Mg(OH)}_2 + \text{CO}_2 \rightarrow \text{MgCO}_3 + \text{H}_2\text{O}$	−37	−38
$\text{Ca}_2\text{SiO}_4 + 2\text{CO}_2 \rightarrow 2\text{CaCO}_3 + \text{SiO}_2$	−81	−64
$\text{Ca}_3\text{SiO}_5 + 3\text{CO}_2 \rightarrow 3\text{CaCO}_3 + \text{SiO}_2$	−148	−92
$\text{Ca}_3\text{Mg(SiO}_4)_2 + 4\text{CO}_2 \rightarrow 3\text{CaCO}_3 + \text{MgCO}_3 + 2\text{SiO}_2$	−399	−686

Looking at these interlinked physicochemical steps, **direct aqueous carbonation** has emerged as the preferred strategy for ex-situ implementation, as it enables the one-step reaction of CO₂ with mineral feedstock in slurry-phase systems [31]. This integrated approach contrasts with *indirect* carbonation routes, which decouple mineral dissolution and carbonate precipitation into separate unit operations. While such indirect strategies can offer improved control over reaction conditions, potentially yielding higher conversion efficiencies and purer products [9], they require additional process steps, chemical reagents, and energy input, particularly for dissolution and separation, limiting their scalability and industrial viability [32].

Calcination Reaction: At elevated temperatures (950-1000°C), the thermodynamically favorable carbonate formation is reversed, decomposing calcium and magnesium carbonates back to their constituent oxides [33]:



This decomposition reaction forms the basis of the widely applied **Loss-on-Ignition (LOI)** method [8]. Palasser et al. [34] reported that in most mineral carbonation systems, including the one investigated in this study, no crystalline water-bearing phases or hydrated silica are formed during the process. Consequently, the mass loss observed upon controlled thermal decomposition can be attributed almost exclusively to CO₂ release, thereby confirming LOI as a robust and reliable gravimetric measure of carbonation extent.

2.1.3 Kinetic Control Mechanisms and Mechanistic Models

To interpret the kinetic behavior observed during aqueous carbonation, two complementary mechanistic models are widely employed for solid-liquid reactions [14]:

- **Classical Nucleation Theory** describes the early-stage precipitation of carbonates as a result of local supersaturation of dissolved alkaline cations and carbonate species. Nucleation is typically heterogeneous [11], occurring on reactive silicate surfaces. This initiates the formation of a product layer composed of carbonates and amorphous silica, which increasingly hinders further mass transport and ion exchange [23]. As the reaction progresses, continued crystal growth and potential agglomeration further modulate the overall reaction kinetics, particularly in strongly supersaturated environments.
- **The Shrinking Core Model** provides a macroscopic framework for solid–liquid reactions, in which spherical particles are assumed to react from the surface inward toward an unreacted core [35]. Depending on the progression of the reaction and the environmental conditions, three distinct rate-limiting regimes can be identified [36]: (i) external mass transfer control, (ii) surface reaction control, and (iii) diffusion through a growing product layer. These mechanisms and their spatial sequence are schematically illustrated in Figure 2.1.

In addition to solid–liquid interfacial reactions, the dissolution and subsequent hydration of CO₂, a prerequisite for the formation of reactive carbonate species, can impose kinetic limitations, particularly under suboptimal pH and temperature conditions. Although frequently considered fast, this step may become rate-limiting in poorly mixed or pressurized systems [37].

The dominant kinetic regime is further influenced by the mineralogical composition of the feedstock. Magnesium-rich materials such as olivine typically exhibit surface-reaction or external mass transfer control [14], whereas calcium-rich slags tend to form dense carbonate shells that impose diffusional resistance even at moderate degrees of conversion [38]. While particle-scale models effectively capture such progression mechanisms, they often fall short in scope. Key phenomena, such as gas–liquid mass transfer, CO₂ dissolution kinetics, or homogeneous nucleation independent of the mineral surface, remain outside their domain. More comprehensive mechanistic frameworks, such as those proposed by Bremen et al. [11] and Wang et al. [14], aim to overcome these limitations by integrating multiple physicochemical subprocesses into unified, system-level models of aqueous carbonation.

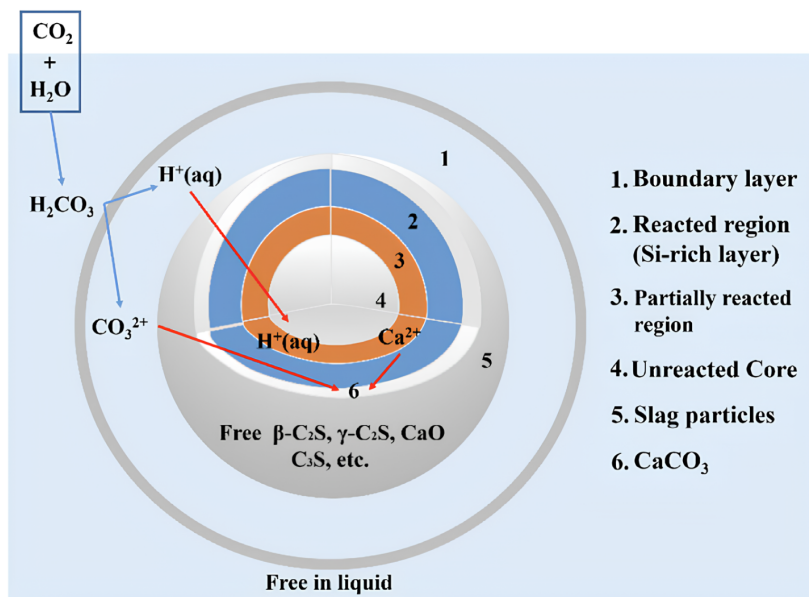


Figure 2.1: Schematic representation of the Shrinking Core Model for solid-liquid reactions. The model differentiates external mass transfer, surface reaction, and diffusion through the product layer as potential rate-limiting steps [39].

2.2 Steel Slag as Feedstock for Carbonation

2.2.1 Industrial Residues and Carbonation Potential

Steel production generates approximately 170–250 million tons of slag annually as a co-product from basic oxygen furnaces (BOF) and electric arc furnaces (EAF), representing a vast underutilized resource for carbon mineralization [40]. Despite their high alkalinity and abundance of reactive oxides such as CaO and MgO, global utilization rates of steel slags remain below 50 %, with substantial fractions stockpiled or landfilled [41, 5].

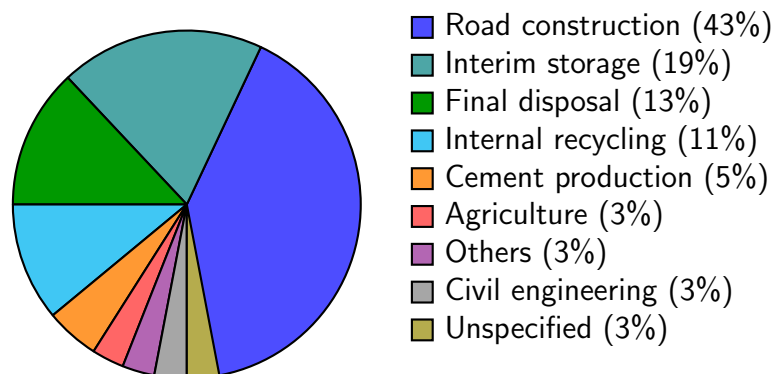


Figure 2.2: Distribution of slag utilization across application routes in Europe (based on share in %) [39].

While European utilization rates of steel slags approach 68 %, primarily through applications in road construction (43 %) and civil engineering (Figure 2.2), these conventional uses predominantly exploit mechanical properties and overlook the chemical reactivity of the material. Building on the previously discussed potential for dual valorization, mineral carbonation enables both permanent CO₂ sequestration and the generation of pozzolanic silica suitable for low-carbon cementitious applications [5]. This pathway not only enhances the environmental impact of slag valorization but also unlocks higher economic value compared to traditional bulk uses, thereby strengthening the case for industrial-scale deployment of carbonation technologies.

2.2.2 Compositional Variability and Material-Specific Behavior

Steel slags exhibit broad compositional and mineralogical variability depending on production routes. BOF slags typically contain 40–50 wt% CaO and reactive phases such as free lime and calcium silicates, whereas EAF slags feature more complex Ca–Mg–Al silicate assemblages [41]. Ferrochrome slags, a specialized EAF subset from ferroalloy production, offer particularly favorable characteristics when calcium-enriched, achieving 40–50 wt% CaO, 9–13 wt% MgO, and 29–40 wt% SiO₂ with minimal iron content [42]. Their sequestration capacity ranges from approximately 410 to 530 g CO₂/kg slag, significantly exceeding conventional steel production slags. This positions them among the most promising industrial residues for mineral carbonation, particularly when combined with low impurity levels and favorable phase composition [9].

These slags belong to a broader class of alkaline industrial residues with carbonation potential, including stainless steel slag, ladle slag, and cement kiln dust. While these materials generally contain elevated levels of alkaline earth oxides, their carbonation performance depends critically on the phase composition rather than bulk chemistry alone [10]. Reactive phases such as free lime (CaO), portlandite (Ca(OH)₂), or periclase (MgO) exhibit markedly higher dissolution kinetics than their crystalline counterparts. However Li et al. [43] report, the phase assemblages and crystallinity are strongly influenced by production parameters, cooling rates, and metallurgical additives, resulting in substantial variability even among materials with similar nominal compositions.

Need for Material-Specific Optimization. This intrinsic variability limits the applicability of standardized carbonation protocols or empirical yield predictions. Instead, material-specific optimization is essential to assess the true reactivity and dual-function potential of

Table 2.2: Typical CO₂ uptake potentials and best reported yields for selected steel slags under aqueous carbonation.

Slag type	CaO + MgO [%wt]	Theoretical potential [g/kg]	Best reported uptake [g/kg]	Representative studies
BOF	32–65	440–740	465	[44, 45, 31]
EAF	27–41	220–480	286	[14, 46, 25]
AOD	36–70	290–580	366	[47, 48]
LF	41–85	330–730	280	[10, 49, 50]
FeCr (EAF)	49–63	410–530	-	[42, 9]

each residue. The development of adaptive, data-driven optimization frameworks becomes particularly important when transitioning from laboratory-scale research to industrial deployment, where process robustness and tolerance to feedstock variability are prerequisites for economic viability.

The ferrochrome slag investigated in this study exemplifies these challenges, while also providing a model system for systematic optimization due to its combination of high calcium content, low impurity levels, and favorable dual-use potential.

2.3 Process Enhancement Strategies

Due to the multi-step nature and intrinsic kinetic limitations of aqueous mineral carbonation, **stirred batch autoclaves** have become the standard platform for systematic process investigations and optimization studies [13]. Their relevance arises from the critical interplay between temperature and pressure, which jointly govern the dissolution kinetics of alkaline solids and the subsequent precipitation of carbonate phases.

Temperature exerts a dual influence on carbonation kinetics: on the one hand, it promotes the dissolution of calcium- and magnesium-bearing phases; on the other hand, it reduces CO₂ solubility and the thermodynamic stability of dissolved carbonate species. This thermokinetic trade-off gives rise to pressure-dependent temperature optima and necessitates precise experimental control [32, 11]. Elevated CO₂ pressures, as governed by Henry’s law, increase the concentration of aqueous carbon species, thereby accelerating both mineral dissolution and carbonate precipitation. Notably, Wang et al. demonstrated that high-pressure conditions favor a transformation in the composition of surface-bound product layers, from dense, amorphous SiO₂ to more porous CaCO₃, which in turn enhances diffusive transport and mitigates passivation effects [51, 14]. This transformation constitutes a mechanistic advantage

that is unique to highly pressurized systems.

By enabling precise control of temperature and pressure, stirred autoclaves offer a well-defined environment to investigate synergistic effects and mitigate transport limitations [13]. Nonetheless, a review by Biava et al. [10] shows that most studies were conducted at moderate CO₂ pressures (1–20 bar), with only 5 of 23 exceeding 10 bar. This contrasts with consistent model-based evidence highlighting significant rate enhancements at elevated pressures (80–150 bar) [11], indicating a critical research gap and substantial potential for process intensification.

Conventional Parameter Optimization

Conventional optimization of mineral carbonation has primarily focused on process parameters such as particle size distribution (PSD), reaction time, temperature, pressure, stirring speed, and solid-to-liquid (S/L) ratio [9]. Among these, **particle size**, **temperature**, and **reaction time** have consistently shown the strongest influence on carbonation yield and underlying mechanisms [9].

Finer particle sizes enhance the specific surface area available for dissolution, thereby accelerating reaction rates and increasing overall conversion [52]. Reaction time determines the dominance of rate-limiting steps, such as surface reaction versus diffusion, over the course of the process, while temperature influences both dissolution kinetics and carbonate stability. These parameters have been extensively studied and represent robust levers for tuning reaction performance [32, 11]. In contrast, operational factors such as stirring speed and solid-to-liquid (S/L) ratio exert limited influence within practical ranges and are often constrained by reactor design [53, 9]. Accordingly, recent studies have shifted focus toward improving gas–liquid mass transfer through advanced reactor configurations. For example, Chang et al. [37] proposed a rotary packed bed to increase interfacial area and CO₂ absorption, while Bremen et al. [11] suggested a continuous flow reactor to minimize diffusion path lengths and enhance conversion efficiency.

Chemical Enhancement Through Additives

Drawing on classical nucleation theory, shrinking core kinetics, and pH equilibria, targeted chemical additives can accelerate mineral carbonation by directly addressing fundamental rate-controlling steps [36, 13]. These enhancement strategies act on distinct mechanistic bottlenecks: pH modifiers optimize dissolution rates and CO₂ speciation [54], calcium-based

additives supply reactive ions and promote nucleation [4], nucleation promoters lower precipitation barriers [55], and ionic strength regulators facilitate mass transfer [3]. Unlike conventional process parameters, these chemical interventions retain their mechanistic function across reactor scales, offering promising levers for transferable process intensification [56].

A consolidated overview of these strategies and their experimental implementations is provided in Table 2.3, highlighting representative additives, system-specific applications, and supporting literature to offer a structured foundation for selecting or combining enhancement mechanisms in future studies. However, the interplay between chemical additives and conventional process parameters (e.g., particle size, reaction time, temperature, pressure) remains poorly understood; as emphasized by Kremer et al. [13], potential synergistic or antagonistic effects have so far received minimal systematic investigation. While sodium salts and pH-buffering additives have proven effective in well-studied olivine systems [3], they exhibit markedly lower efficacy for steel slags, as reported by Matus et al. [57]. Conclusive evidence remains scarce, particularly for alternative feedstocks [58, 3], thereby hindering the development of integrated optimization strategies that effectively combine chemical and physical enhancement approaches.

2.4 Optimization Challenges and Research Gaps

Despite substantial research efforts, aqueous mineral carbonation remains a highly complex and intrinsically non-linear process. This complexity stems from the interplay of interdependent factors, including material heterogeneity, reactor-specific constraints, competing chemical equilibria, and parameter interactions with antagonistic trade-offs. For example, Chang et al. [37] demonstrated that elevated Ca^{2+} concentrations can simultaneously enhance silicate dissolution and induce premature carbonate saturation, illustrating the conflicting roles of single variables. Similarly, pH adjustment exerts opposing effects on dissolution kinetics and carbonate precipitation equilibria [9]. These multidirectional dependencies complicate mechanistic interpretation and hinder predictive control, especially across heterogeneous feedstocks and varying reactor environments. Under such conditions, conventional deterministic approaches offer limited utility. To derive actionable insights and identify high-performance process configurations, adaptive, data-driven optimization methods are required, capable of learning from noisy data and efficiently navigating complex design spaces. The following subsections outline the key sources of uncertainty and process variability that motivate this methodological shift.

Table 2.3: Overview of chemical enhancement strategies for aqueous carbonation, including representative additives and supporting literature.

Strategy	Representative Additives or Methods	Reference
pH modification and buffering	NaHCO ₃ addition in olivine systems	Stopić et al. (2018) [59]
	NaCl + NaHCO ₃ at elevated temperatures	Gadikota et al. (2014) [60]
	NaHCO ₃ with organic acids	Matus et al. (2021) [57]
	pH control for enhanced silicate dissolution	Khudhur et al. (2022) [61]
Calcium supplementation	Ca(OH) ₂ addition to steel slag systems	Huijgen et al. (2006) [4]
Nucleation enhancement	Seeding with carbonate crystals	Rashid et al. (2023) [55]
	Magnesite seeds to reduce olivine passivation	Kremer et al. (2022) [13]
	Ceramic bead addition	Kremer et al. (2022) [13]
Ionic strength control	NaCl to enhance CO ₂ transfer	Cheng et al. (2020) [62]
	Combined NaCl + NaHCO ₃ systems	Gadikota et al. (2014) [60]
	NaCl addition in serpentine carbonation	O'Connor et al. (2005) [3]
Chelation effects	EDTA for cation complexation	Biava et al. (2024) [63]
	Oxalic and ascorbic acid combinations	Matus et al. (2021) [57]
	Organic ligands for olivine dissolution	Wogelius & Walther (1991) [64]

Feedstock-Specific Reactivity and Variability. Material-specific behavior poses a significant barrier to generalization in aqueous carbonation. The dissolution kinetics, nucleation behavior, and ultimate carbonation yield of slag-derived materials are determined not only by bulk oxide composition, but by the accessibility, crystallinity, and phase assemblage of reactive mineral components [10]. Even minor variations in microstructure or mineralogy can lead to pronounced differences in CO₂ uptake under otherwise identical conditions [65].

Table 2.4 exemplifies this sensitivity based on published experimental data, underscoring the limitations of standardized process protocols. These findings emphasize the need for residue-specific optimization strategies guided by empirical feedback and detailed material characterization.

Table 2.4: Selected experiments with comparable process conditions but significantly different reaction yields. Reported values refer to the carbonation extent, i.e., the percentage of the theoretical maximum CO₂ binding capacity based on CaO and MgO content. Slag types: EAF = Electric Arc Furnace slag; BOF = Basic Oxygen Furnace slag; AOD = Argon Oxygen Decarburization slag [10].

Slag	T [°C]	pCO ₂ [bar]	Time [h]	Particle size [μm] (D ₅₀)	S/L [g/L]	Reaction yield [%]
EAF	20	0.15	40	72	10	1.7
EAF	25	5.00	3	63	5	59.5
BOF	100	2.00	4	43.1	5	53.0
BOF	100	10.0	24	50.2	5	40.3
AOD	60	2.00	1.5	<74	8	50.0
AOD	90	6.00	1.0	39.3	16	44.6

Limitations of Mechanistic Models and Black-Box System Behavior. Classical models such as the Classical Nucleation Theory and the Shrinking Core Model offer valuable abstractions of individual kinetic phenomena but fail to capture the integrated behavior of real-world aqueous carbonation systems. These models typically assume idealized particle geometries, a single rate-limiting step, and neglect critical effects such as evolving porosity, agglomeration, or the formation of passivating mixed-phase surface layers [35]. Even more advanced mechanistic frameworks that couple transport and reaction kinetics [11, 12] often prove insufficient in systems characterized by strong additive–matrix interactions, multiphase transformations, or high ionic strength. As Bremen et al. [11] emphasize, these limitations are particularly pronounced under electrolytic or high-salinity conditions, where model calibration becomes intractable. The result is a system that behaves effectively as a “black box,” necessitating empirical modeling approaches and adaptive optimization techniques to inform

experimental planning and guide process development.



Industrial Constraints and Methodological Misalignment. The industrial implementation of aqueous carbonation hinges on achieving high reaction efficiencies within stringent operational constraints. These typically include short residence times (≤ 3 h), minimal additive usage, and simple reactor designs to ensure scalability and economic viability [57]. However, most academic studies deviate markedly from these conditions, employing extended reaction durations [66], low CO₂ pressures [38], or highly reactive model feedstocks with limited relevance to industrial residues [31]. This methodological misalignment limits the transferability of research findings to real-world applications. Reported high conversion efficiencies often rely on excessive reaction times (>24 h) [42] or atypically low Si/Ca–Mg ratios that are unrepresentative of actual slag compositions [31, 67]. Bridging this gap requires a shift in research priorities toward more realistic operating conditions and the adoption of optimization strategies that explicitly incorporate industrial constraints.

Intrinsic Variability and Measurement Uncertainty. Aqueous carbonation experiments are inherently prone to variability arising from both material and operational factors. Feedstock heterogeneities, such as differences in mineral grain size, porosity, and spatial phase distribution, lead to localized variations in reactivity and ion release, which are not reliably captured by bulk compositional analysis [37, 68]. This introduces substantial noise into experimental outcomes. Reproducibility is further challenged by system-level effects, including microclimatic fluctuations, nonuniform mixing, and reactor-specific geometries. Autoclave systems, for instance, may exhibit intra-reactor temperature gradients exceeding 50 °C despite nominally isothermal conditions [69]. Likewise, the impact of solid–liquid ratio or stirring speed can vary significantly with reactor volume, impeller design, and slurry rheology [53].

The combined effect of these sources of noise is a degradation of signal fidelity across experimental campaigns. This not only limits the robustness of statistical inference but also hampers the efficacy of conventional optimization strategies. In response, probabilistic and uncertainty-aware methods, such as Gaussian process modeling and Bayesian experimental design, offer a principled framework for navigating this complexity.

2.5 From Traditional DoE to Adaptive Optimization

2.5.1 Limitations of Traditional Experimental Design

Given the multifactorial complexity, intrinsic variability, and industrial constraints outlined above, it is essential to critically reassess the limitations of traditional experimental design methodologies. Although classical approaches have underpinned process development in aqueous mineral carbonation for decades, they are increasingly ill-suited for navigating the high-dimensional, nonlinear, and uncertain parameter spaces characteristic of real-world carbonation systems.

Classical Approaches and Their Constraints

Conventional design-of-experiment (DoE) frameworks, including one-factor-at-a-time (OFAT), full factorial designs, and space-filling sampling schemes, remain prevalent in mineral carbonation research due to their simplicity and legacy in chemical engineering. However, these methods generally allocate sampling effort uniformly across the input space, regardless of emerging performance trends or prior knowledge. This static sampling strategy is fundamentally misaligned with optimization objectives in resource-constrained, data-scarce industrial environments [70].

OFAT methods, while intuitive, systematically fail to detect interaction effects among variables, a critical limitation given the well-documented interdependence between temperature, CO₂ pressure, pH, and additive dosage in influencing reaction kinetics [71]. Moreover, their sequential structure prevents concurrent exploration of multidimensional input spaces and leads to highly inefficient use of experimental resources [17].

Factorial designs attempt to address these limitations by systematically exploring combinations of multiple variables. For instance, Polettini et al. [67] employed a 3³ factorial design to assess the effects of temperature, CO₂ partial pressure, and residence time in BOF slag carbonation. Similarly, Baciocchi et al. [26] and Chang et al. [37] used factorial matrices to capture second-order interactions and synergistic effects. However, factorial designs suffer from exponential scaling: the number of experimental trials increases combinatorially with the number of variables. As Lazić [72] illustrates, even a modest 2³ full factorial design can entail over 50 trials when accounting for replication and blocking, rendering such strategies impractical in high-dimensional or cost-intensive domains.

Latin Hypercube Sampling (LHS) and other space-filling methods offer improved input coverage but remain non-adaptive: they treat all regions of the design space as equally informative. While LHS can be useful for initial model calibration or uncertainty propagation, it lacks the ability to prioritize promising subspaces based on emerging data trends. In practice, such methods often serve as initialization steps for more sophisticated, adaptive frameworks like Bayesian Optimization [18, 73].

Response Surface Methodology: Advances and Limitations

Response Surface Methodology (RSM) offers a more model-centric approach by fitting low-order polynomial functions to describe system response as a function of experimental inputs. Studies by Wang et al. [52] and Irfan et al. [16] demonstrate the use of Box-Behnken and central composite designs in the optimization of aqueous carbonation processes.

However, RSM faces critical limitations in dealing with the nonlinearities and irregularities intrinsic to high-pressure, slag-based carbonation systems:

- **Assumed polynomial structure:** Standard RSM formulations rely on second-order polynomial approximations, which are poorly suited for capturing multimodal landscapes, discontinuities, or phase-switching behavior [71].
- **Lack of adaptivity:** Although sequential RSM updates are possible, the framework lacks formal mechanisms for dynamically steering experiments based on posterior uncertainty or expected improvement [74].
- **Inadequate treatment of experimental noise:** RSM is not inherently suited to capture stochastic variability arising from measurement uncertainty and process fluctuations in heterogeneous carbonation systems [75].
- **Dimensional scalability:** As the number of input variables increases, the number of required design points grows rapidly, leading to infeasibility in settings where each experiment incurs significant cost or time [70].

These limitations collectively constrain RSM's utility in guiding experimentation for industrial carbonation processes characterized by uncertain kinetics, variable feedstocks, and stringent operational constraints.

2.5.2 Motivation for Bayesian Optimization

In light of these limitations, Bayesian Optimization (BO) has emerged as a powerful and data-efficient framework for adaptive experimentation in complex, nonlinear systems [20, 76]. Unlike traditional DoE, BO operates within a sequential decision-making paradigm: it maintains a probabilistic surrogate model, typically a Gaussian Process (GP), of the unknown objective function and uses this model to iteratively select the most informative or promising experimental configuration [77]. By incorporating both mean predictions and uncertainty estimates, BO balances exploration (sampling regions with high model uncertainty) and exploitation (sampling regions with high expected performance) [18]. This makes it particularly well-suited for aqueous carbonation systems, where each experiment is costly, noisy, and potentially sensitive to nonlinear interaction effects [19]. The next section details the formulation and implementation of BO in this context, and outlines how it addresses the methodological and practical gaps identified in traditional approaches.

2.5.3 Bayesian Optimization: Framework and Advantages

The framework operates on three core principles [77]:

1. **Probabilistic surrogate modeling:** A statistical model, typically a Gaussian Process (GP), captures both predicted responses and associated uncertainties across the parameter space .
2. **Acquisition function optimization:** Mathematical criteria balance exploration (sampling in high-uncertainty regions) with exploitation (sampling near predicted optima) to guide experiment selection.
3. **Sequential adaptation:** Experimental strategies dynamically evolve based on accumulating evidence, enabling efficient navigation of complex response surfaces.

This approach is inherently model-free, requiring no assumptions about functional forms of response surfaces, while providing principled uncertainty quantification that enables risk-aware decision-making under experimental constraints. Figure 2.3 shows a graphic of the exact process of Bayesian Optimization as DoE strategy. Mathematical details are found in Appendix A.1.

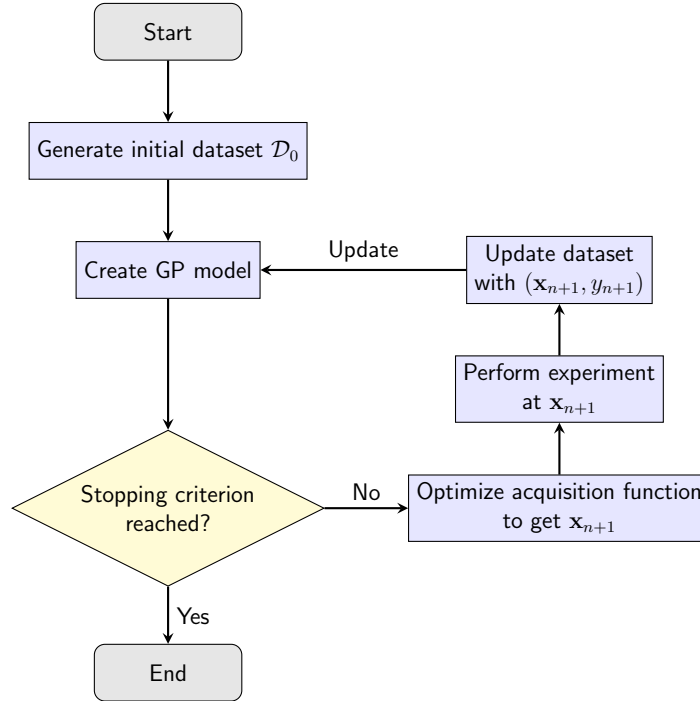


Figure 2.3: Bayesian Optimization process: A surrogate model (Gaussian Process) is iteratively updated with new experiments and used to select the next evaluation point via an acquisition function.

Gaussian Process Regression as Surrogate Model

Gaussian Processes serve as the probabilistic foundation for Bayesian Optimization, providing non-parametric function approximation with calibrated uncertainty estimates. A GP is completely specified by its mean function $m(\mathbf{x})$ and covariance function $k(\mathbf{x}, \mathbf{x}')$, defining a distribution over possible functions rather than assuming specific functional forms.

For a dataset of n experimental observations $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$, where $y_i = f(\mathbf{x}_i) + \epsilon_i$ represents measured responses with Gaussian observation noise $\epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2)$, the GP posterior distribution at new point $\mathbf{x}^*$ follows:

$$p(f(\mathbf{x}^*)|\mathcal{D}, \mathbf{x}^*) = \mathcal{N}(\mu(\mathbf{x}^*), \sigma^2(\mathbf{x}^*))$$

The predictive mean $\mu(\mathbf{x}^*)$ provides the best estimate of the response, while the predictive variance $\sigma^2(\mathbf{x}^*)$ quantifies model uncertainty.

A crucial aspect of modeling with Gaussian Processes is the choice of kernel, which defines the

smoothness and structure of the modeled function. Beyond the standard Squared Exponential kernel, **Matérn kernels** offer a more flexible representation of physical processes. They generate functions that are not infinitely differentiable, thereby better capturing the inherent "roughness" of real experimental data. The parameter ν in the Matérn kernel controls the smoothness of the function; values such as $\nu = 1.5$ and $\nu = 2.5$ have proven particularly suitable for experimental data with moderate noise levels.

Acquisition Function Strategies

The decision of where to conduct the next experiment is made using an acquisition function $\alpha(\mathbf{x})$. This function carefully balances two key objectives:

- **Exploitation:** Exploring regions where the model predicts high performance, aiming to improve the current best result.
- **Exploration:** Sampling in areas where the model is highly uncertain, to reduce uncertainty and potentially discover new, unpredicted optimal regions.

A widely used example is Expected Improvement (EI), which quantifies the expected gain from evaluating a new point $\mathbf{x}$ relative to the current best observation $f(\mathbf{x}^+)$:

$$\alpha_{\text{EI}}(\mathbf{x}) = \mathbb{E} [\max(0, f(\mathbf{x}) - f(\mathbf{x}^+))] \quad (2.7)$$

The next experimental point $\mathbf{x}_{n+1}$ is then chosen by maximizing this acquisition function:

$$\mathbf{x}_{n+1} = \arg \max_{\mathbf{x}} \alpha(\mathbf{x}) \quad (2.8)$$

Additional settings

In Gaussian Process (GP) modeling, **prior distributions** encode assumptions about the objective function before any data is observed. They play a crucial role in regularizing the model, guiding hyperparameter learning, and incorporating domain knowledge—particularly when experimental data are scarce or noisy. Common prior choices reflect sensible assumptions about the expected smoothness, signal amplitude, or observation noise. State-of-the-art GP implementations frequently employ **Gamma** or **log-normal** priors for kernel hyperparameters such as the length scale and output scale, as these distributions enforce positivity while providing flexibility to encode domain-specific ranges or soft constraints [78]. In multi-dimensional problems, separate priors can be assigned per input variable using **Automatic Relevance**

Determination (ARD). This allows the model to infer individual parameter sensitivities. Shorter inferred length scales imply a stronger influence of the corresponding input dimension on the output (see Appendix A.1). This Bayesian treatment of model structure and uncertainty underpins the efficiency of Bayesian Optimization (BO). As illustrated in Figure 2.4, each new experiment iteratively refines the GP model, which updates the acquisition function to select the next most promising point. This loop enables sample-efficient navigation of complex design spaces under uncertainty.

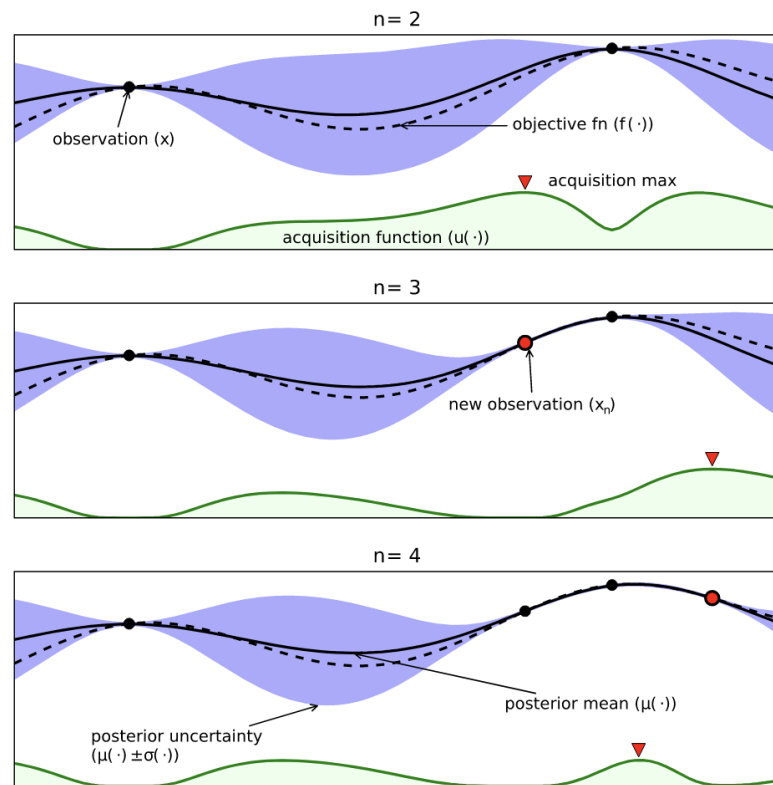


Figure 2.4: Progress of Bayesian Optimization: The surrogate model (dashed line with shaded uncertainty) is iteratively refined as new observations (black/red dots) are made, guided by the acquisition function (green line) to explore uncertain or promising regions [18].

BO for experimental applications

From an experimental design perspective, BO offers several key advantages:

- **Sequential adaptation:** New experiments are selected based on the current state of knowledge, allowing real-time refinement of the experimental strategy.

- **Data efficiency:** BO typically achieves comparable or superior results to classical designs with substantially fewer experiments [79].
- **Uncertainty quantification:** Predictions are accompanied by confidence intervals, enabling risk-aware decision-making and targeted exploration of uncertain regions.
- **Constraint integration:** Practical process limits (e.g., temperature, pressure, additive dosage) can be incorporated directly into the optimization routine.
- **Transferability:** The framework is agnostic to the underlying material system and can be applied across different slag compositions and reactor configurations without structural redefinition.

Several recent studies have demonstrated the effectiveness of BO in related domains. Shields et al. [79] applied BO to optimize organic synthesis conditions, achieving superior results with fewer trials compared to expert chemists. Similarly, Schweidtmann et al. [73] used BO to guide flow chemistry experiments under process constraints, illustrating its flexibility in real-world chemical engineering contexts.

In summary, the characteristics of the framework render Bayesian Optimization particularly attractive for the development of mineral carbonation processes. The methodology aligns with industrial constraints, limited time, expensive experiments, and process variability, while supporting systematic identification of high-performance operating windows and gain strong system understanding.

3 Materials and Methods

This chapter outlines the complete experimental and modeling methodology employed to investigate and optimize aqueous carbonation of ferrochrome slag. It is structured as follows: Section 3.1 outlines the detailed scope of this study. Section 3.2 describes the materials and feedstock characterization, including chemical, mineralogical, and physical properties as well as the investigated additives. Section 3.3 presents the high-pressure batch reactor setup, standardized operating protocols, and quality assurance measures. Section 3.4.1 details the analytical methods used to quantify CO₂ uptake via Loss on Ignition and its surrogate metric based on dry mass gain, including correction procedures accounting for additive contributions. The adaptive Bayesian optimization framework and Gaussian Process modeling strategy are introduced in Section 3.5, covering parameter selection, constraint handling, and acquisition strategies. The final subsections summarize the planning phase, the initial sampling approach, and the iterative optimization campaign, including the key assumptions underlying the experimental design and implementation. Methods for interpreting the underlying chemical processes are also presented. The methodological framework and rationale for the retrospective benchmarking analysis are detailed in Section 3.6.

3.1 Scope of the Study

This study focuses on the data-driven design of experiments for optimizing the aqueous mineral carbonation of ferrochrome slag. The experimental system is based on direct aqueous carbonation conducted in stirred batch autoclave reactors. The scope explicitly excludes any pretreatment steps (e.g., milling, purification) as well as material characterization before or after the reaction (e.g., XRF, XRD, SEM). The industrial feedstock is used as received, relying solely on the specifications provided by the producer without thermal or chemical activation. The response variable of interest is the carbonation reaction yield of the slag, which is quantified using a gravimetric Loss-on-Ignition method corrected for additive contributions. To ensure operational feasibility and resource efficiency, a batched design is employed in which three experiments are conducted and proposed simultaneously in each optimization iteration.

Additionally, LOI values are approximated via a linear surrogate model based on the dry mass gain after reaction. The optimization follows a single-objective Bayesian Optimization approach without relying on prior data or mechanistic assumptions, due to the limited and inconsistent evidence available in the literature and the absence of existing BO applications in this specific context; to regularize model behavior and anchor the regression near zero reaction time, five pseudo-experiments with zero yield at $t = 0$ were appended at the end of the campaign. The study focuses on high-pressure conditions (100 bar) to capture relevant kinetic behavior and investigates the effect of selected additives and chemical enhancement strategies on the carbonation performance. To ensure industrial relevance and product compliance with potential use as supplementary cementitious material, all experimental parameters are constrained to reflect economic and regulatory limits. In particular, additive dosages are bounded by calcium-based composition limits and chloride thresholds in the final product [80]. The overarching objective is to establish a transferable, reproducible framework for data-efficient experimentation that can be extended to other industrial feedstocks. Beyond the identification of robust, high-performance operating windows, the surrogate Gaussian process model is used to derive mechanistic insight into the role of chemical additives especially in light of their still poorly understood effects on slag reactivity. The scope is thereby not limited to empirical performance gains but extends to hypothesis generation and system understanding.

3.2 Materials and Characterization

3.2.1 Ferrochrome Slag Characterization

The feedstock used in this study is a calcium-rich ferrochrome slag sourced from Elektrowerk Weisweiler (Eschweiler-Weisweiler, Germany). The material represents an industrially pre-treated variant with reduced chromium content, typically applied in low-value applications such as road base layers or general backfilling [81]. The material is thus representative of the industrial residues with high dual valorisation potential discussed in Section 2.2 and is therefore suitable for a case study.

Chemical and Mineralogical Composition

X-ray fluorescence (XRF) analysis revealed a high combined CaO and MgO content of 59.5 wt%, classifying the slag as a high-potential feedstock for mineral carbonation [38].

Additionally, its SiO_2 content of 32.2 wt% supports its use as a supplementary cementitious material (SCM).

Table 3.1: Chemical composition of the ferrochrome slag (wt%).

Component	Content (wt%)
CaO	47.03
MgO	12.45
SiO_2	32.23
Al_2O_3	8.37
Fe_2O_3	0.03
TiO_2	0.01
Na_2O equivalent	0.05
SO_3	0.08
Loss on ignition	0.17

X-ray diffraction (XRD) with Rietveld refinement identified merwinite ($\text{Ca}_3\text{Mg}(\text{SiO}_4)_2$) as the dominant crystalline phase (64 wt%), followed by gehlenite ($\text{Ca}_2\text{Al}_2\text{SiO}_7$; 15 wt%) and $\gamma\text{-C}_2\text{S}$ (13 wt%). The presence of merwinite is particularly beneficial for aqueous carbonation due to its high reactivity stemming from its open crystal structure and weak Ca–O bonding [12]. Similar slag systems have demonstrated carbonation yields above 60 % even under moderate reaction conditions [25].

Physical Properties and Carbonation Potential

Laser diffraction analysis showed a moderate to high particle size distribution with $D_{90} = 131 \mu\text{m}$, representative of typical industrial conditions that balance grinding cost and reactivity [11]. While finer particles generally favor carbonation kinetics, the chosen distribution maintains industrial relevance. Loss-on-ignition (LOI) analysis confirmed a minimal volatile content ($<0.2 \text{ wt}\%$) [34], indicating the absence of pre-existing carbonation and validating LOI as an accurate measure of reaction yield. The theoretical maximum CO_2 uptake, derived from the reactive oxide content, is estimated at 505.0 g CO_2/kg slag (Equation 3.4.3).

Taken together, the high reactive oxide content, favorable mineralogy, and representative physical properties underline the material's suitability for CO_2 mineralization studies and support the development of transferable data-driven optimization strategies.

3.2.2 Chemical Additives and Reagents

A total of seven chemical additives were investigated during the experimental campaign, representing established laboratory reagents commonly used in mineral carbonation studies. The selected additives were calcium oxide (CaO), calcium hydroxide ($\text{Ca}(\text{OH})_2$), calcium chloride (CaCl_2), calcium carbonate (CaCO_3), calcium sulfate dihydrate ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$), sodium bicarbonate (NaHCO_3), and sodium chloride (NaCl). The rationale for their selection is detailed in Section 3.5. All additives were of common laboratory-grade purity and handled under standardized conditions.

Additional specifications of the chemical additives and the slag feedstock, including product details and physical characteristics, are provided in Appendix A.2.

3.3 Experimental Setup and Procedures

3.3.1 High-Pressure Autoclave Configuration

Carbonation experiments were performed using a parallel reactor setup comprising three high-pressure batch autoclaves (model BR-300, Berghof Products + Instruments GmbH). Each reactor vessel is constructed from stainless steel 316Ti, with a working volume of approximately 390 mL and rated for up to 200 bar and 300 °C. This configuration allows parallel experimentation with individually controlled temperatures and shared pressure supply, increasing throughput while maintaining precise process control.

Each autoclave is equipped with a magnetically coupled overhead stirrer (up to 2000 rpm) and a pitched-blade impeller to ensure homogeneous suspension and effective solid–liquid contact throughout the reaction. Temperature control is provided by independent electrical heating mantles regulated via a Berghof BRC control unit featuring PID logic with auto-tuning. The system supports real-time digital logging of temperature and pressure profiles for each reactor independently. All heating mantles were programmed to follow identical PID parameters, and once the target temperature was reached, a thermal stability of $\pm 2^\circ\text{C}$ was maintained. Reaction time was monitored and documented with an accuracy of ± 1 minute, as both the initiation and termination of the reaction (see Section 3.3.2) were manually controlled.

While temperature, stirring speed, and reaction time are controlled individually, pressure is applied uniformly to all reactors via a shared CO_2 gas manifold. High-purity CO_2 (4.8 grade,

99.998%, NIPPON GASES) is supplied from a cylinder with internal eductor tube. Downstream pressure regulation is achieved using a high-pressure piston pump from MAXIMATOR operated via voltage control, enabling target pressures between 60 and 150,bar with an accuracy of ± 0.5 ,bar, using compressed air as the working medium. The reactor setup is shown in Figure 3.1.

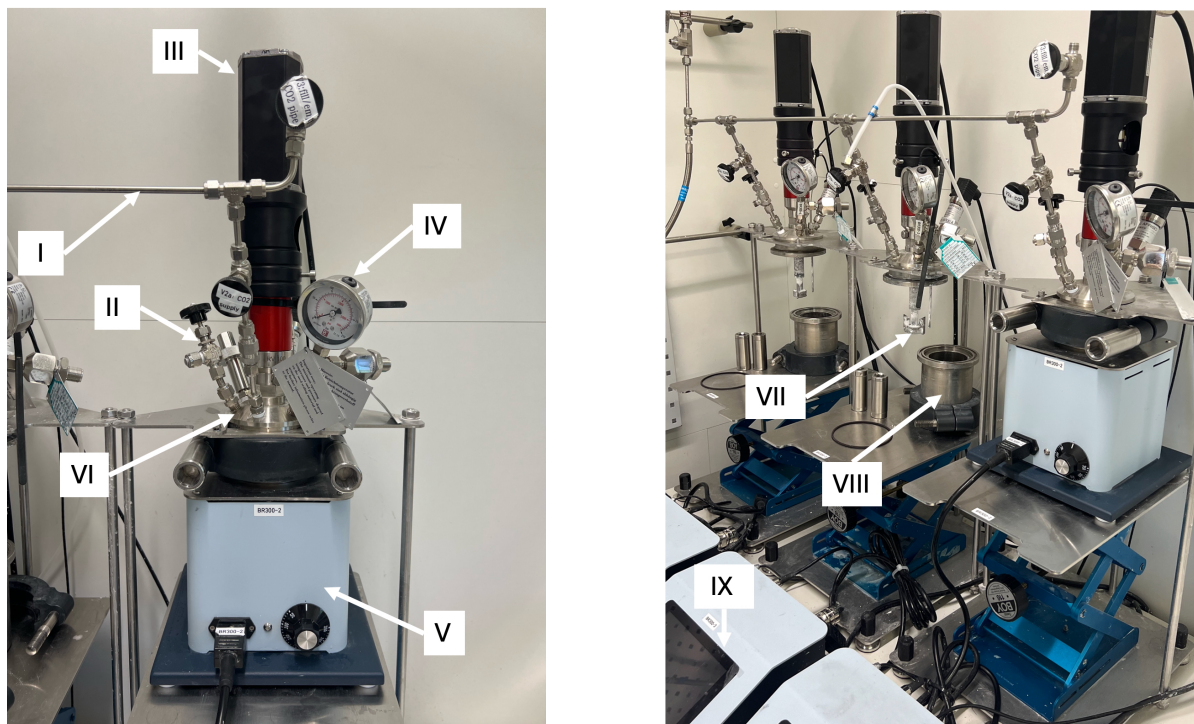


Figure 3.1: Experimental setup for high-pressure carbonation using a triple-parallel reactor system. **(a)** Detailed view of a single reactor unit: (I) CO₂ inlet line, (II) venting valve, (III) overhead stirrer motor, (IV) analog pressure gauge, (V) heating mantle, (VI) reactor hood with thermocouple and digital pressure sensor. **(b)** Overview of the triple-reactor setup during operation of one unit: (VII) overhead stirrer with thermocouples, (VIII) stainless steel reactor vessels, (IX) BRC control unit.

3.3.2 Standardized Experimental Protocol

Pre-Reaction Preparation Prior to each experiment, the required quantities of ferrochrome slag and chemical additives were weighed using an analytical balance (VWR, readability: 0.001 g). The slag feedstock mass was maintained at 50.000 g per experiment to ensure consistent solid loading and comparable reaction conditions. Chemical additives, including CaO, Ca(OH)₂, CaCO₃, and CaCl₂, were added proportionally based on the target dosing levels specified by the experimental design and mixed with the dry feedstock. For sodium-based

additives such as NaCl and NaHCO_3 , solutions were prepared using tap water to reflect realistic background ionic conditions relevant for industrial implementation. Although not fully deionized, it is assumed that the tap water composition remained stable throughout the experimental campaign, ensuring consistent chemical conditions across all relevant experiments. In this study we used a standard S/L ratio of 0.4 g/mL.

Salt dissolution was facilitated through magnetic stirring at ambient temperature (22–25 °C) on standard laboratory stir plates for 10–15 minutes until complete homogenization was achieved. The total water volume was calculated to achieve the specified solid-to-liquid (S/L) ratio for each experiment, accounting for the combined mass of slag and any solid additives. This approach ensures that the S/L ratio accurately reflects the total solids content rather than just the slag fraction.

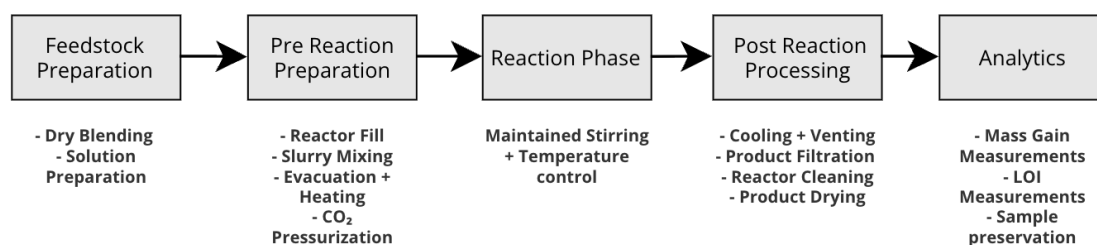


Figure 3.2: Block flow representation of the high-pressure carbonation process, detailing the sequential operations from feedstock preparation through reaction execution to downstream processing and analysis.

Reaction Execution Protocol The slag–additive mixture was subsequently introduced and mixed with the solution in the reactor vessel, while continuous stirring prevented particle settling and ensured homogeneous solid–liquid contact. This in situ slurry formation reflects standard practice in industrial and continuous systems, where mixing is maintained from the onset of the reaction. For this study we used an elevated continuous stirring speed at 500 rpm for all of our experiments.

After sealing the reactor using the screw-based closure system, the vessel was evacuated to eliminate ambient air prior to CO_2 introduction. This ensured a defined gas-phase composition and avoided dilution effects that could reduce CO_2 solubility or alter carbonation kinetics. Heating and stirring were initiated simultaneously to promote uniform temperature distribution and stable particle suspension.

The official reaction time was defined as beginning once both target temperature and CO_2 pressure had been reached and remained stable for at least one minute. This definition

compensates for minor variations in heat-up and pressurization rates across runs. CO₂ was introduced once the internal temperature approached within 15–20, °C of the target value, followed by pressurization to the designated setpoint via piston pump. CO₂ pressure was maintained at 100 bar throughout all experiments. Due to the strongly exothermic nature of the carbonation reaction (see Table 2.1, CO₂ was deliberately introduced prior to reaching the target temperature. This approach allowed the heat release from rapidly reacting phase components, which often caused temperature increases of up to 10–15, °C, to contribute to the remaining heating phase, thereby preventing overshooting of the predefined target temperature.

Post-Reaction Processing Upon completion of the predefined reaction duration, the CO₂ supply was shut off and the reactors were rapidly cooled by immersion in an external water bath. This quenching step was implemented to minimize continued carbonation during the cooling phase, ensuring that the measured CO₂ uptake corresponds strictly to the defined reaction time. Once the reactors returned to temperature close to ambient condition, after about 15 minutes, internal CO₂ pressure was released gradually through manual venting. The resulting slurry was separated via vacuum filtration using glass Büchner funnels equipped with VWR Grade 410 qualitative filter paper (Ø 110 mm, nominal retention ~2 µm). In line with comparative studies [13], a cellulose-based slow-filtration medium was selected due to its fine retention capacity, allowing efficient separation of small carbonate crystallites while maintaining acceptable filtration rates. The filtercake was subsequently dried in a ventilated laboratory oven at 120 °C for 12 hours to ensure complete moisture removal without decomposing carbonate phases. Dried samples were weighed using analytic balance and sealed in airtight polyethylene containers to prevent atmospheric carbonation or moisture uptake prior to analytical characterization.

Quality Assurance and Reproducibility Measures Standardized time logging during heating, pressurization, and cooling phases was employed to minimize procedural variability. While all reactor vessels, sampling tools, and weighing equipment were thoroughly rinsed with tap water before and after each experiment to avoid visible contamination and cross-experiment carryover, the feedstock material was not pre-sieved. Mass measurements were conducted using analytical balances with a readability of 0.001 g to ensure precise quantification of CO₂ uptake. Photographic documentation of the reactor interior and the recovered solid product was performed after each run to monitor fouling progression and potential morphological differences in reaction products.

3.4 Analytical Methods and Data Processing

3.4.1 Gravimetric Determination of CO₂ Uptake via Loss on Ignition

Quantification of CO₂ uptake resulting from the carbonation reaction and its derived yield was performed using the Loss on Ignition (LOI) method established in Section 2.1.2, while all intermediate steps followed standardized procedures as described in the literature [34]. Approximately 1–2 g of each dried sample were weighed. To prevent any re-absorption of moisture before analysis, these small samples were stored in a desiccator. For the thermal treatment, samples were transferred into ceramic crucibles. Prior to each use, these crucibles were individually pre-fired (ignited at 1000 °C) and subsequently pre-weighed. The samples in their crucibles were then subjected to thermal treatment in a muffle furnace at 1000 °C for 2 hours under ambient air conditions. The calcination temperature of 1000 °C was chosen based on commonly used thermogravimetric procedures to ensure complete decomposition of carbonate species (see Equation 2.1.2) [8].



(a) External view of the muffle furnace. The desiccator used for pre/post-calcination sample storage is visible on the right.



(b) Interior of the muffle furnace with crucibles containing carbonate samples during the calcination process.

Figure 3.3: Muffle furnace setup for Loss on Ignition (LOI) analysis of carbonated slag samples.

Gravimetric CO₂ uptake was calculated using the following expression:

$$\text{CO}_2^{\text{measured}} [\text{g}] = m_{\text{dry}} - m_{\text{residue}} \quad (3.1)$$

$$\text{CO}_2^{\text{uptake}} [\text{wt}\%] = \frac{\text{CO}_2^{\text{measured}}}{m_{\text{dry}}} \cdot 100 \quad (3.2)$$

where m_{dry} is the mass of the dried sample prior to calcination, and m_{residue} is the remaining mass after ignition.

As we proved absence of carbonate species in the feedstock material we assert that subsequent CO_2 uptake is solely due to reaction-induced carbonate formation. All LOI measurements were conducted as single determinations due to time and capacity constraints. While this introduces some uncertainty, the presence of internal controls and consistent mass balances support the overall reliability of the results. Figure 3.3 presents the used muffle oven from Heraeus instruments from the inside and outside, its control panel and the used desiccator.

Systematic Correction for Additive Contributions The measured LOI values require systematic correction to account for mass contributions from chemical additives, ensuring accurate quantification of slag-specific carbonation [13]. In this context, a key assumption was applied: **For all calcium-based additives (see Section 4.4.1), complete conversion to their respective carbonate forms was assumed.** This assumption is supported by the high solubility and known reactivity of these compounds under the employed hydrothermal conditions [82] and aligns with established practice in related mineral carbonation studies.

The theoretical CO_2 contribution from any carbonate-forming additive i can be calculated as:

$$\text{CO}_2^{\text{additive}, i} [\text{g}] = \frac{m_{i, \text{added}} \cdot M_{\text{CO}_2}}{M_i} \quad (3.3)$$

where $m_{\text{CaO added}}$ is the mass of added calcium oxide, and M_{CO_2} and M_{CaO} are the respective molar masses. The corrected CO_2 uptake is calculated as:

$$\text{CO}_2^{\text{corrected}} [\text{g}] = \text{CO}_2^{\text{measured}} - \sum_i \text{CO}_2^{\text{additive}, i} \quad (3.4)$$

3.4.2 Surrogate Metrics for Real-Time Optimization

Since efficient LOI analyses require batch processing and are typically performed only after accumulating sufficient sample quantities (12–16 samples), immediate CO_2 uptake values were not available during the experimental progression. To enable timely feedback for the Bayesian optimization algorithm, the dried solid mass obtained after filtration and oven drying was used as a surrogate metric until LOI results became available.

Preliminary experiments revealed a strong linear correlation ($R^2 = 0.995$, $\text{RMSE} = 2.26\%$) between dried mass increase and LOI-derived CO_2 uptake, enabling provisional assessment of experimental performance within 24 hours of completion. The use of surrogate metrics represents a practical compromise between the algorithmic need for rapid feedback and the time requirements of high-precision analytical methods. This approach ensured efficient progression of the optimization campaign while maintaining analytical accuracy for final evaluation (see Section 3.5.2 for further rationale). Different from LOI analysis, yield estimation based on dried mass gain must account for contributions from both calcium-based additives and retained salts. **For sodium-based additives such as NaCl and NaHCO_3 , as well as chloride originating from CaCl_2 , a retention factor of 28 % was applied.** This value is derived from systematic moisture measurements conducted during previous experimental campaigns and accounts for the partial retention of sodium salts within the filter cake after aqueous processing. Under the assumption of a closed system and complete removal of residual moisture, the CO_2 uptake was estimated from the increase in dried mass relative to the initial feedstock mass, corrected for the mass of all added chemicals:

$$\text{CO}_2^{\text{surrogate}} [\text{g}] = m_{\text{dry}} - m_{\text{feedstock}} - m_{\text{additive, corrected}} \quad (3.5)$$

Here, m_{dry} is the total dried solid mass recovered after the reaction, $m_{\text{feedstock}}$ is the initial mass of slag introduced into the reactor, and $m_{\text{additive, corrected}}$ corresponds to the stoichiometrically adjusted CO_2 contribution of the added chemicals as described above.

3.4.3 Carbonation Reaction Yield Calculation

Corrected CO_2 uptake values were used to calculate the carbonation reaction yield, providing a normalized metric for comparing performance across different experimental conditions. The reaction yield η is defined as the percentage of the theoretical maximum CO_2 uptake based on the reactive calcium and magnesium content:

$$\eta [\%] = \frac{\text{CO}_2^{\text{corrected}} [\text{g}]}{\text{CO}_2^{\text{theoretical max}} [\text{g}]} \cdot 100 \quad (3.6)$$

The theoretical maximum CO_2 uptake was derived from the stoichiometric reaction capacity of calcium and magnesium oxides present in the slag in Section 3.2.1:

$$\text{CO}_2^{\text{theoretical max}} [\text{g}] = \left(\frac{w_{\text{Ca,slag}}}{M_{\text{Ca}}} + \frac{w_{\text{Mg,slag}}}{M_{\text{Mg}}} \right) \cdot M_{\text{CO}_2} \cdot m_{\text{slag}} \quad (3.7)$$

where $M_{\text{Ca}} = 40.08 \text{ g/mol}$ and $M_{\text{Mg}} = 24.31 \text{ g/mol}$ are the molar masses of calcium and magnesium, respectively. Only calcium and magnesium were considered in this calculation, as they represent the dominant reactive components in the ferrochrome slag employed. Other potential contributors such as iron or manganese were neglected due to their low concentrations in the feedstock (Table 3.1) and their typically slower carbonation kinetics compared to alkaline earth metals [9].

3.5 Bayesian Optimization Framework

This section outlines the implementation of the adaptive optimization strategy developed for ferrochrome slag carbonation. The framework integrates strategic objectives, parameter selection rationale, constraint handling, and the Gaussian Process model architecture. These components jointly enable efficient navigation of the complex, noisy design space while ensuring rigorous uncertainty quantification throughout the experimental campaign. The resulting framework is designed to be modular and transferable to future carbonation studies.

3.5.1 Implementation Strategy and Objectives

As established in Section 2.5.2, Bayesian Optimization offers a principled approach for experimental design under uncertainty, particularly suited for black-box systems such as slag carbonation. BO enables adaptive learning from limited, noisy data, facilitating both efficient process optimization and mechanistic hypothesis generation.

Our methodology follows a five-phase structure inspired by state-of-the-art BO workflows for experimental sciences [83, 19, 18]. Each phase contributes to maximizing information gain within the constraints of available resources:

- **Planning:** Definition of the optimization problem, identification of critical factors, and allocation of the experimental budget.
- **Screening:** Systematic additive evaluation under baseline conditions to identify effective enhancement mechanisms and constrain the optimization space.
- **Initial Sampling:** Generation of a foundation dataset using space-filling design to initialize the GP model with representative coverage.
- **Model Training and Adaptive Optimization:** Iterative GP updates and acquisition-guided experiment selection until convergence or budget exhaustion, with continuous monitoring of model uncertainty.
- **Validation:** Independent experiments for assessing predictive performance and confirming process improvements.

This structured approach ensures a robust transition from initial process understanding to optimized operating windows and validated generalization performance. It simultaneously addresses the dual goals of maximizing carbonation yield and improving scientific insight into additive-process interactions.

3.5.2 Planning and Foundation Phase

The planning phase establishes the optimization problem definition, identifies critical influencing factors, and allocates experimental resources within the practical constraints of a bachelor's thesis scope and available reactor setup.

Problem Definition and Stopping Strategy This study pursues two complementary objectives that leverage the strengths of probabilistic surrogate modeling:

- 1. Process Optimization:** Identification of optimal process conditions for maximizing CO₂ uptake through systematic exploration of the design space.
- 2. Mechanistic Interpretation:** Development of a validated Gaussian Process surrogate model to enable mechanistic interpretation and hypothesis generation despite limited data.

Response Variable and Measurement Strategy Carbonation reaction yield was selected as the primary response variable, determined via Loss on Ignition analysis with systematic additive corrections (see Section 3.6). To ensure timely feedback for the adaptive optimization algorithm, a surrogate measurement strategy using dried mass gain (see Section 3.4.2) was implemented, with LOI updates incorporated as analytical results became available. System reproducibility was validated using historical olivine carbonation data obtained under identical reaction conditions. In line with common literature practice [17], an intraclass correlation coefficient (ICC) was calculated and yielded a value of 0.973, indicating excellent consistency between replicate runs (see Appendix A.3). Based on these findings, single-reactor measurements were deemed sufficiently reliable, allowing the full experimental budget to be allocated within the Bayesian optimization framework.

Parameter Space Definition Four continuous process parameters were selected for optimization: temperature, reaction time, and the concentrations of two selected chemical additives. Temperature and reaction time were chosen as key kinetic variables, given their fundamental influence on dissolution rates, nucleation dynamics, and precipitation equilibria [11]. Additionally, these parameters could be reliably and independently varied within the existing parallel reactor setup, which features synchronized gas feed and temperature control across three units. The reaction pressure was fixed at 100 bar to ensure comparability with established high-pressure carbonation protocols. This choice is supported by studies such as Stropic et al. [59], who demonstrated enhanced conversion in olivine systems under comparable pressure conditions due to increased CO₂ solubility and driving force for carbonate formation. The additive concentrations were selected based on a systematic screening strategy informed by four mechanistic hypotheses, aligned with literature-based approaches discussed in Section 2.3: (i) calcium supplementation, (ii) pH buffering, (iii) nucleation enhancement, and (iv) ionic strength modification. These mechanisms are hypothesized to accelerate carbonation kinetics or improve conversion efficiency [3, 13]. The specific roles and the seven chosen compounds are summarized in Table 3.2.

Table 3.2: Additive categories and hypothesized mechanistic roles in slag carbonation.

Functional Role	Additives	Mechanistic Rationale
Reactive calcium supplementation	CaO, Ca(OH) ₂ , CaCl ₂	Direct calcium provision via different chemical routes. All doses standardized to 0–2 wt% elemental Ca to enable mechanistically meaningful comparisons.
pH modification and buffering	NaHCO ₃	Intermediate-pH buffering effects; hypothesis: pH control enhances silicate dissolution and carbonate speciation.
Nucleation enhancement	CaCO ₃ , CaSO ₄ ·2H ₂ O	Heterogeneous nucleation templates; alternative surfaces for carbonate growth; potential for surface-catalyzed rate enhancement.
Ionic strength control	NaCl	Pure ionic strength modifier; isolation of transport effects from pH or calcium-related influences.

Dosing Constraints and Rationale To ensure economic and regulatory viability, the total amount of calcium-based additives was limited to 2 wt% Ca, and NaHCO₃ was restricted to 1.0 mol/L for solubility reasons. For CaCl₂, an upper limit of 0.45 wt% Ca (corresponding to approximately 0.8 wt% CaCl₂) was imposed in accordance with common specifications for supplementary cementitious materials, which typically tolerate up to 0.1 wt% Cl[−] in the final product [80]. Consistent with this specification, the NaCl concentration was limited to 0.09 mol/L.

Experimental Budget Allocation A total of 63 experiments was strategically allocated across the four-phase methodology [17]: screening phase (15 experiments), initial sampling (15), adaptive optimization (27), and independent validation (6), as summarized in Table 3.3. The experimental campaign was planned over a period of seven weeks, targeting an average throughput of nine experiments per week.

Batch Optimization Design The adaptive optimization phase employed batch Bayesian optimization, with the algorithm proposing three experimental candidates simultaneously to fully utilize the parallel autoclave setup. This batched approach maximized experimental throughput while accounting for inter-experimental correlations via the underlying Gaussian

Table 3.3: Experiment allocation across the four-phase optimization framework.

Phase	Description	Experiments
Screening	Additive evaluation and mechanistic hypothesis testing	15
Initial Sampling	Space-filling sampling for model initialization	15
Adaptive Optimization	Iterative model-guided experimentation	27 (9×3)
Independent Validation	Post-optimization confirmation experiments	6
Total		63

Process model. All experiments, including the initial screening, orthogonal Latin Hypercube sampling, and adaptive optimization, were conducted in parallel batches of three, reflecting the fixed capacity of the reactor system and ensuring consistent procedural execution throughout the campaign.

Screening Experiments

In accordance with Montgomery et al. [17], the screening phase employed a one-factor-at-a-time approach to evaluate the seven selected additives described above. All screening experiments utilized reference conditions adapted from established olivine carbonation studies (180°C temperature, 100 bar CO₂ pressure, 180 min reaction time, S/L ratio of 0.4 g/mL) to ensure comparability with existing carbonation literature [11, 59]. The experimental protocol employed all three autoclaves in parallel configuration for each screening run. Two reactors tested individual target additives while the third served as an additive-free baseline reference. For consistent evaluation across all additives, standardized concentrations of 1 wt% Ca (for calcium-based additives) and 1.0 mol/L for the sodium salts were employed throughout the OFAT screening (see Section 4.2 for details). For CaCl₂ and NaCl₂, these elevated dosages exceed practical application thresholds but were deliberately chosen to isolate mechanistic effects of ionic strength enhancement independent of regulatory constraints, providing clear performance indicators for subsequent optimization. Based on screening performance and mechanistic complementarity, CaO (reactive Ca source) and CaCl₂ (ionic strength enhancer) were selected for combination tests. Moderate concentrations (1 wt% Ca as CaO, 0.23 wt% Ca as CaCl₂) were chosen to probe potential synergies under practically relevant conditions. The complete screening campaign comprised nine additive-testing experiments and six baseline reference measurements. This screening phase aimed to establish prelimi-

nary performance indicators and define the design space for subsequent Bayesian optimization rather than generate statistically comprehensive data. The results informed the selection of promising additives and concentration ranges for the adaptive experimental campaign.

Parameter Selection and Bounds The parameter selection for Bayesian optimization was informed by initial screening results (see Section 4.2) and mechanistic insights from the literature, aiming to balance process relevance, experimental controllability, and industrial applicability[59, 57]. Temperature was limited to 100–200 °C. This window is proven to capture the trade-off between reaction kinetics and the thermodynamic stability of carbonates at elevated temperatures in high pressure systems [14]. A reaction time of 30–180 min was chosen to span regimes from chemical reaction controlled to dissolution-controlled and near-equilibrium behavior while remaining within industrially feasible residence times [9, 15]. Among all tested additives, CaO and CaCl₂ exhibited the highest yield improvements, reflecting their distinct functional roles and demonstrating synergistic potential in combined application under the tested screening specifications. The final design space in Table 3.4 is given by this, while the constraints that were already used in the screening are retained.

Table 3.4: Optimization parameters and operational bounds.

Parameter	Minimum	Maximum	Units	Rationale
Temperature	100	200	°C	Kinetic vs. thermodynamic balance
Reaction time	30	180	min	Industrial feasibility window
CaCl ₂ addition	0	0.45	wt% Ca	Chloride limit screening
CaO addition	0	2	wt% Ca	Economic feasibility threshold
Total Ca addition	0	2	wt% Ca	Economic feasibility threshold

As already mentioned above, CaO and CaCl₂ dosages will be reported in terms of equivalent elemental calcium content (wt% Ca as CaO or wt% Ca as CaCl₂). This representation facilitates direct comparison between additive types while maintaining transparency with respect to the total calcium constraint.

Constraint Implementation The Bayesian optimization framework therefore incorporates a single explicit constraint reflecting practical dosing limits (Section 3.5.2):

$$w_{\text{Ca, CaO}} + w_{\text{Ca, CaCl}_2} \leq 2.0\% \quad (3.8)$$

This constraint was enforced by restricting the sampling domain in all experimental phases, excluding parameter combinations violating the additive sum limit to ensure only chemically and economically feasible formulations were explored.

Initial Sampling Strategy and Foundation Experiments

As elaborated by Rasmussen et al. [77], a structured initial dataset is essential for effective Bayesian optimization, enabling realistic kernel learning and providing a robust foundation for model-driven experimentation. In the context of surrogate-based optimization under constrained experimental budgets, initial sampling must ensure sufficient coverage of the input space, avoid clustering, and promote statistical stability during Gaussian Process training.

To this end, several sampling strategies were evaluated for their spatial efficiency and suitability under noise and budget constraints, including random sampling, Sobol sequences, and Latin Hypercube Sampling (LHS). An orthogonal array design based on LHS offered the best balance between stratification and space-filling performance, as detailed in Appendix A.4, and was therefore selected in alignment with state-of-the-art practices [17].

Design literature commonly recommends 2^n or $5 \times n$ points for n continuous variables, which yields 16–20 points for $n = 4$. Accordingly, 15 orthogonal samples were generated and supplemented with 5 representative points from the screening phase, resulting in a 20-point foundation dataset for GP initialization (see Table A.14). Five data points from the screening phase were strategically incorporated into the Bayesian optimization dataset to complement the orthogonal design while maintaining experimental diversity. The selected experiments comprised: (i) three baseline reference measurements to establish additive-free performance, (ii) one CaO-enhanced experiment to capture single-additive effects, and (iii) one CaO-CaCl₂ combination experiment to represent synergistic interactions. This selection ensured representation of key experimental conditions within the defined optimization parameter space while avoiding systematic bias toward specific additive configurations. The remaining screening experiments were excluded from the optimization and training dataset as they either investigated additives outside the selected four-parameter optimization framework.

Figure 3.4 shows all pairwise projections of the four-dimensional Latin Hypercube design across temperature, reaction time, CaO concentration, and CaCl₂ concentration. The matrix demonstrates uniform space-filling coverage and balanced parameter variation, ensuring well-conditioned covariance matrices critical for stable Gaussian Process inference under experimental uncertainty.

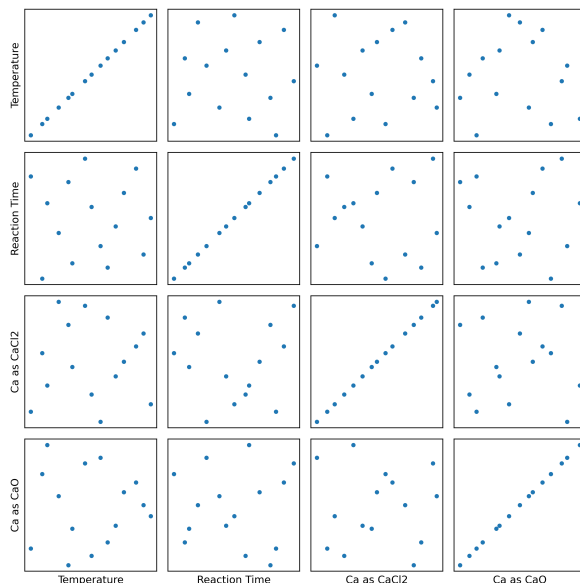


Figure 3.4: Two-dimensional projections of the initial Latin Hypercube sampling design, illustrating space-filling behavior across the four primary input dimensions: temperature, reaction time, and calcium-based additives (CaO and CaCl₂).

3.5.3 Adaptive Learning Strategy

From this foundation and their LOI result, the Gaussian Process model was trained and adaptive optimization commenced with systematic model development to navigate the four-dimensional parameter space while accounting for experimental noise and measurement uncertainty.

Gaussian Process Model Architecture and Implementation

The GP framework utilized GPyTorch’s ExactGP model with the SingleTaskGP wrapper from the botorch library, providing probabilistic predictions of carbonation yield at untested parameter combinations $\mathbf{x} = [T, t, w_{\text{Ca,CaO}}, w_{\text{Ca,CaCl}_2}]$ while quantifying predictive uncertainty via posterior variance. Hyperparameters were optimized via marginal log-likelihood maximization using the ADAM optimizer under standardized GPyTorch settings. These applications represent the current best practice for flexible, modular Bayesian optimization frameworks and are well-established in recent scientific literature [19, 18]. Initial configurations employed radial basis function (RBF) kernels (see Appendix A.1.1 for mathematical insights) with isotropic lengthscales and a homoskedastic Gaussian likelihood to model experimental reproducibility noise [78]. A zero-mean prior was used, consistent with the assumption of limited prior knowledge about the response surface. All input variables were internally

standardized to zero mean and unit variance before training, following standard practice in Gaussian Process regression to improve numerical stability and kernel hyperparameter learning. This normalization ensures consistent scaling across parameters with different physical units and enhances the model's ability to generalize from limited data. Weakly informative, dimension-dependent log-normal priors were applied for the observation noise σ_n and the lengthscales ℓ to maintain robust uncertainty estimates under limited training data [77]. The model did not employ a fixed observation noise, but inferred σ_n from the data via marginal likelihood optimization. This configuration was selected based on documented performance across varying problem dimensionalities by Hvarfner et. al [78] and therefore was assumed to be suitable as a first Bayesian optimization implementation in mineral carbonation contexts. Model predictions were discretized according to experimental resolution: reaction time to minute accuracy, temperature to degree accuracy, and additive quantities to milligram precision (see Section 3.3.2).

Optimization Workflow and Real-Time Integration

Each iteration followed a structured workflow maximizing information integration and decision speed while accommodating analytical constraints. Previous batch experiments underwent drying (120 °C, 9 h) and weighing protocols yielding dried mass estimates within 12 h of reaction completion. This rapid surrogate measurement enabled continuous model updates, avoiding delays from LOI analysis (3–5 days) that would compromise adaptive capabilities within the 7-week campaign. Surrogate yield estimates and their regression correlation to LOI were immediately integrated through automated retraining pipelines. The custom Python optimization platform integrated data preprocessing, constraint handling, and real-time visualization for experimental decision-making. Actual measured process parameters were used for all model updates. PID-controlled temperature measurements showed negligible deviation from candidate values. Reaction time deviations due to operational delays were systematically documented and updated. Additive quantities exhibited minor deviations (± 3 mg), and were updated accordingly. Yields from mass gain and LOI measurements were treated equivalently for model training. Each new result triggered automated workflows including: (i) model retraining, (ii) outlier detection via standardized residuals ($\pm 2.5\sigma$), and (iii) acquisition function evaluation. Visualization tools provided contour plots, design space coverage, uncertainty maps, and predicted optima. Outlier detection was performed using standardized residuals defined as

$$r_i = \frac{y_i - \hat{y}_i}{\hat{\sigma}_i},$$

where y_i denotes the observed yield, $\hat{y}_i$ the predicted mean, and $\hat{\sigma}_i$ the predicted standard deviation of the Gaussian Process posterior. Observations with $|r_i| > 2.5$ were flagged for potential exclusion or diagnostic evaluation, ensuring model robustness against anomalous measurements or experimental artifacts. Upon availability, LOI results triggered secondary model updates with refined targets, often revealing minor biases in the surrogate model and informing updated calibrations.

Progressive Model Enhancement Strategy

To balance model flexibility with data availability, model complexity was gradually increased in parallel with the growing experimental dataset. After 30 experiments (initial design plus 10 experiments), the Gaussian Process surrogate transitioned from an isotropic to an Automatic Relevance Determination (ARD) kernel. This change was motivated by the increasing data density, which enabled the robust estimation of individual parameter sensitivities, in line with recommendations by Rasmussen et al. [77, 78]:

$$k_{\text{ARD}}(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp \left(-\frac{1}{2} \sum_{d=1}^4 \frac{(x_d - x'_d)^2}{\ell_d^2} \right) \quad (3.9)$$

To ensure realistic modeling behavior and avoid implausible lengthscales extremes, Gamma priors were applied to the individual ARD lengthscales ℓ_d . In addition, the kernel function was changed from a squared exponential (RBF) to a Matérn kernel with smoothness parameter $\nu = 2.5$. This decision was motivated by two considerations: first, the Matérn-2.5 kernel allows for limited differentiability of the response surface, which better reflects the expected physical behavior of the system under study (e.g., regime transitions, non-smooth effects of additive interactions). Second, empirical model comparisons showed slightly improved generalization and reduced overfitting in high-noise regions when using Matérn-2.5 instead of the overly smooth RBF kernel. Lower smoothness variants of the Matérn kernel ($\nu = 0.5$ and 1.5) were also tested but exhibited excessive posterior oscillations and instability, making them unsuitable for the given application. This progressive refinement strategy aligns with best practices in Bayesian optimization, whereby model complexity is adapted to the available data in order to improve predictive accuracy and acquisition quality [18, 20]. A detailed overview of the employed model architecture is provided in Appendix A.6.

Pseudo-Experimental Anchoring at Zero Reaction Time To address physically implausible extrapolations observed in preliminary response surface analysis, such as predicted

non-zero carbonation at $t = 0$, five pseudo-experiments with varied input parameters and a fixed yield of 0% were retrospectively added at zero reaction time for final model training. This intervention was implemented only after completion of the experimental campaign when boundary condition issues became apparent. The pseudo-experiments were not used during the adaptive optimization process to avoid biasing sequential experimental design, but were incorporated exclusively for final model analysis and mechanistic interpretation. While minor carbonation during initial pressurization cannot be categorically excluded, these effects are expected to fall within the model uncertainty range. The retrospective anchoring acts as a soft constraint that nudges the surrogate towards chemically consistent behavior in edge regions without forcing strict adherence to zero yield at $t = 0$. This approach aligns with established practice in Gaussian Process modeling, where synthetic anchor points regularize model behavior in underconstrained regimes without introducing hard priors [77, 84] (see Table A.13). This post hoc intervention reflects a limitation of the original experimental design, which did not anticipate boundary condition regularization needs. Future studies should benefit from including short-time measurements explicitly in the initial design or formulating time-informed prior mean functions to capture physically motivated boundary behavior systematically from the outset [85].

Acquisition Strategy Evolution and Implementation

The acquisition strategy was adapted to the maturity of the dataset and the characteristics of the underlying response function. In the early stages of the campaign, qLogNEI (log-transformed q-Noisy Expected Improvement) was employed. This acquisition function is particularly well-suited for noisy chemical systems with strongly noisy behavior and multiplicative uncertainty, as is typical for carbonation yields. The logarithmic transformation mitigates the influence of outliers and stabilizes variance across the design space, while the batch formulation (q) leverages the parallel autoclave setup by proposing multiple informative candidates per iteration. This makes qLogNEI an effective choice for early-stage exploration under limited data availability and uncertain response surfaces (see Appendix A.1.2).

Exploration-Exploitation Balance through Hybrid Acquisition After iteration 32 (foundation + 4 batches), convergence diagnostics revealed clustering in narrow subspaces particularly in the temperature dimension (e.g., ranges $<15^{\circ}\text{C}$) and repeated time settings despite persistently high uncertainty in the additive dimensions (see Section 4.5.2). To counteract premature convergence and reinforce global exploration, the acquisition strategy was adjusted to a hybrid formulation combining qLogNEI and qGIBBON in a 2:1 ratio.

qGIBBON, a recent entropy-based acquisition function designed for robust performance in noisy environments, prioritizes candidates that yield the highest expected information gain about the location of the global optimum. This shifts focus more toward undersampled, uncertain regions and complements the inherently exploitative behavior of qLogNEI [86]. Due to the lack of native batch support in qGIBBON, a custom greedy selection algorithm was implemented in Python to enable parallel batch proposal. This hybrid strategy ensured both convergence efficiency and uncertainty-aware design space coverage, particularly in additive-related subspaces that showed slower information gain under purely exploitative strategies. The transition towards a hybrid acquisition strategy reflects the evolving need to balance exploitation in promising subspaces with renewed exploration in uncertain or undersampled regions. This trade-off modulation was essential to avoid premature convergence and to ensure interpretable coverage of critical process domains (see Section 4.5).

All acquisition batches respected design constraints, including a cumulative calcium bound ($w_{\text{CaO}} + w_{\text{CaCl}_2} \leq 2.0 \text{ wt\%}$). Acquisition function optimization was performed using the gradient-based L-BFGS-B algorithm, which supports box constraints and is well suited for continuous optimization problems [77]. A modular platform design allows future extensions to multi-objective formulations, alternative acquisition functions, and transfer learning for novel feedstock materials.

Convergence Monitoring via GP Learning Curves and Posterior Surface Analysis

Due to the non-parametric nature of Gaussian Processes, model convergence cannot be assessed through fixed parameter metrics alone. Instead, the evolution of model performance was tracked through learning curves based on leave-one-out cross-validation metrics (e.g., RMSE, R^2), posterior uncertainty shrinkage, and the stability of acquisition-driven optima. Additionally, two-dimensional posterior mean and variance plots along selected parameter slices provided interpretable visualizations of response surface evolution. These methods are widely adopted in BO contexts for monitoring convergence, assessing uncertainty calibration, and supporting mechanistic interpretation [20, 87].

3.5.4 Statistical Analysis and Validation Protocol

System Consistency Monitoring To assess potential temporal drift arising from observed changes in stirrer fouling behavior, three baseline experiments (180°C, 180 min, no additives) were strategically incorporated during iterations 8 and 9 of the optimization campaign. These baseline tests were performed inline within the campaign without time delays to monitor

system consistency in real-time. Statistical evaluation was performed using two-sample t-tests comparing these results with initial reference experiments, complemented by standardized residual analysis to assess GP model performance over time. Despite their primary purpose as quality control measures, these baseline experiments were included in the adaptive learning training set to maintain comprehensive data coverage for GP model development.

Independent Validation Strategy Six independent validation experiments were reserved for post-optimization to assess model predictive accuracy and generalization capability without bias from training data inclusion. Validation point selection follows best practice in the active learning literature [83], employing a dual-criteria approach targeting both high-performance regions (posterior mean maximization) and high-uncertainty regions (posterior variance maximization). This strategy enables comprehensive evaluation of the surrogate model across both exploitative and exploratory domains. Three validation experiments were selected to replicate the best seen experiments, i.e., those that yielded the highest observed carbonation yields during the optimization campaign. This allowed assessment of experimental repeatability and evaluation of model robustness under optimal yet previously tested conditions. The remaining three experiments specifically probed undersampled regions with the largest predicted posterior uncertainty to assess extrapolation behavior and the reliability of uncertainty quantification in critical, yet poorly explored, areas of the design space. Model performance was evaluated along two complementary axes: (i) *prediction accuracy*, assessed via the absolute deviation from the posterior mean, and (ii) *uncertainty calibration*, evaluated through empirical coverage probabilities of the 95 % confidence intervals. Together, these analyses ensured that both pointwise accuracy and uncertainty quantification were reliable, thereby validating the model's suitability for informed decision-making in variable industrial settings.

The results of this validation procedure are presented and analyzed in Section 4.5.4.

Modeling assumptions

A comprehensive overview of the key modeling assumptions, architectural choices, implementation decisions and final model settings is provided in Appendix A.5. This summary supports transparency, reproducibility, and facilitates critical evaluation of the surrogate modeling framework.

3.5.5 Chemical Process Interpretation

To extract mechanistic insights and process understanding from the optimized GP model, an analysis framework was implemented combining response surface interpretation with kinetic modeling. The final GP model, with its calibrated hyperparameters, served as the foundation for generating two-dimensional contour plots and one-dimensional slice plots using the posterior mean predictions. These visualizations incorporated GP uncertainty estimates to assess prediction reliability across the parameter space.

Kinetic Analysis Framework

Building upon the GP-based response surface, a post hoc kinetic analysis was conducted to mechanistically interpret the observed parameter dependencies, with a particular focus on the characteristic temporal evolution of carbonation yield. Three kinetic control mechanisms were systematically evaluated via linear regression analysis using GP-predicted time–conversion trajectories across the experimental parameter space [12] and is based upon the Shrinking Core Model (see Section 2.1.3):

- **Mass Transport Control:** $t = k_1 \cdot \alpha$
- **Diffusion Control:** $t = k_2 \cdot \left[1 - \frac{2\alpha}{3} - (1 - \alpha)^{2/3}\right]$
- **Chemical Reaction Control:** $t = k_3 \cdot \left[1 - (1 - \alpha)^{1/3}\right]$

where α denotes the fractional carbonation reaction yield and k_1 , k_2 , and k_3 are the corresponding apparent rate constants.

These models represent distinct rate-limiting regimes typically encountered in heterogeneous carbonation systems: mass transport control implies that ion transport (e.g., Ca^{2+} , Mg^{2+} from solid particles and CO_3^{2-} , HCO_3^- from solution) limits the overall reaction rate; diffusion control suggests that the formation of a product layer imposes resistance to reactant transport; chemical reaction control indicates that surface dissolution kinetics of reactive silicate phases govern the overall conversion rate. Model discrimination was performed via comparative regression analysis, using the coefficient of determination (R^2) as the primary selection criterion. This allowed identification of the dominant kinetic regime under varying process conditions and provided mechanistic context for the parameter interactions and the chemical interpretations derived from the GP response surfaces.

The used approach draws on established methodologies in mineral carbonation research [14],

yet must be interpreted with caution in the present context. Specifically, the analysis relies on GP-predicted time-conversion profiles rather than direct time-series measurements. The underlying assumption is that the GP posterior mean, trained on experimentally validated data and constrained by domain knowledge, provides a sufficiently accurate and continuous surrogate of the true time-conversion behavior across the design space. While this approximation enables the principled exploration of kinetic regimes under practical budget limitations, it does not constitute a rigorous kinetic study in the classical sense [38]. Nonetheless, this surrogate based framework offers a tractable means to generate mechanistic hypotheses, identify characteristic rate regimes, and inform future targeted experimentation under resource constraints.

3.6 Retrospective Benchmarking Framework

To systematically evaluate the efficiency of the implemented Bayesian Optimization strategy and validate its superiority over traditional experimental design approaches, a comprehensive benchmarking study, comparable to the methodology of Rummukainen et al. [88], was conducted using a synthetic ground truth function derived from experimental data collected in this study, which served as a virtual test environment for evaluating different optimization strategies under controlled, reproducible conditions.

The analysis pursued three primary objectives: (i) validation of BO performance against established design-of-experiments (DoE) methodologies, (ii) optimization of acquisition function strategies and initial sampling approaches for future campaigns, and (iii) quantification of performance penalties associated with batched versus sequential experimental designs. This retrospective analysis does not aim to offer a universally applicable proof of BO's superiority but rather provides system-specific insight for carbonation processes with comparable complexity and noise levels. It enables a quantitative evaluation of preferable acquisition strategies, the trade-off between initial sampling and adaptive iterations, and the expected convergence behavior in comparable noisy optimization problems. To simulate realistic experimental decision-making and evaluate the efficiency of different optimization strategies, the experimental conditions proposed by both classical DoE approaches and Bayesian Optimization were emulated using the synthetic ground truth function as surrogate evaluator. Each experimental design was treated as a function call with added noise, representing a single virtual experiment. To ensure robust statistical comparison and suppress the influence of random initialization effects, all Bayesian Optimization loops were repeated using 20 independent random seeds per configuration. For comparability, a fixed total budget of 47

function evaluations was applied across all strategies, reflecting the number of experiments performed during the actual Bayesian optimization campaign (20 foundation experiments + 27 adaptive optimization) (see Section 3.5.2). As a result, the number of Bayesian iterations varied depending on the number of initial design points (e.g., 5, 10, or 20), with the remaining budget allocated to acquisition-based optimization steps.

This framework enabled rigorous benchmarking across strategies while maintaining alignment with practical experimental constraints and data availability.

Synthetic Ground Truth Function Development

A third-degree polynomial function was constructed based on the complete experimental dataset, incorporating key parameter interactions, non-linear temperature dependencies, and additive effects observed in the carbonation system:

$$f_{\text{synthetic}}(\mathbf{x}) = \beta_0 + \sum_{i=1}^4 \beta_i x_i + \sum_{i=1}^4 \sum_{j=i}^4 \beta_{ij} x_i x_j + \sum_{i=1}^4 \sum_{j=i}^4 \sum_{k=j}^4 \beta_{ijk} x_i x_j x_k + \epsilon \quad (3.10)$$

where coefficients were determined through regression analysis of experimental data ($R^2 = 0.83$). This polynomial formulation served as a realistic surrogate for the true process landscape, enabling controlled evaluation of optimization strategies without experimental variability.

Comparative Benchmarking Methodology

Traditional DoE Strategy Comparison To benchmark Bayesian Optimization against classical design approaches, three one-shot strategies, orthogonal Latin Hypercube Sampling, random sampling, and full factorial designs, were evaluated under equivalent conditions [17]. Each method selected 47 sampling points from the synthetic ground truth function, representing virtual experiments in the 4D design space. Subsequently, identical Gaussian Process models were trained on the resulting datasets and optimized to identify the predicted global optimum. This unified post-processing isolates the influence of the sampling strategy by eliminating model architecture as a confounding factor (see Appendix A.9.2 for implementation details). Traditional DoE strategies allocated all samples simultaneously based on space-filling or systematic principles, whereas BO employed sequential sampling, starting from 5, 10, or

20 initial points, followed by acquisition-guided optimization. Performance was assessed by evaluating the true function value at the predicted optimum, quantifying how effectively each strategy exploited the available evaluation budget to identify high-yield conditions.

Acquisition Function Strategy Optimization Three acquisition functions were evaluated in direct comparison: qLogNoisyExpectedImprovement (noise-explicit formulation), qGIBBON (information-theoretic approach), and UCB (classical confidence bound)(see Appendix A.1.2). Each acquisition function was simulated across different initial sample sizes (5, 10, 20) via orthogonal LHS to investigate the influence of initialization strategy on overall BO performance and convergence behavior. Initial sample sizes of 5, 10, and 20 points were chosen to reflect commonly used heuristics in Bayesian optimization, which suggest using approximately $5d$ to d^2 design points, balancing initial space-filling coverage with sufficient budget for adaptive refinement. All simulations employed single-point sampling (batch size = 1), with the remaining experimental budget allocated to sequential acquisition-guided optimization. UCB was implemented using an α parameter of 2; variation of α was not investigated in this study.

Sequential versus Batched Design Assessment The performance of pure sequential design (single experiment per iteration) was compared with batched designs (3 experiments per iteration) to assess the impact of model update frequency and parallelization constraints on overall BO effectiveness. Sequential sampling enables maximum learning from each experimental result through frequent model updates, while batched approaches reduce experimental overhead but limit adaptive learning opportunities. This comparison quantifies the trade-off between experimental efficiency and optimization performance under practical laboratory constraints.

Experimental Realism and Noise Modeling

To replicate realistic experimental conditions, observation noise was introduced according to the posterior variance distribution observed in the actual experiments:

$$f_{\text{observed}}(\mathbf{x}) = f_{\text{synthetic}}(\mathbf{x}) + \epsilon_{\text{noise}} \quad (3.11)$$

where $\epsilon_{\text{noise}} \sim \mathcal{N}(0, \sigma^2)$ and $\sigma = 4\%$ based on the final GP model's noise estimate (see

Section 4.5.3). This enabled robust evaluation of each optimization strategy's resilience under noisy and uncertain data conditions.

Regret Analysis Optimization performance was primarily evaluated using the regret metric, which quantifies the difference between the known global optimum of the synthetic function and the true function value at the location predicted as optimal by each strategy's final surrogate model:

$$\text{Average Regret} = \frac{1}{N} \sum_{i=1}^N |f_{\text{true}}^* - f_{\text{true}}(\mathbf{x}_{\text{predicted},i}^*)| \quad (3.12)$$

where f_{true}^* represents the true global optimum of the synthetic ground truth function, $\mathbf{x}_{\text{predicted},i}^*$ is the location identified as optimal by the i -th independent run's final Gaussian Process model, $f_{\text{true}}(\mathbf{x}_{\text{predicted},i}^*)$ is the true function value at that predicted location, and $N = 20$ represents the number of independent replications per configuration. This metric directly measures how effectively each optimization strategy guides the discovery process toward high-performing parameter combinations, with lower regret values indicating superior optimization performance.

Model Quality Assessment Surrogate model performance was evaluated using two complementary metrics: global R^2 and $\text{RMSE}_{\text{high}}$. The R^2 score quantifies the overall variance explained by the GP model across the design space, capturing global trend fidelity. In contrast, $\text{RMSE}_{\text{high}}$, the root mean square error within the top 20% of true yield values, assesses local accuracy in the most relevant performance regime, where precise predictions are critical for identifying optima and guiding decisions. This dual evaluation provides a robust measure of both generalization and practical optimization utility in carbonation process applications (implementation details see Appendix A.9.2).

4 Results and Discussion

4.1 Chapter Overview

The following chapter presents the experimental and modeling results of the carbonation campaign and provides a structured interpretation of the underlying chemical and process phenomena and an assessment of the Bayesian optimization framework effectiveness. Section 4.2 outlines the baseline performance and additive screening results, establishing the foundation for process optimization. The optimization results obtained via Bayesian optimization are presented in Section 4.3, including the identification of global optima and parameter sensitivities. Section 4.4 analyzes the modeled response surfaces to derive mechanistic insights, covering individual parameter effects, kinetic regimes, and synergistic interactions. This includes a detailed assessment of mass transport limitations and their experimental validation through scale-up studies. The role of physical model constraints is evaluated in the final part of this section. Section 4.5 assesses the performance and convergence behavior of the Bayesian optimization strategy, the quality and calibration of the surrogate model, and the robustness of the final predictions. Independent validation experiments and a detailed analysis of experimental uncertainties are included. The chapter concludes with retrospective benchmarking simulations in Section 4.6, comparing the applied optimization strategy against classical design-of-experiment approaches to derive generalizable insights for adaptive experimentation in comparable chemical systems.

4.2 Baseline Performance and Additive Screening

Preliminary carbonation experiments under reference conditions (180 °C, 100 bar, 180 min, S/L = 0.4 g/mL) established a baseline carbonation yield of 30.2 ± 2.9 % or 152.5gCO₂/kg of slag . This result is lower than the typical results found in the literature for similarly reactive slags [9].

Table 4.1: Additive screening results under standardized conditions (180 °C, 180 min, 100 bar, S/L = 0.4 g/mL). For calcium-based additives, dosages are expressed as calcium weight percent relative to feedstock mass. Enhancement values in parentheses represent absolute percentage point improvements relative to the additive-free reference condition.

Additive	Dosage	Reaction Yield (%)
None (reference)	–	30.2 ± 2.9
<i>Calcium-based additives</i>		
CaO	1 wt% Ca	46.4 (+16.2)
Ca(OH) ₂	1 wt% Ca	43.0 (+12.8)
CaCl ₂	1 wt% Ca	42.1 (+11.9)
CaCO ₃	1 wt% Ca	30.5 (+0.3)
<i>Sodium-based additives</i>		
NaCl	1 M	36.3 (+6.4)
NaHCO ₃	1 M	26.4 (–3.8)
<i>Other additives</i>		
CaSO ₄ ·2H ₂ O	1 wt% Ca	25.1 (–5.1)

Note: Applied NaCl concentration exceeds the permissible Cl[–] content for cement applications by a factor of 11. CaCl₂ concentration by a factor of 2.2 (see Section 3.5.2).

CaO achieved the most significant improvement through its dual role as a Ca²⁺ source and strong base that maintains favorable dissolution conditions. **Ca(OH)₂** showed comparable but slightly lower enhancement, likely due to differences in particle reactivity compared to in-situ formed Ca(OH)₂ from CaO hydration. **CaCl₂** also provided high improvement through combined calcium supply and ionic strength modification effects. CaCl₂ was tested at higher concentrations in the screening experiments, although typical SCM applications restrict the Ca content to a maximum of 0.45 wt% (see Section 3.5.2).

At a concentration of 1.0 mol/L, **NaCl** yielded a substantial improvement in reaction yield (+6.4 pp), indicating that increased ionic strength may indeed enhance carbonation kinetics, likely by promoting CO₂ solubility and ion mobility. However, this dosage exceeds the permissible chloride content in supplementary cementitious materials by a factor of approximately 11. Assuming linear scaling, reducing the concentration to the application-relevant threshold of 0.090 mol/L results in a corrected yield increase of only +0.6 pp, in agreement with previous studies reporting marginal effects of NaCl in steel slag systems [89]. pH modifiers like **NaHCO₃** decreased yields (–3.8 pp), indicating that bicarbonate buffering at suboptimal pH inhibits silicate breakdown or carbonate precipitation [90].

Nucleation templates (**CaCO₃**, **CaSO₄·2H₂O**) showed negligible to negative effects. While

CaCO_3 typically facilitates nucleation by reducing energy barriers [91], its minimal impact (+0.3 pp) suggests ineffectiveness in this feedstock. $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ reduced yields (−5.1 pp), likely due to competitive precipitation or sulfate-induced crystallization inhibition, as reported by Zarga et. al [92].

These results underscore the need for feedstock-specific optimization, as observations from ultramafic systems (e.g., olivine) cannot be directly transferred to slag systems, highlighting strong interactions with specific mineralogy and surface chemistry. The screening results demonstrate clear distinctions between enhancement mechanisms, with calcium-based additives showing superior performance compared to solely ionic strength modifiers in compliant dosages. These results serve as initial indicators rather than statistically validated performance metrics. While enhancement effects for CaO , $\text{Ca}(\text{OH})_2$, and CaCl_2 appear distinct, statistical significance would require dedicated testing.

Based on these findings, **CaO and CaCl_2 were selected as the primary optimization variables** for the Bayesian optimization campaign. These additives represent complementary enhancement strategies: CaO promises sustained calcium release, while CaCl_2 promises rapid calcium availability and increased ionic strength.

4.3 Description of Optimization Results

The GP model trained through sequential Bayesian optimization suggests identification of the global optimum. Optimization was performed through systematic maximization of posterior mean values while accounting for model uncertainties derived from the Bayesian framework (see Section 4.5).

Table 4.2: Identified global optimum with uncertainty quantification from GP model predictions

Parameter	Optimal Value	95 % Confidence Interval
Maximum Yield (%)	53.2	[50.2, 56.2]
Process Conditions:		
Temperature (°C)	157	[142, 169]
Reaction Time (min)	180	[164, 180]
CaO Addition (wt % Ca)	2.00	[1.36, 2.00]
CaCl ₂ Addition (wt % Ca)	0.00	[0.00, 0.34]
Prediction Uncertainty	1.9 %	

The identified global optimum is located at high reaction times and the upper bound of Ca

addition as CaO, with 0 wt % Ca provided as CaCl₂. The optimal temperature was found to lie at a moderate level of 157 °C. The reaction yield prediction of 53.2 % (269 g CO₂/kg slag) at this point is associated with narrow uncertainty bounds ($\pm 1.9\%$), indicating high model confidence in the vicinity of the optimum.

4.4 Chemical Process Understanding

The Bayesian optimization campaign not only identified optimal process conditions but also generated a comprehensive dataset suitable for mechanistic interpretation of ferrochrome slag carbonation. This section aims to extract chemical process insights from the GP-modeled response surfaces, parameter sensitivities, and inferred reaction kinetics. The analysis seeks to establish a quantitative ranking of parameter importance, explore individual effects and synergistic interactions, and develop hypotheses regarding dominant rate-controlling mechanisms under varying conditions. All interpretations are made in the context of GP posterior uncertainty, particularly near the design space boundaries where extrapolation becomes less reliable. These findings do not represent definitive mechanistic proof but rather serve as a basis for formulating targeted hypotheses that merit further experimental validation.

4.4.1 Individual Parameter Effects and Mechanisms

Reaction Time: Saturation Kinetics and Transport Limitations Among all investigated variables, reaction time exhibits the highest sensitivity and follows a characteristic saturation profile, modulated by the presence of calcium-based additives over the investigated range of 30–180 minutes (Figure 4.1). In additive-free systems (0 wt % Ca, Curve 3), carbonation yield increases rapidly within the first 60 minutes, after which the reaction decelerates markedly, reaching approximately 37 % after 180 minutes. This trend aligns with previous findings in steel slag carbonation [48, 38], and suggests a transition from reaction-controlled to diffusion-limited kinetics, likely caused by the progressive formation of passivating product layers on slag particle surfaces (see Section 2.3). Mechanistically, the response surface reveals two distinct kinetic regimes:

Initial Phase (0–60 min): Carbonation progresses rapidly, governed primarily by the intrinsic dissolution kinetics of reactive silicate phases such as Ca₂SiO₄ and Ca₃Mg(SiO₄)₂, with minimal diffusional hindrance and high surface accessibility. The fast release of Ca²⁺ and Mg²⁺ into the aqueous phase supports elevated reaction rates, consistent with the behavior of merwinite-dominated feedstocks [93, 45].

Intermediate Phase (60–180 min): As product layers accumulate, diffusive resistance increases, slowing ion transport and leading to reduced carbonation rates. Under additive-free conditions (Curve 3), yield increases from 26 % to 37 %, a relative improvement of approximately 42 %, in agreement with the transition to diffusion-limited behavior reported by Di Giovanni et al. [9].

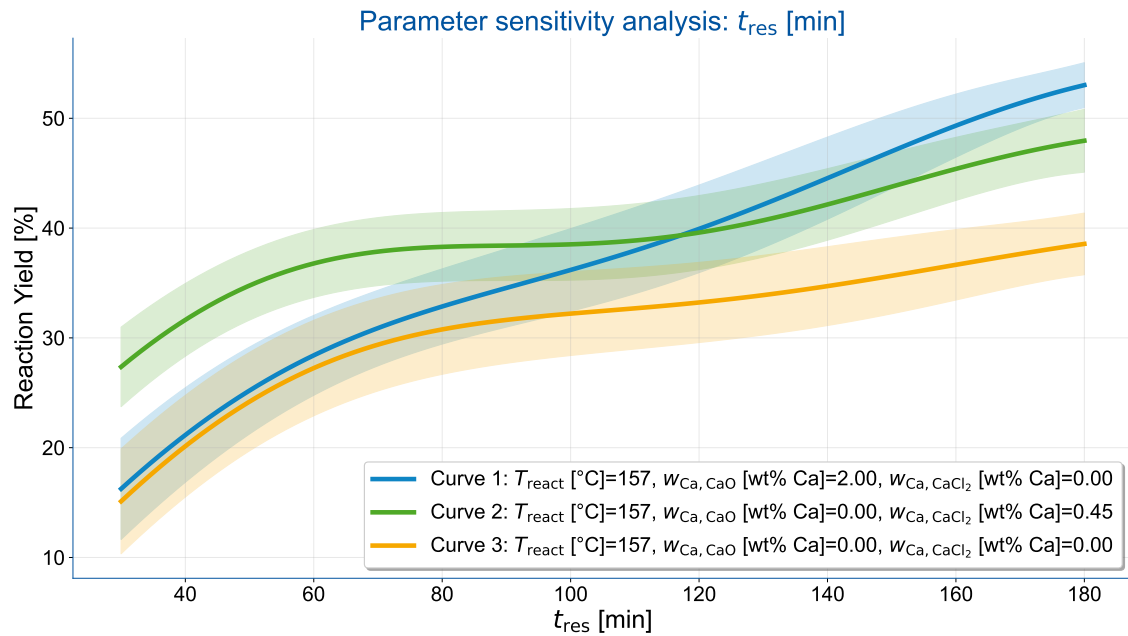


Figure 4.1: Reaction time slice analysis at optimal temperature (157 °C), comparing additive conditions. Without additives (Curve 3): saturation at ~37 % after 180 min. With CaO (2 wt % Ca): sustained linear progression to 53 %. With CaCl₂ (0.45 wt % Ca): enhanced early yields but limited long-term progression. Error bars represent GP model uncertainty ($\pm 1\sigma$).

Temperature: Competing Thermodynamic and Kinetic Effects Temperature is the second most influential parameter, exhibiting a bell-shaped response profile with a robust high-yield regime observed between 145–165 °C under otherwise optimal process conditions (Curve 1, Figure 4.2) [11]. Yields in this region ranged from 30.2 % to 53.2 %. Notably, compared to the sharply defined maxima reported by Bremen et al., the observed optimum in this study appears broader and less peaked, indicating a wider operational window.

The optimum at 157 °C results from the interplay of opposing thermodynamic and kinetic effects. On the one hand, higher temperatures enhance the dissolution kinetics of Ca^{2+} and Mg^{2+} from the slag matrix, thereby accelerating leaching and supporting rapid nucleation and precipitation, in line with the observations of Smeets et al. [91]. On the other hand, elevated temperatures reduce the solubility of CO_2 in water (as governed by Henry's law) and shift the carbonic acid speciation equilibrium away from bicarbonate (HCO_3^-), thereby diminishing the

concentration of reactive carbon species. This trade-off defines a thermokinetic “sweet spot,” wherein sufficiently high leaching rates must be balanced against reduced carbon availability to maintain high reaction efficiency.

Additionally, the exothermic nature of the carbonation reaction may lead to increased rates of reverse decomposition reactions at excessive temperatures, partially offsetting kinetic benefits. Higher temperatures can also influence product morphology: as shown by Wang et al. [12], they may favor the formation of dense, amorphous CaCO_3 layers over porous crystalline calcite, leading to surface passivation and reduced accessibility.

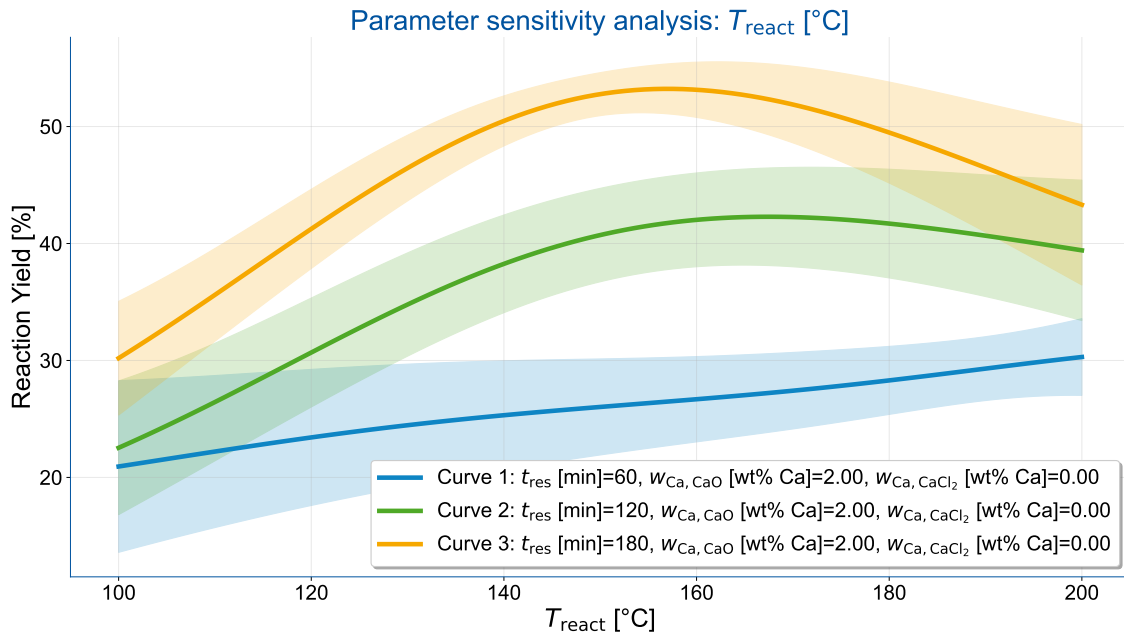


Figure 4.2: Temperature slice analysis at reaction times of 60, 120, and 180 min under optimal additive conditions (2 wt % Ca as CaO, 0 wt % Ca as CaCl_2). Bell-shaped response with optimum at 157°C , showing increasing yield magnitude with reaction time while maintaining consistent optimal temperature. Error bars represent GP model uncertainty ($\pm 1\sigma$).

The identified optimum at 157°C is consistent with mechanistic expectations but deviates from findings for other materials and conditions. For instance, Bremen et al. [11] reported higher optima ($175\text{--}210^{\circ}\text{C}$) for Mg-rich forsterite at 150,bar. This shift can be attributed to differences in feedstock composition (Ca-rich vs. Mg-rich), particle size ($D_{50} < 54.5, \mu\text{m}$ vs. $< 25, \mu\text{m}$), and particularly system pressure. Lower pressure (100,bar vs. 150,bar) reduces the solubility of CO_2 in the aqueous phase, limiting carbonation kinetics and thereby shifting the temperature optimum to lower values in this study.

Further, studies on other slag types reveal considerable variability: Kim et al. [46] found temperature optima around $40\text{--}80^{\circ}\text{C}$ for EAF slags, while Matus et al. [57] reported maxima

near 175 °C for ladle and EAF slags at 40–80 bar. These discrepancies underscore the critical importance of material-specific behavior, particle size, and pressure in defining optimal temperature ranges for carbonation processes.

Calcium-Based Additives: Hypothesized Mechanisms of Nucleation Promotion and pH Buffering Calcium-based additives lead to notable, albeit secondary, improvements in carbonation yield relative to thermal parameters. Under optimized conditions (157 °C, 180 minutes), the yield increases from 39.5 % (no additive) to 53.2 % at 2 wt % Ca (added as CaO), corresponding to a relative improvement of 33.7 % (Figure 4.3). The presence of a clear minimum at zero additive further suggests that all tested formulations exert a net positive effect. These findings point to underlying mechanistic pathways by which Ca-based additives may enhance performance, although dedicated validation remains pending. Further slice analyses are provided in Figure A.3a for CaO and in Figure A.3b for CaCl₂.

CaO – Working Hypothesis: Soluble Calcium Source and Nucleation Substrate. The observed yield increase associated with CaO addition may be explained by its behavior during slurry formation and reaction. Upon contact with water, CaO rapidly hydrates to form calcium hydroxide:



This Ca(OH)₂ then dissolves partially:



Given its higher solubility compared to calcium silicates ($K_{\text{sp}} = 5.5 \times 10^{-6}$), it is hypothesized that Ca(OH)₂ acts as a sustained Ca²⁺ source, supporting continuous carbonate precipitation even in later stages of the reaction [45]. This may help prevent localized depletion of Ca²⁺, which otherwise limits reaction progression. In parallel, hydroxide release increases the slurry pH, likely shifting the CO₂ speciation equilibrium toward carbonate ions (CO₃²⁻) [94]. This may promote supersaturation, increase ionic strength, and reduce diffusional resistance within the slurry [95]. Whether this pH effect persists throughout the entire reaction or is localized to the early phase remains to be validated. In addition to chemical effects, the solid Ca(OH)₂ phase may serve as a heterogeneous nucleation substrate. It is hypothesized that the dispersed Ca(OH)₂ particles increase the number of energetically favorable nucleation sites for CaCO₃, thereby accelerating initial crystallization and promoting more uniform precipitation, consistent with the predictions of population balance models [11]. This dual-function hypothesis is supported by earlier findings from Huijgen et al., who reported enhanced CO₂

uptake in slags with higher intrinsic portlandite ($\text{Ca}(\text{OH})_2$) content [4]. However, the exact contributions of pH modulation versus nucleation enhancement remain to be disentangled experimentally. Preliminary in situ pH observations during autoclave runs provide limited but suggestive evidence. Additive-free systems exhibited distinctly acidic conditions, while setups containing 2 wt % Ca as CaO maintained significantly higher pH throughout the reaction. These trends, albeit based on a small number of measurements, support the notion that CaO, and its in situ transformation to $\text{Ca}(\text{OH})_2$, acts as a pH buffer, potentially stabilizing conditions favorable for continuous carbonate formation [96].

In summary, it is hypothesized that CaO acts through a dual mechanism, (i) ensuring sustained release of Ca^{2+} and OH^- to maintain favorable chemical conditions, and (ii) promoting nucleation via solid-phase templating. Both pathways plausibly contribute to the observed yield increase, although further targeted studies are required to confirm their relative importance and dynamics.

CaCl_2 – Hypothesized Ionic Strength Mechanism: The addition of CaCl_2 leads to a marked increase in carbonation yield at low concentrations (< 0.25 wt % Ca), with a plateauing effect observed at higher dosages. Mechanistically, this effect may be attributed to two interacting phenomena [82, 97]: the complete dissociation of CaCl_2 provides fully soluble Ca^{2+} ions without requiring hydration, and the concurrent release of Cl^- ions increases the ionic strength of the aqueous phase:



This dual release is hypothesized to enhance carbonation by promoting rapid Ca^{2+} supersaturation and improving ion transport. Elevated ionic strength may facilitate Ca^{2+} diffusion through the slurry and disrupt early CaCO_3 surface layers, thereby enabling continued access to reactive calcium sources within the slag particle core [97, 98]. These effects could explain the steep performance increase observed at low CaCl_2 dosages in the contour plots. The immediate and complete Ca^{2+} availability is consistent with the findings of Moon et al. [82], who reported enhanced initial carbonation rates for CaCl_2 compared to $\text{Ca}(\text{OH})_2$ or CaO. Further pH measurements would help to establish whether the reduction in Ca^{2+} due to the suspected strong carbonate precipitation at the start of the reaction leads to a subsequent drop in pH. Unlike CaO-derived $\text{Ca}(\text{OH})_2$, CaCl_2 does not provide a solid-phase nucleation template, as it is entirely soluble. However, the rapid generation of high Ca^{2+} concentrations may induce a high degree of supersaturation with respect to CaCO_3 , which in turn accelerates both heterogeneous nucleation on existing slag surfaces and early-stage crystal growth. This mechanism could support the formation of initial carbonate phases even

in the absence of added nucleators. At elevated CaCl_2 dosages, however, negative effects may emerge. High ionic strength environments are known to reduce the solubility of CO_2 in water, a salting-out effect, which decreases the concentration of dissolved carbonate species and offsets the benefits of Ca^{2+} excess [95]. This could explain the plateau or slight decline in performance observed at higher additive loadings, suggesting the existence of an optimal concentration window for effective use [97].

In summary, CaCl_2 appears to enhance early-stage carbonation through immediate Ca^{2+} availability and ionic strength-mediated improvements in mass transport and surface activation. While it does not act as a nucleation substrate or pH buffer, the induced supersaturation likely accelerates nucleation and crystal growth in the initial reaction phase. However, its effect is transient and dosage-dependent, and sustained enhancement throughout the full reaction duration appears limited. When carefully dosed, CaCl_2 may serve as an effective kinetic promoter in calcium-rich slag carbonation. Notably, once Ca^{2+} is depleted through precipitation, Cl^- remains in solution without contributing to pH stabilization. Unlike $\text{Ca}(\text{OH})_2$, CaCl_2 does not buffer the system; on the contrary, the accumulation of Cl^- may contribute to a gradual decline in pH during later reaction stages, potentially inhibiting continued CO_3^{2-} formation and limiting overall conversion.

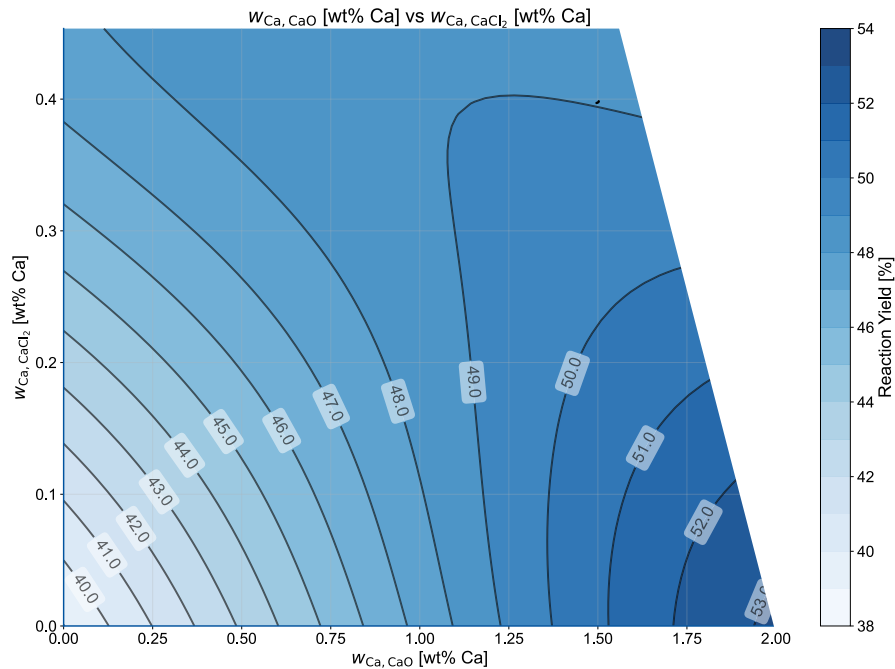


Figure 4.3: Additive interactions under optimal thermal conditions (157°C , 180 min). Yields range from 39.5 % (no additives) to 53.2 % (2 wt % Ca as CaO). Higher sensitivity to CaCl_2 compared to CaO at equivalent calcium loadings. White region excluded by 2 wt % total calcium constraint.

Optimal Additive Configuration Under the imposed 2 wt % total calcium constraint, the optimal configuration favors exclusive CaO addition (2.0 wt % Ca as CaO, 0.0 wt % as CaCl₂), suggesting that CaO provides more sustained reactivity through its dual role as a Ca²⁺ reservoir and heterogeneous nucleation substrate. Its buffering capacity and solid-phase dispersion support prolonged supersaturation and precipitation, particularly at elevated loadings. In contrast, CaCl₂ enhances early-stage kinetics via immediate Ca²⁺ release and increased ionic strength but exhibits a plateauing effect at higher concentrations. As shown in Figure 4.3, CaCl₂ sensitivity is high at low dosages, yet overall yield gains are limited compared to CaO. Moreover, mixed additive systems show non-additive interactions: elevated ionic strength may suppress Ca(OH)₂ solubility and reduce its nucleation effectiveness. This aligns with findings by Atkins et al. [36], who note that solubility in multicomponent systems is governed by the most soluble species, here, CaCl₂, potentially dominating the aqueous chemistry and hindering CaO's sustained contribution. While CaO provides a structural platform for nucleation, entirely soluble additives like CaCl₂ lack this function. Other nucleators, such as MgCO₃, have been successfully applied in magnesium-rich systems [13], indicating that additive interactions are highly material-specific.

In summary, the superiority of pure CaO reflects its combined chemical and structural roles, whereas CaCl₂ acts as a short-lived kinetic initiator. These findings underscore the importance of mechanistic compatibility in additive design and suggest limited synergy between soluble and solid-phase calcium sources under fixed composition constraints.

4.4.2 Parameter Interactions and Synergistic Effects

Time–Additive Interactions: Overcoming Transport Limitations The time–additive interaction reveals how calcium-based additives enable continued carbonation beyond the 60-min saturation threshold observed in non-additive systems (Figure 4.1). In particular, the system with 2 wt % Ca as CaO exhibits an increase from 26 % to 53 % yield between 60–180 min, corresponding to a 104 % relative improvement during this late-stage period, while the additive-free baseline plateaus at approximately 37 %.

CaO Temporal Profile: CaO-dosed systems follow a two-phase trajectory: initially (0–60 min), they mirror the baseline (Curve 1 in Figure 4.1). A clearer divergence occurs beyond 80 min, where yields continue rising, albeit at reduced slope, despite the onset of product-layer diffusion barriers. This behavior suggests that the early, rapid carbonation enabled by the dispersed Ca(OH)₂ addition not only facilitates an initial, immediate Ca²⁺ supply but also promotes spatially distributed nucleation. This leads to the formation of a more permeable CaCO₃ layer

on the slag particles, effectively reducing passivation via thinner layers and shorter diffusion paths, thereby sustaining the accessibility of calcium from the underlying slag phases even beyond 80 min [99, 100]. GP model extrapolations suggests that this trend continues towards the edge of the reaction time at 180 min. Whether this is to be understood as a boundary anomaly of the GP remains to be investigated. A contour plot for this interaction is shown in Figure A.4.

CaCl₂ Temporal Profile: In contrast, CaCl₂ leads to markedly higher yields during the initial 60 min due to immediate Ca²⁺ release and ionic strength enhancement [82] (Curve 2 in Figure 4.1). This facilitates ion mobility and early nucleation, yielding a sharp initial rise. Beyond 60 min, the curve flattens less than in the baseline, indicating some continued benefit from enhanced mass transport. However, without pH buffering or a solid nucleation substrate, CaCl₂ cannot sustain high nucleation density, leading to slower progression compared to CaO in later stages.

In summary, CaCl₂ provides strong early-stage enhancement but lacks the structural and chemical persistence of CaO. While moderate slope retention beyond 60 min suggests ongoing ionic-strength-driven transport benefits, CaO remains superior for sustaining reactivity under diffusion-limited conditions.

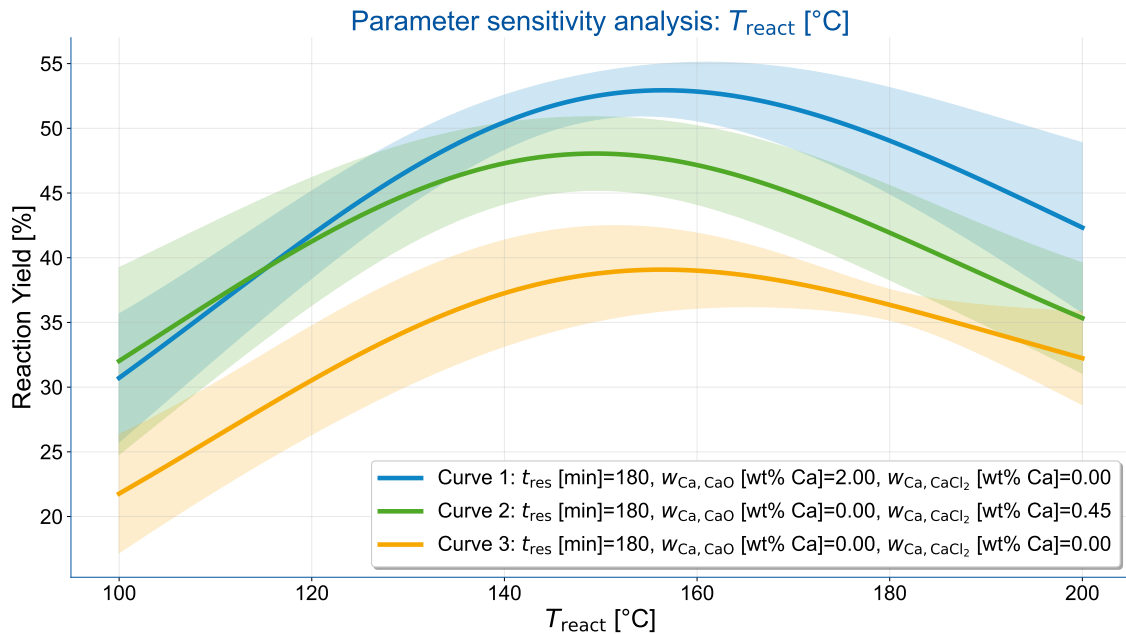


Figure 4.4: Temperature slice analysis at reaction times different additive compositions under optimal reaction time (180 min). Bell-shaped response surface remains, additional Ca as CaCl₂ enhances reaction conversion at lower temperatures, decreases at higher temperatures.

Temperature–Additive Interactions The addition of CaO preserves the characteristic bell-shaped temperature response while uniformly enhancing overall yields, without shifting the optimum at 157 °C. This suggests that CaO's effect, via nucleation promotion and sustained Ca^{2+} availability, amplifies reaction intensity within the existing thermokinetic regime, rather than modifying the underlying temperature–conversion balance. In contrast, CaCl_2 addition induces a noticeable shift of the optimal temperature toward more moderate values. This can be attributed to the compounded influence of reduced CO_2 solubility at elevated temperatures and salting-out effects associated with increased ionic strength. The latter may enhance carbonation kinetics at lower temperatures by promoting ion mobility and reactivity, effectively reshaping the temperature–conversion profile [98] (see also Figure A.5).

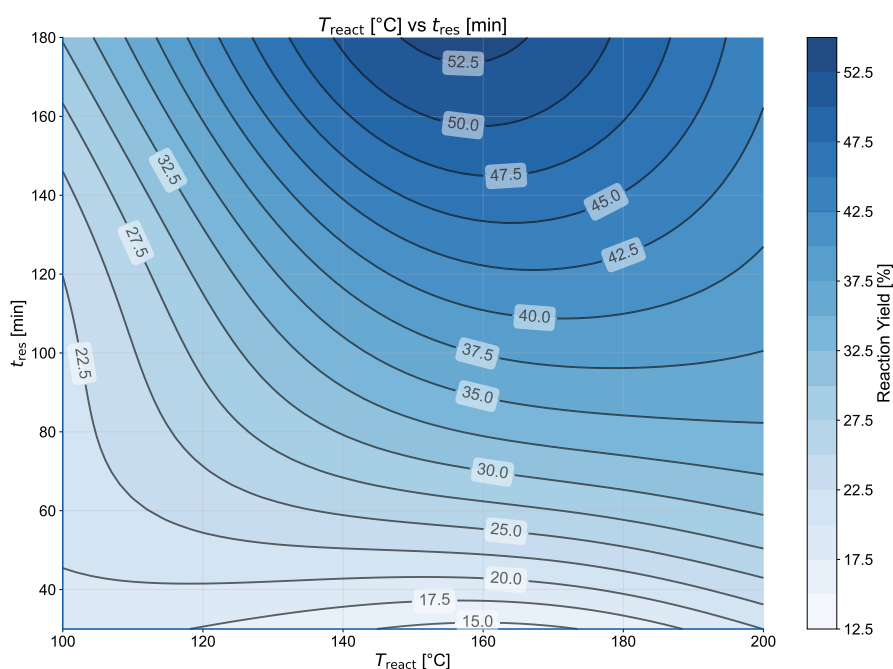


Figure 4.5: Temperature-time interactions under optimal additive conditions (2 wt % Ca as CaO, 0 wt % Ca as CaCl_2). Global optimum at 157°C and 180 min with robust high-yield region (>50 %) spanning 145–165°C and 160–180 min. Temperature independence at short times (<60 min) transitions to pronounced sensitivity at extended times.

Temperature–Time Interactions The two-dimensional response surface (Figure 4.5) reveals a non-linear interaction between temperature and reaction time under optimized additive conditions. At short durations (30–60 min), carbonation yield remains largely insensitive to temperature, indicating that limited reaction time constrains overall conversion regardless of thermal input. Temperature sensitivity emerges progressively with increasing duration, particularly in the low-temperature regime where kinetics dominate. At extended reaction

times (>120 min), a distinct temperature optimum forms and converges toward the global maximum at 157°C by 180 min. This pattern reflects the combined influence of accelerated feedstock dissolution at higher temperatures and CO_2 availability governed by speciation equilibria. The resulting high-yield regime spans approximately $145\text{--}165^{\circ}\text{C}$ and 160–180 min, defining a robust operational window for industrial deployment. Within this space, yields consistently exceed 52.5 %, offering practical process flexibility and tolerance to moderate deviations in time or temperature.

4.4.3 Transport and Reaction Limitations

Rate-Limiting Mechanism Identification To further substantiate the presence of transport limitations identified in the reaction time analysis, carbonation kinetics were evaluated using the shrinking core model framework, as outlined in Section 3.5.5. Regression analyses reveal that the prevailing rate-limiting mechanism varies notably with additive presence, deviating from patterns commonly reported for steel slag carbonation. Under optimal conditions (157°C , 180 min, 2 wt % Ca as CaO), the shrinking core model variant assuming mass transport control yields the highest coefficients of determination ($R^2 > 0.990$). The surface-reaction-controlled model performs nearly as well, often within an R^2 margin of 0.001–0.002, suggesting that the system operates near the boundary between transport- and reaction-limited regimes. A full table of R^2 values for all conditions is provided in Appendix A.7.

In the absence of additives, diffusion-controlled kinetics dominate, a result consistent with established observations for steel slag carbonation [9, 38]. However, upon addition of calcium-based additives, especially CaO, the kinetic behavior begins to resemble that of olivine systems, where surface-reaction-limited kinetics are frequently observed [101, 11].

This apparent mechanistic shift indicates that under additive-rich conditions, particularly with CaO, silicate dissolution is sufficiently accelerated and a sustained Ca^{2+} reservoir is established, such that proton or CO_2 transport to the reactive surface becomes the limiting step. Consequently, additives appear to suppress the typical onset of diffusion limitation, enabling continued reaction progression well beyond the saturation point observed in additive-free systems. The observed performance enhancement thus arises primarily from improved mass transport phenomena rather than a fundamental alteration in surface reaction kinetics.

Scale-Up Evidence for Transport Limitations The predominance of mass transport limitations is further supported by scale-up experiments. Under otherwise identical conditions, carbonation yields increased from 53.2 % in the small autoclaves (390 mL) to 80 % in a 5.7 L

reactor, a 26.8 % improvement in absolute yield attributable to enhanced mixing and more favorable hydrodynamics. This pronounced scale dependence indicates that suboptimal flow and limited suspension quality in the small reactor impose significant performance constraints. In particular, **solid–liquid mass transfer and ion transport within the bulk liquid** appear to dominate: sedimentation beneath the impeller (Figure 4.6) revealed stagnant zones with reduced particle–fluid interaction and interfacial exchange. While **gas–liquid mass transfer** may also play a role, its relative importance appears lower under the given conditions. The observed differences are more plausibly explained by weaker turbulence, insufficient radial flow, and less effective solid suspension in the small reactor. In contrast, the cooling-coil configuration of the larger system promotes secondary circulation and improved fluid dynamics, enhancing the renewal of liquid at reactive particle surfaces. These findings are consistent with mechanistic interpretations and the experimental results of Chang et al. [44], who emphasize the influence of hydrodynamic contact conditions. The high performance observed in the larger reactor also aligns with expected reactivity potentials based on feedstock composition and phase assemblage (Section 3.2). Comparable slags have achieved yields above 60 % under less stringent conditions [25], supporting the conclusion that observed small-scale limitations are reactor-induced rather than material-inherent.

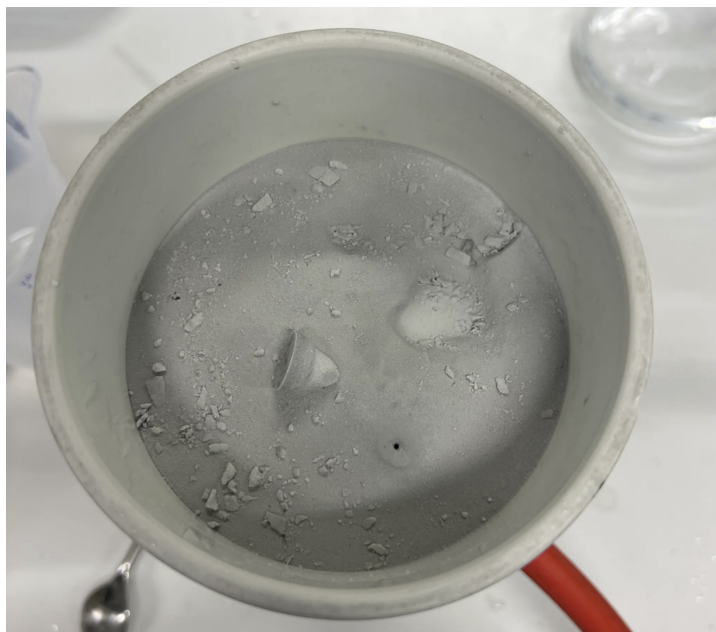


Figure 4.6: Observed sediment cone beneath the stirrer in the small-scale reactor. The conical shape indicates a persistent flow shadow and supports the hypothesis of localized mass transfer limitations.

Model Limitations Despite the robustness of the kinetic interpretation, the analysis must be viewed with appropriate caution due to the inherent uncertainties in the GP-based predic-

tions. The R^2 values associated with competing kinetic models often differ only marginally (typically by 0.01–0.03), suggesting that the system may reside within a multimechanistic transition regime. In such cases, predictive noise or local model variance could obscure the true rate-limiting mechanism or induce apparent shifts in model preference.

Nonetheless, the convergence of multiple independent lines of evidence, including the kinetic model fitting, qualitative trends in the time-dependent response plots, and reactor scale-up experiments, provides a coherent mechanistic picture. Even considering the narrow margins in statistical fitting, these findings collectively support the identification of a transport-limited carbonation regime under the evaluated conditions.

4.4.4 Physical Constraint Validation and Impact Assessment

Temporal Validity of Model Behavior To ensure physically realistic extrapolation, the Gaussian Process model was trained with boundary constraints using pseudo-experiments at $t = 0$ (0 % yield). This soft correction notably impacted early-time predictions (< 80 min), where data sparsity led the unconstrained model to predict implausible yields close to zero reaction time (unconstrained: Figure 4.7; constrained Figure 4.1). Such behavior would distort gradient estimates and mechanistic interpretation. In contrast, the constrained model respected chemical expectations, captured steeper early gradients, and preserved plausible curvature. For well-sampled regions (> 80 min), both models show comparable trends, confirming the validity of subsequent kinetic and process analyses.

Methodological Validation and Learning Efficiency In line with Rasmussen and Williams [77], the use of boundary conditions constitutes a meaningful structural prior that enhances model generalization. Here, the $t = 0$ anchor point improved early-phase gradient learning and stabilized model behavior in undersampled regions. Without this correction, the Bayesian Optimization routine would likely misallocate sampling effort or overestimate early reactivity. All kinetic regime classifications, parameter interaction effects, and transport limitation conclusions were derived from the constrained model, ensuring that interpretations remain grounded in physical plausibility. Since predictions in well-sampled domains remained unchanged, the constraint implementation is validated as a localized and non-intrusive correction.

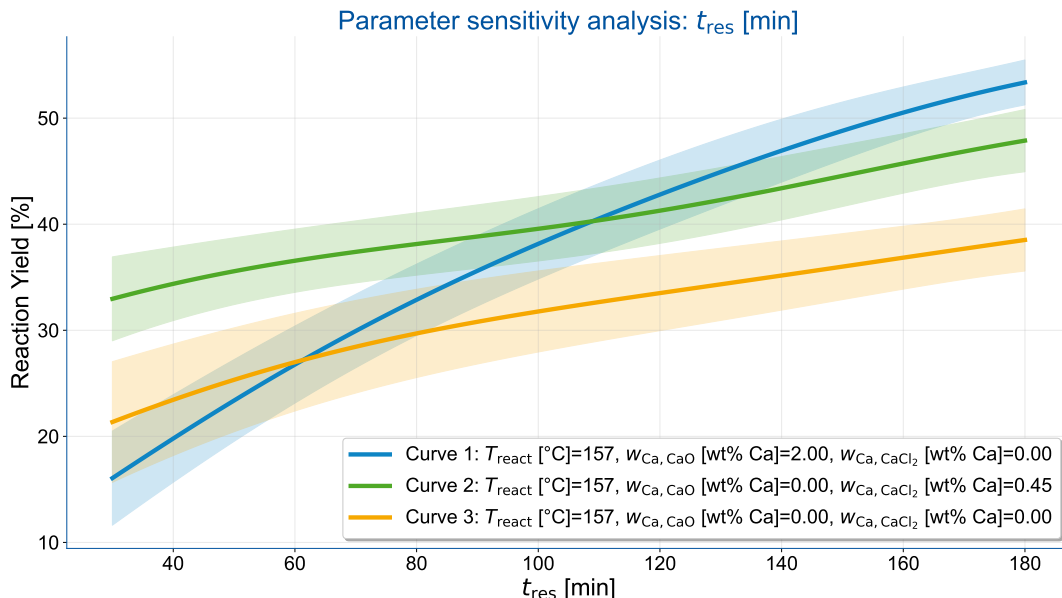


Figure 4.7: Predicted carbonation yield versus reaction time for the unconstrained GP model (constrained in Figure 4.2). The model fails to capture the expected zero-yield boundary at $t = 0$ and underestimates gradient steepness in the early phase, indicating insufficient physical plausibility in sparsely sampled regions.

4.4.5 Summary of Chemical Process Understanding

The combined experimental and model-based analysis establishes a coherent mechanistic interpretation of ferrochrome slag carbonation under the investigated conditions. Temperature (157°C) and reaction time (180 min) emerge as primary optimization levers, while calcium-based additives, especially CaO at 2 wt% Ca, provide substantial secondary enhancement through distinct mechanisms: CaO promotes sustained Ca^{2+} availability and heterogeneous nucleation; CaCl_2 delivers immediate ionic strength effects and high early Ca^{2+} concentrations.

Mass transport was identified as the dominant rate-limiting mechanism under optimized conditions, particularly in additive-enhanced systems. This was validated through scale-up experiments, where improved mixing and reactor hydrodynamics yielded an additional 26.8 % increase in carbonation efficiency. These findings indicate that further intensification should primarily address engineering limitations, such as flow and transfer characteristics, rather than purely chemical factors.

Importantly, the interpretations presented here, derived from GP-based surrogate modeling and systematic experimentation, are not intended as conclusive mechanistic proofs but as targeted hypotheses. They offer a robust and plausible framework to explain the observed

process behavior and provide a scientifically grounded basis for optimization.

These hypotheses now warrant focused experimental validation, especially in relation to transport facilitation, additive roles, and reactor design. The major insights are summarized in Table 4.3.

Table 4.3: Summary of Major Hypotheses on Ferrochrome Slag Carbonation Mechanisms

Category		Major Hypothesis
General Interpretation		GP-modeled insights are interpreted as mechanistic hypotheses requiring experimental validation.
Reaction Time		The process transitions from surface-reaction to diffusion-limited kinetics around 60 min, driven by CaCO_3 product layer formation in the absence of additives.
Temperature		The temperature optimum (157°C) reflects a balance between accelerated leaching and reduced CO_2 solubility/speciation at higher temperatures.
CaO Additive		CaO provides sustained Ca^{2+} and OH^- release and nucleation sites, reducing diffusion limitations by promoting thinner product layers.
CaCl_2 Additive		CaCl_2 enhances early-stage yield via immediate Ca^{2+} release and ionic strength increase but lacks pH buffering or nucleation templates. While its effect is transient, elevated ionic strength may moderately support sustained reaction under diffusion-limited conditions.
Additive Interactions		Pure CaO is preferred over mixed systems due to its persistent effects and the saturation of CaCl_2 benefits at low dosages.
Rate Limitation		Under optimized conditions, the process operates at the interface of mass transport and surface-reaction control, with additives facilitating transport.
Scale-Up		Small-scale efficiency is constrained by gas-liquid and solid-liquid transport limitations; large-scale setups reveal the process's full potential.

4.5 Bayesian Optimization Performance Evaluation and Strategy Assessment

This section presents a detailed evaluation of the Bayesian optimization framework introduced in Section 3.5, applied to the adaptive experimental optimization of direct aqueous carbonation. Building upon the Gaussian Process model architecture, acquisition strategy, and noise management protocols outlined previously, we assess the practical performance of the BO campaign with respect to process improvement, convergence dynamics, and experimental robustness. The analysis is structured around three central themes: (i) the technical effectiveness of BO in navigating the noisy and multidimensional design space, (ii) the quality and reliability of the underlying surrogate model as validated through cross-validation and independent test experiments, and (iii) the experimental constraints, noise sources, and mitigation strategies that critically shaped campaign success. Retrospective benchmarking against classical experimental design methods is used to contextualize the relative benefits and limitations of the adaptive approach. The findings underscore not only the potential of BO for accelerated optimization in chemical systems, but also the importance of carefully aligned methodological choices, especially under data scarcity and operational uncertainty.

4.5.1 Initial Sampling Results and Foundation Assessment

The initial dataset of 20 systematically sampled experiments provided the foundation for Gaussian Process model training and baseline performance assessment. Observed carbonation yields ranged from the untreated reference (30.2 %) to a maximum of 46.7 %, representing a 54.6 % relative improvement. The foundation GP model achieved a leave-one-out cross-validation score of $R^2 = 0.526$, indicating moderate predictive capability under sparse data conditions. The predicted optimum was 44.7 ± 5.0 % at 177°C, 180 min, 1.59 wt % Ca as CaO, and 0.052 wt % Ca as CaCl₂, demonstrating substantial uncertainty.

Table 4.4: Relative improvement of key GP model metrics from foundation to final model.

Metric	Foundation Model	Final Model	Relative Change
R^2 (LOOCV)	0.526	0.903	+71.7%
Uncertainty at optimum ($\pm$)	0.0498	0.0189	−62.1%
Best observed yield (%)	46.7	55.8	+19.5%
Design space coverage	0.65	0.82	+26.2%
Predicted optimum yield (%)	44.7	53.1	+18.8%

Limitations and Implications Comparison with the final optimization outcome revealed clear limitations: the initial model underestimated the optimal yield by 8.5 % and mispredicted key parameter locations, especially temperature and additive composition. While reaction time was captured reasonably, the additive profile diverged from the final optimum (2.0 wt % Ca as CaO, 0.0 wt % Ca as CaCl₂). These discrepancies highlight the inherent limitations of non-adaptive design strategies when applied to complex parameter spaces. Despite providing adequate model initialization and boundary validation, the foundation phase lacked sufficient resolution for robust prediction. Adaptive Bayesian optimization was essential to refine parameter estimates, reduce uncertainty, and approach global optima.

4.5.2 Bayesian Optimization Algorithm Performance

Technical Performance and Convergence Analysis

The Bayesian optimization algorithm successfully identified the best performing process conditions within the explored parameter space, improving the carbonation yield from the baseline 30.2 % to 53.2 % under laboratory conditions, while a solid understanding of the parameter space was obtained with an overall design space coverage of over 82 %. Figure 4.8 shows the optimization trajectory of our experimental campaign. Here we can see how the best observed value compares to the best model predicted value over the experimental iteration. We start here after our initial sampling with experiment 20 (see Section 3.5 for the exact sequence of the campaign).

Convergence Behavior and Influence of Experimental Noise The optimization algorithm demonstrated steady performance gains, surpassing the best result from the initial sampling phase as early as experiment 4 and continuing to improve until experiment 33. After a plateau phase (experiments 34–43), a final maximum yield of 55.8 % was achieved in experiment 47, corresponding to a 19.0 % improvement over the foundation maximum (46.7 %). The GP posterior mean trajectory exhibited appropriate responsiveness to new data while maintaining calibrated uncertainty estimates. The consistent underestimation of best observed yields by the posterior mean reflects expected behavior in well-calibrated models, particularly when optimal results lie near the upper confidence interval boundaries. Despite significant experimental noise, evident in the large uncertainty bands and deviations between predicted and observed values, the optimization proceeded effectively. All observed improvements remained within the model's posterior uncertainty bounds, confirming that experimental variability was adequately captured. Standardized residuals showed no systematic

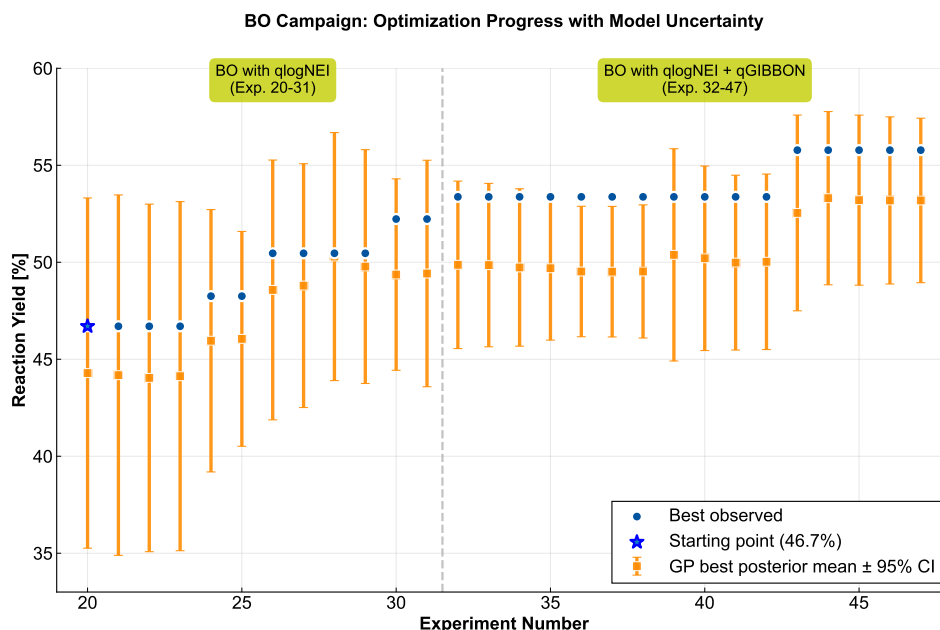
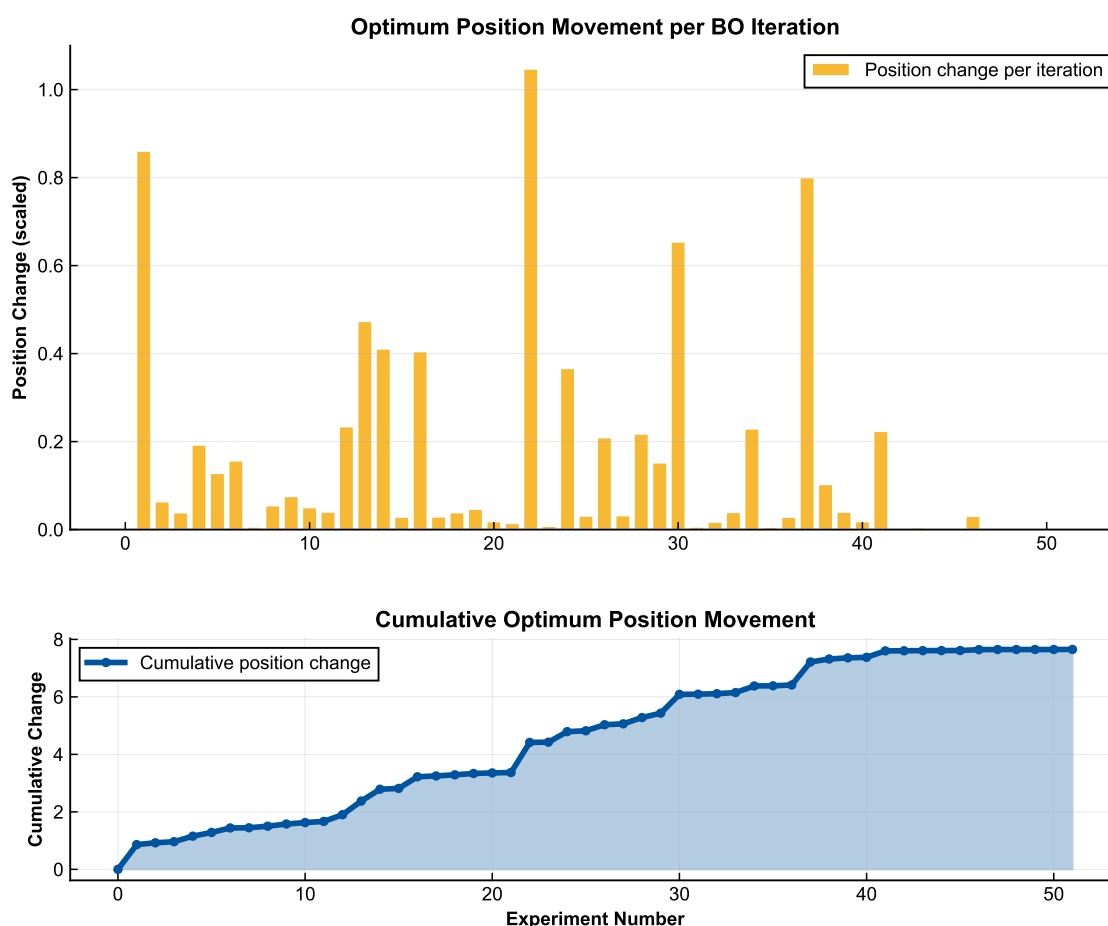


Figure 4.8: BO Campaign: Optimization progress with model uncertainty showing the evolution of reaction yield over 47 experiments. The plot illustrates two distinct phases: initial optimization using qLogNEI (experiments 20–31) and enhanced optimization with qLogNEI + qGIBBON (experiments 32–47). The shaded area represents the 95 % confidence interval of the Gaussian process model.

bias, supporting the model’s ability to explain experimental variation without drift. The algorithm retained full functionality under noisy conditions. Model updates remained responsive and the acquisition function performance stable, ensuring reliable exploration–exploitation trade-offs throughout the campaign.

Convergence Assessment Challenges Experiment 47 marked the end of the experimental budget. However, proving formal convergence under noisy conditions proved challenging. A prolonged plateau phase (experiments 30–42) initially suggested convergence, which was disrupted by a late-stage improvement. Classical convergence criteria based on best observed values were insufficient due to pronounced experimental variability. Optimal posterior mean predictions promise to offer a more robust convergence indicator in noisy environments, but even this metric did not provide definitive convergence. The relative distance between consecutively predicted best points emerged as a more reliable convergence indicator: the final four experiments exhibited the lowest relative distances observed across all 47 data points (minimum distances of 0.0002–0.004), indicating potential convergence within the experimental constraints (Figure 4.9). This pattern was confirmed by validation experiments, which showed clear flattening behavior despite maintaining approximately 4 % absolute mean deviation. These findings suggest that convergence was likely achieved within the experimental

campaign, although absolute certainty remains unattainable. Nonetheless, the resulting optimum can be robustly interpreted and communicated within the established uncertainty bands. The optimization thus achieved not only significant process improvements but also generated a reproducible understanding of system behavior under uncertainty, while highlighting the need for complementary metrics beyond classical yield tracking for assessing convergence in noisy chemical systems.



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Figure 4.9: Optimum position stability during the Bayesian optimization campaign. The plot shows the scaled position change of the predicted optimum between consecutive iterations. Large spikes indicate significant shifts in the model's predicted optimal location, while values approaching zero indicate convergence. The algorithm demonstrates stabilization in the final iterations (>43), suggesting successful convergence to a stable optimum position.

Acquisition Function Strategy Evolution

The transition from pure `qLogNoisyExpectedImprovement` to combined `qGIBBON + ExpectedImprovement` after experiment 32 proved essential for comprehensive design space explo-

ration. Prior to this modification, the algorithm showed signs of premature convergence in the most sensitive parameter dimensions: 8 of the last 10 experiments before the strategy change were conducted at 180 min, and 9 of 10 fell within 10°C of the eventually preferred temperature. During this phase, the model concentrated primarily on additive exploration while maintaining minimal variation in thermal parameters. The strategy transition increased variation of thermal parameter exploration shown in Figure 4.10 and insured an elevation in design space coverage from 70 % to over 82 %, enabling the detailed chemical interpretation and response surface analysis presented in Section 4.3. This enhanced coverage proved crucial for understanding parameter interactions and validating mechanistic hypotheses, though it came at the cost of focused optimization in the most promising regions.

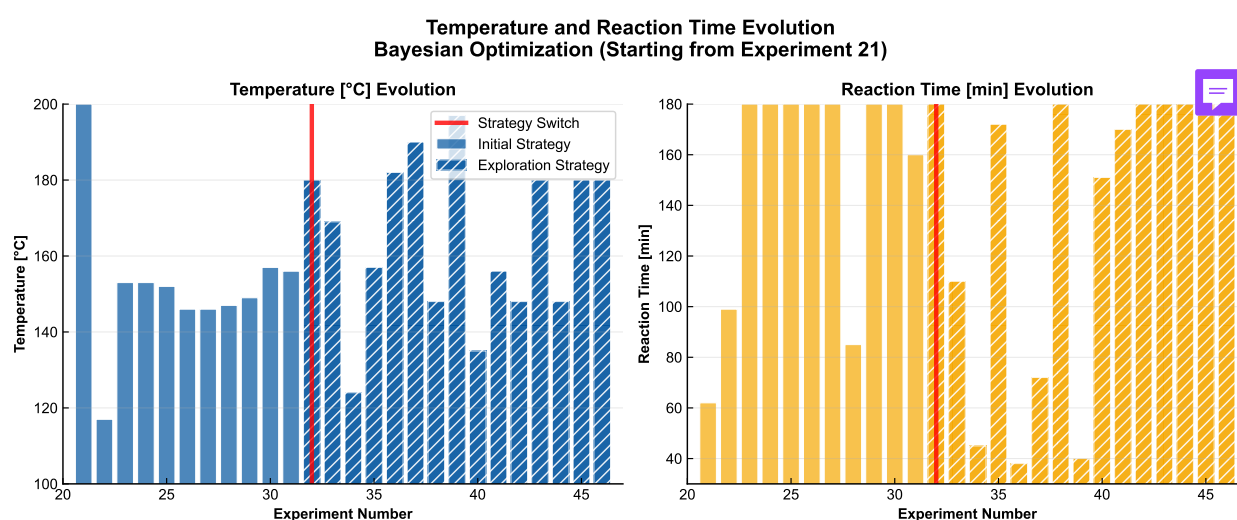


Figure 4.10: Temperature and reaction time parameter evolution during Bayesian Optimization iterations. The acquisition function strategy switch at iteration 32 (red vertical line) broadened exploration in both temperature and reaction time dimensions, demonstrating enhanced sampling diversity in the most critical process parameters.

Parameter-Specific Optimization Dynamics

Temperature and Time Optimization: The optimization revealed distinct convergence patterns across parameter dimensions, shown representatively in Table 4.5. For reaction time, the model correctly identified the most effective value (180 min) immediately after initial sampling and maintained this prediction throughout all iterations, questioning the necessity of BO for exploiting this parameter. Temperature optimization proved more challenging due to non-convex behavior from competing thermodynamic effects, but the model established reasonable predictions within $\pm 15^\circ\text{C}$ of the best performing temperature relatively early in

the campaign (from iteration 2 on), with final convergence to $\pm 2^{\circ}\text{C}$ for the last eight experiments.

Additive Optimization Challenges: Additive optimization proved substantially more challenging, requiring nearly the entire experimental budget to identify the most effective combination of 2 wt % Ca as CaO and 0 wt % Ca as CaCl₂ within the tested parameter space. After initial sampling, the model could not constrain the preferred additive compositions due to their lower effect magnitude compared to thermal parameters and the high experimental noise. Systematic preference for the final additive configuration only emerged from experiment 40 (iteration 7) onward and stabilized for the remaining experiments and validation studies.

This differential optimization performance directly reflects the parameter sensitivity hierarchy as discussed above: temperature and time, exhibiting the highest sensitivities, enabled clear signal separation from noise, while additive effects required extensive sampling to distinguish genuine trends from experimental variability.

Table 4.5: Best predicted parameter combinations during selected Bayesian Optimization iterations. (*With data cleaning and final model specifications)

Iteration	T [°C]	t [min]	CaO [wt % Ca]	CaCl ₂ [wt % Ca]	Predicted Yield [%]
0	177	180	1.59	0.052	44.66 ± 4.98
2	149	180	0.266	0.373	45.95 ± 3.45
4	146	180	0.982	0.428	49.37 ± 2.52
6	153	180	1.66	0.301	49.69 ± 1.84
8	157	180	2.00	0.00	50.22 ± 2.45
Validation	157	180	2.00	0.00	53.18 ± 1.89*

4.5.3 Model Validation and Final Performance

GP Model Calibration and Performance Metrics

Despite the challenging noise environment mentioned above, the final GP model demonstrated robust performance after systematic dataset refinement. Outlier detection using standardized residual analysis identified two experimental points (experiments 0 and 2) with anomalous behavior exceeding ± 2.5 standard deviations, which were excluded as likely measurement artifacts rather than genuine system behavior.

Following dataset refinement, the final GP model achieved strong performance across all validation criteria. Table 4.6 presents comprehensive assessment results demonstrating the model's predictive capability.

Table 4.6: Final GP model validation metrics demonstrating robust performance despite noise challenges.

Metric	Value
RMSE (LOOCV, normalized)	0.0378
MAE	0.0318
R^2 (LOOCV)	0.9028
R^2 (Training)	0.9788
Average CI size	0.132
95 % coverage	95.7 %
Observation noise (avg)	0.0337
Intrinsic noise (avg)	0.0187

The high R^2 values exceeding 0.9 demonstrate strong predictive capability across the four-dimensional parameter space. The normalized RMSE of 3.78 % indicates good prediction accuracy relative to the experimental yield range, while the excellent 95.7 % confidence interval coverage confirms well-calibrated uncertainty quantification essential for reliable process guidance.

Gaussian Process Calibration and System Consistency Comprehensive residual analysis confirmed good calibration of the final GP model despite experimental noise. Standardized residuals followed an approximately normal distribution centered near zero and showed no systematic patterns across predicted values or input parameters, indicating unbiased model behavior and accurate reconstruction of the underlying response surface. This suggests that the observed variability primarily reflects true process noise rather than structural model deficiencies, supporting confidence in the derived parameter relationships and optimal conditions, even under the uncertainty constraints discussed in Section 4.5.5. In parallel, statistical analysis of inline baseline experiments revealed no significant temporal drift ($p = 0.14$, $\alpha = 0.05$), confirming the consistency of experimental conditions throughout the campaign (see Section 3.5.4). Residuals over time also showed no systematic deviations, indicating that temporal effects were adequately captured by the model. Accordingly, all test baseline experiments were retained in the training set. Nonetheless, continued monitoring of potential drift is recommended to ensure sustained system reliability in future studies (see Section 3.5.4).

4.5.4 Independent Validation and Temporal Stability

The predictive performance of the final Gaussian Process model was evaluated after the campaign using six independent validation experiments conducted after the completion of the adaptive optimization campaign (Table 4.7).

Table 4.7: Comparison of GP-predicted and experimentally observed yields in independent validation experiments.

Group	Predicted	Observed	Prediction
	Yield (%)	Yield (%)	Error (%)
High-Performance Region	53.07	46.98	+6.09
	51.50	42.07	+9.43
	49.00	50.18	−1.18
High-Uncertainty Region	31.62	24.79	+6.83
	36.66	35.86	+0.80
	25.83	23.49	+2.35

In high-performance regions, observed yields were consistently lower than predicted values, with an average deviation of -4.0% and a maximum deviation of -9.4% . Five out of six validation runs fell below the predicted mean, indicating a systematic overestimation. However, most validation points remained within the average uncertainty band of the GP predictions, suggesting that the model's confidence estimates appropriately captured the observed variability. In contrast, experiments in high-uncertainty regions showed smaller absolute deviations (typically $< 3\%$), aligning more closely with their wider predicted confidence intervals. Although this appears counterintuitive, as densely sampled regions are expected to yield more accurate predictions, a possible explanation lies in the temporal separation of the validation campaign. In contrast to the consistency monitoring experiments discussed above, all validation experiments were not performed inline but conducted several days after the main optimization and involved intermediate use of the reactor for other materials. Subtle changes in surface conditions, reactor fouling, or calibration drift may have shifted the baseline system response. Since high-performing setups operate closer to kinetic optima, even minor perturbations can cause disproportionately large yield drops. Conversely, regions of inherently low reactivity are less sensitive to such variations. When normalized to their respective target values, the relative deviations in both regions converge to a comparable range (approximately 10%). This supports the hypothesis that the observed overprediction in high-performing areas could reflect temporal system drift rather than structural model error. Nevertheless, this interpretation remains tentative and should be re-evaluated through follow-up campaigns with stricter process control and rolling validation. Despite these limitations, the model pre-

served the relative yield ranking and accurately located the optimal design region, thereby confirming its suitability for optimization and mechanistic interpretation (Figure 4.9).

4.5.5 Experimental Challenges and Noise Management

While the adaptive optimization campaign successfully identified high-performance parameter combinations, several experimental limitations affected model fidelity, surrogate validity, and optimization efficiency. A retrospective analysis of these challenges is essential for interpreting model performance and deriving methodological improvements.

Surrogate Accuracy and LOI Deviations Despite a strong correlation between dried mass gain and LOI-derived yields ($R^2 = 0.982$), the RMSE increased from initially expected 2.34 % (based on earlier experiments) to 5.12 %. Likely causes include differences in CO₂ retention, incomplete drying, or formation of secondary phases. Nevertheless, the surrogate method was indispensable to maintain optimization momentum under time constraints. Its limitations likely contributed to inefficiencies and stagnation phases, particularly during surrogate-to-LOI transition delays.

Sedimentation and Flow Regime Effects Persistent sedimentation beneath the stirrer was observed, likely due to incomplete suspension of slag particles. Although higher stirring speeds could have improved dispersion, excessive turbulence risked splashing and instrumentation damage. A modified impeller design or flow disruptors, such as the cooling coil used in the larger reactor setup, would have been preferable but could not be implemented mid-campaign without compromising reproducibility. Resulting flow shadows may have introduced intra-batch heterogeneities (Figure 4.6).

Cross-Contamination and Reactor Cleaning Progressive cross-contamination occurred via carbonate scaling on reactor internals, especially the stirrer. Water rinsing proved insufficient, and visible deposits were removed mechanically between runs. Although acidic cleaning (e.g., with dilute acetic acid) would have ensured more consistent baseline conditions [13], it was avoided to preserve experimental consistency.

Reactor Variability and Feedstock Inhomogeneity Minor differences in PID tuning and jacket performance led to variable heat-up and pressurization profiles, introducing devi-



(a) Stirrer after moderate scaling.



(b) Stirrer after heavy scaling accumulation.

Figure 4.11: **Fouling** variability on the stirrer over campaign progression. Residual layers caused cross-contamination between experiments; mechanical cleaning was only partially effective.

ations in effective reaction time. The feedstock was used as received without sieving, under the assumption of bulk uniformity. However, varying PSDs between batches likely affected dissolution and precipitation kinetics [10].

Dose Variability and Parameter Tracking Additive dosing exhibited minor deviations ($\pm 1\text{--}3\text{ mg}$), which were accounted for by updating all models with the actually measured values rather than acquisition suggestions. While this may have slightly reduced optimization efficiency due to input noise, it ensured accurate representation of experimental conditions.

Noise Monitoring and Mitigation Posterior noise estimates remained within plausible ranges (1.5–4 %). Outliers exceeding $\pm 2.5\sigma$ were excluded via standardized residual analysis. Additionally, high-gradient point pairs were temporarily excluded when they induced unrealistic slopes, thereby improving noise learning and overall model stability.

4.6 Retrospective Benchmarking Results

To complement the experimental findings and validate the implemented Bayesian Optimization strategy under controlled conditions, a series of retrospective simulations was conducted using the synthetic third-degree polynomial ground truth function described in Section 3.6 (see Appendix A.9.1 for the full function definition). These simulations enable direct comparison of different acquisition functions, design paradigms, and initialization strategies under

realistic noise assumptions. The following results provide a quantitative evaluation of the performance boundaries, convergence behavior, and robustness of Bayesian Optimization in a multi-parameter chemical process system. They serve to generalize the insights gained from the adaptive experimental campaign and inform the design of future optimization studies in comparable experimental settings. This retrospective analysis does not intend to provide universal proof of BO's superiority, but rather delivers system-specific guidance for carbonation processes with analogous complexity and noise levels.

Sequential Design Outperforms Traditional Strategies Simulations confirmed that the implemented Bayesian Optimization framework consistently outperforms classical one-shot DoE strategies under identical conditions. Each strategy was evaluated using an identical budget of 47 function calls across 20 independent seed runs (see Figure 4.12). As performance metric, the true yield at the optimal input position suggested by the GP model, i.e., the point with the highest predicted mean, was used (see Section 3.6). The best-performing BO configuration, using 20 initial samples and qGIBBON with one experiment per iteration, achieved an average performance of 54.0 %, corresponding to 98.0 % of the theoretical maximum of the ground truth (55.1 %). Beyond superior average performance, Bayesian Optimization demonstrated higher consistency with performance spanning 51.2-55.0 % compared to traditional methods showing broader and lower spans of 42.3-52.4 % (see Table A.11 for details). Latin Hypercube Sampling (LHS) and Random search reached only 91.1 % and 88.6 % of the optimum, respectively, while full factorial design slightly outperformed these methods by achieving 92.3 % on average. The result illustrates the value of sequential experimentation, even when all strategies share the same total number of experimental evaluations and surrogate modeling approach. Notably, the uncertainty (posterior standard deviation) of the GP model at the best-predicted point was markedly lower for the sequential design (1.8 %) compared to the best-performing factorial strategy (4.6 %). This highlights that in noisy environments, adaptive learning through iterative sampling and systematic information gain yields not only higher performance but also more robust and reliable model predictions.

Batched Design Limits Iterative Efficiency The impact of batched execution was quantified by comparing sequential BO (one experiment per iteration) with batched BO (three experiments per iteration) using qGIBBON. While both approaches outperformed traditional designs, the batch configuration converged more slowly and reached a lower average final yield (52.5 %) than the single-experiment version (54.0 %). With only nine effective model updates, the batched setup was less capable of exploiting intermediate learning, underscoring the inherent dependency of BO on frequent model refinement. Notably, the predictive

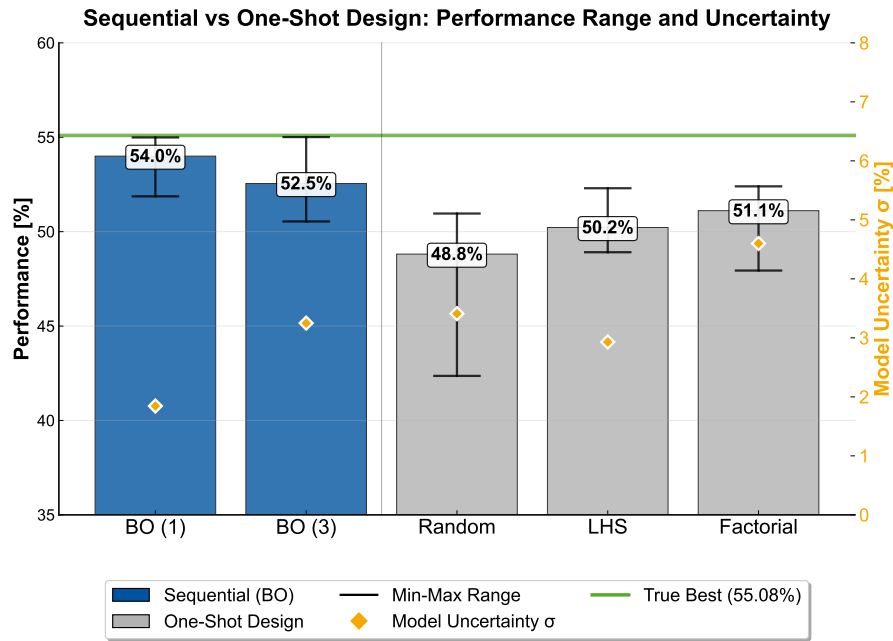


Figure 4.12: Sequential Bayesian Optimization outperforms one-shot experimental designs in both yield and reliability. BO (1) using qGIBBON achieves a yield of 54.0% with an uncertainty of $\sigma = 1.9\%$, compared to one-shot methods ranging from 48.8% to 51.1% with higher uncertainties ($\sigma = 2.9\%$ to 4.6%). The true optimum is 55.1%.

uncertainty at the final recommended point remained with 3.2% significantly higher in the batched setting due to the reduced number of model updates. This observation illustrates the comparatively lower information gain per experiment and highlights the relative inefficiency of batch-based designs in noise-prone environments where frequent re-evaluation and adaptation are essential. Nevertheless, batched BO still outperformed one-shot strategies by a margin. Especially in terms of performance consistency, batched BO demonstrated superior reliability (50.5-55.0%) compared to DoE methods (42.3-52.4%), highlighting the inherent advantages of adaptive sampling even under parallelization constraints.

Acquisition Function Comparison The systematic comparison of acquisition strategies demonstrates that both qGIBBON and qLogNoisyExpectedImprovement are well-suited for achieving high-yield optimization while maintaining reliable model accuracy (Figure 4.13 and Figure 4.14). Contrary to initial expectations, qGIBBON exhibited outstanding performance, particularly in terms of optimization effectiveness. The method achieved the lowest average regret of 1.03% with moderate initialization (10 initial LHS points), consistently outperforming qLogNEI across all tested configurations. Both qGIBBON and qLogNEI demonstrated excellent surrogate model fidelity, with R^2 values exceeding 0.9 and high predictive accuracy in high-performing regions ($\text{RMSE}_{\text{high}} \leq 2.4\%$). The number of initial sampling points

moderately influenced performance: 10–20 initial points yielded the best trade-offs between early exploration and the remaining adaptive sampling budget. In contrast, the UCB strategy failed to deliver competitive results under noise, exhibiting 3–4 $\times$ higher regret and substantially degraded model quality. This can be attributed to its architectural inability to distinguish between epistemic and aleatoric uncertainty, resulting in inefficient sampling focused on high-variance regions rather than genuinely promising areas. Overall, qGIBBON emerges as the preferred acquisition function for future optimization campaigns, offering a compelling balance between optimization performance and model reliability. For regret minimization, moderate initialization with 10 LHS points proved optimal (1.03 %), while 20 initial points yielded superior model quality ($R^2 = 0.925$, $\text{RMSE}_{\text{high}} = 2.37\%$) and lower variability (see Table A.11) at the cost of slightly increased regret (1.08 %). This suggests that yield-oriented campaigns benefit from maximized learning efficiency through moderate initialization, whereas model-building efforts may justify larger initial datasets. Together with the findings on sequential and batched execution, these results highlight the critical role of noise-aware acquisition functions and adaptive sampling in achieving robust, high-quality process optimization under uncertainty.

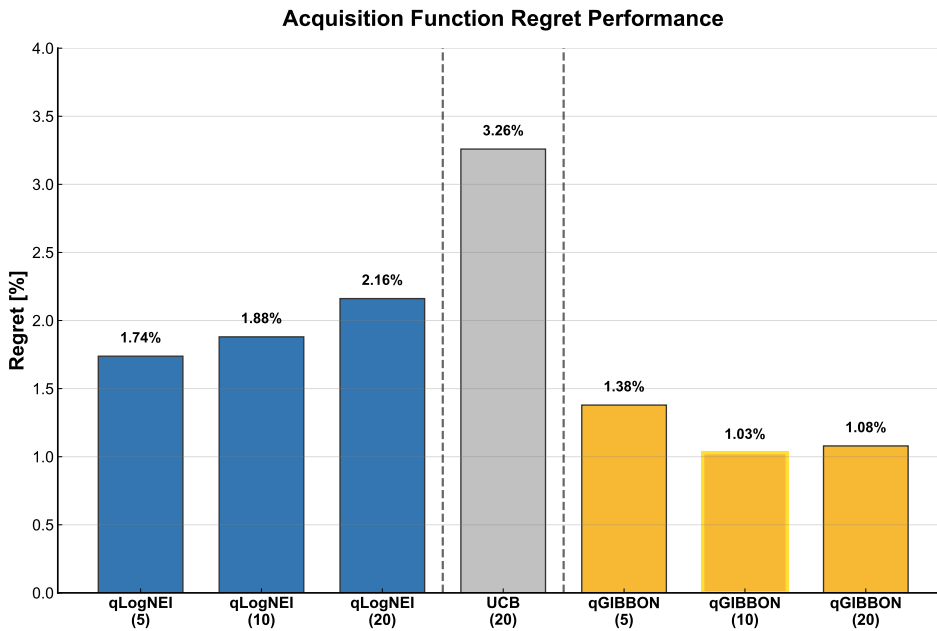


Figure 4.13: Acquisition function regret comparison: qGIBBON outperforms other strategies with consistently low regret (1.03–1.38%) across all iteration budgets, while qLogNEI shows moderate performance (1.74–2.16%) and UCB exhibits highest regret (3.26%) at 20 iterations.



Figure 4.14: Model quality comparison: qGIBBON achieves superior predictive accuracy (R^2 up to 0.925) with low RMSE in high-performing regions (2.4–3.1%), while UCB shows poor model quality ($R^2 = 0.684$, RMSE = 7.5%). qLogNEI demonstrates moderate performance with improving R^2 values (0.891–0.917) across iteration budgets.

4.6.1 Summary and Strategic Implications

The adaptive experimental campaign confirmed that Bayesian Optimization can substantially enhance process performance in noisy, multi-parameter chemical systems, achieving a 76 % improvement over baseline yields and providing robust uncertainty quantification across the design space. While thermal parameters converged quickly and could likely be optimized via simpler mechanistic approaches, the optimization of additive compositions clearly benefited from BO's non-parametric flexibility and uncertainty-aware exploration. Sequential design enabled efficient learning, with retrospective simulations indicating a 5.7 % relative performance improvement and substantially reduced result variability (91.0–99.9 % vs. 76.8–95.1 %) compared to the best-performing classical DoE strategy. The transition from exploitation-based to information-theoretic acquisition strategies (qGIBBON) was essential for improving both posterior quality and global search performance. Despite experimental limitations, including surrogate deviations, sedimentation effects, and reactor variability, the final GP model demonstrated strong predictive capability ($R^2 = 0.90$, RMSE = 3.8 %) with well-calibrated uncertainty estimates. Independent validation confirmed model generalizability, while also underscoring the importance of temporal consistency in experimental systems.

In summary, the results establish BO as a powerful and reliable framework for process opti-

mization under uncertainty, provided that sufficient initial sampling, model diagnostics, and experimental standardization are ensured. These findings not only validate the current campaign but also offer transferable guidance for the design of future adaptive optimization efforts.

5 Summary and Outlook

5.1 Summary of Work and Key Results

This thesis developed and applied a data-driven experimental strategy to optimize the direct aqueous autoclave carbonation of ferrochrome slag under high-pressure conditions (100 bar). A Bayesian Optimization framework, based on Gaussian Process regression, was implemented to efficiently explore a four-dimensional design space comprising temperature, reaction time, CaO addition, and CaCl₂ addition. A total of 63 autoclave experiments were conducted in batched iterations, with CO₂ uptake quantified via gravimetric Loss-on-Ignition measurements. The response surface modeling enabled both optimization and mechanistic interpretation of the observed process behavior.

The maximum predicted CO₂ uptake was 53.2 % (269 g CO₂/kg slag), representing a relative increase of 76 % over the baseline yield of 30.2 %. This optimum was achieved at 157 °C, 180 min, 2 wt % Ca as CaO, and 0 wt % Ca as CaCl₂, subject to the assumptions made (see Appendix A.5). The experimental data and surrogate response surfaces enabled the formulation of several mechanistic hypotheses:

- **Calcium supplementation via CaO** improves reactivity not by immediate ion release, but by sustaining a reactive environment over longer timescales, serving as a heterogeneous nucleation site promoting thinner product layers.
- **Chloride-based additives** accelerate early-stage conversion but do not sustain long-term kinetics due to rapid Ca²⁺ release and pH destabilization.
- **Mass Transfer Limitations:** The autoclave system showed mass transfer limitations, likely due to coarse slag particles. Under optimized conditions, the process operated at the interface of transport and surface-reaction control, with additives enhancing mass transfer.
- **Synergistic Interactions:** Combined additive systems did not show performance ben-

efits beyond CaO alone, suggesting interference or saturation-limited effects at the tested dosages.

These findings contribute to a deeper understanding of the roles and limitations of chemical enhancement strategies in calcium-dominated carbonation systems and provide mechanistic guidance for future process intensification.

5.2 Adaptive Modeling and Experimental Strategy

The Gaussian Process surrogate demonstrated strong predictive performance despite substantial experimental noise, achieving a leave-one-out cross-validation R^2 of 0.90 and a normalized RMSE of 3.8 %. Posterior uncertainty estimates were well-calibrated, with a 95.7 % coverage of observed values and an average confidence interval width of 13.2 percentage points. Residual analysis and consistency tests confirmed the absence of systematic bias, indicating that experimental variability was appropriately captured. Within 43 adaptive learning experiments, the surrogate-guided exploration successfully identified high-yield parameter combinations, improving the carbonation yield by 76 % over baseline conditions. Sequential sampling allowed efficient refinement of the response surface, particularly in additive dimensions where low signal-to-noise ratios initially obscured trends. Yield-based convergence criteria proved unreliable in this noisy environment; instead, the stabilization of the predicted optimum location emerged as a more robust convergence proxy. The transition from exploitation-based to information-theoretic acquisition strategies (notably qGIBBON) significantly improved global exploration and posterior model quality. Simulations on a synthetic ground truth function derived from the gathered data confirmed that the BO framework outperformed the best-performing classical Design-of-Experiments (DoE) strategy in retrospective benchmarking by 5.4 % in relative performance, while also yielding models with significantly lower predictive uncertainty and prediction variability. A moderate initial sampling size of 10–20 LHS points proved optimal for balancing early design space coverage and adaptive flexibility, with qGIBBON consistently delivering the lowest regret and highest surrogate fidelity. Overall, the results validate Bayesian optimization as a powerful and transferable tool for model-based experimental campaigns in noisy, high-dimensional chemical systems. When combined with rigorous model diagnostics and adaptive sampling, the GP-based framework enables both efficient process improvement and mechanistic insight under uncertainty.

5.3 Scientific Significance

This study demonstrates, for the first time, the application of a Bayesian Design-of-Experiments approach to the aqueous carbonation of ferrous slag. The resulting framework accounts for experimental uncertainty and system complexity, offering a reproducible basis for future studies aimed at optimizing similar reactive mineral systems. Furthermore, the combined use of Bayesian Optimization and Gaussian Process modeling facilitated the formulation of mechanistic hypotheses regarding the role of CaO and CaCl₂ in modulating reaction kinetics, while generally confirming their enhancing effect. This contributes to a better understanding of chemical enhancement strategies in calcium-dominated slag systems, a domain that remains comparatively underexplored in the literature and important for viable process scale-up.

5.4 Critical Reflection

The effectiveness of the Bayesian optimization framework was limited by substantial experimental noise arising from process-internal variability, the use of dry mass gain as a surrogate for LOI, and uncertainties due to scaling (i.e., precipitate buildup), sedimentation effects, and incomplete mixing. These factors reduced model accuracy and constrained optimization efficiency. The systematic overprediction observed in independent validation experiments also highlights the importance of ensuring temporal process stability and strict cleaning routines in long-term campaigns. While the GP response surfaces enabled the formulation of mechanistic hypotheses, particularly regarding additive effects and rate-limiting regimes, these must be interpreted in light of model uncertainty and do not constitute a full kinetic analysis. Nevertheless, the consistent uncertainty calibration and preserved relative ranking of predicted optima support the scientific validity of the model-assisted interpretations presented.

5.5 Outlook and Future Directions

To substantiate the mechanistic hypotheses developed in this study, particularly regarding the nucleation and buffering effects of CaO and the mass transport modulation by CaCl₂, further experimental investigations are required. Time-resolved pH measurements are of particular interest, as they could offer in situ insight into the dynamic evolution of the carbonate system and the buffering capacity of different additives. In principle, such online measurements can be implemented similarly to other spectroscopic or sensor-based methods, including Raman

spectroscopy, and would help resolve temporal changes that static endpoint measurements cannot capture. Beyond real-time monitoring, retrospective characterization techniques such as SEM–EDX for nucleation mapping, phase-resolved XRD for carbonate speciation, and cross-sectional imaging for diffusion layer analysis remain essential to validate hypothesized reaction pathways. Elemental mapping of reacted particles could provide insight into diffusion barriers or additive-induced precipitation fronts, particularly relevant for systems involving CaCl_2 . From a methodological perspective, the current Bayesian optimization framework offers promising capabilities, yet future studies should explore its extension to multi-objective settings, incorporating additional targets such as product quality or process economics. The integration of mechanistic priors through grey-box modeling, as well as the application of multi-fidelity or transfer-learning-based BO strategies, may further enhance efficiency and generalizability. Specifically, the framework could be transferred to other alkaline residues or upscaled systems. The observed sensitivity to mass transport limitations, further optimization of the experimental setup is also warranted. This may include increasing stirrer speed, adjusting impeller geometry, or narrowing the particle size distribution to mitigate sedimentation effects and improve homogeneity within the reactor. Such refinements would not only reduce experimental noise but also yield more representative kinetic data, strengthening both the optimization and mechanistic interpretation.

List of Abbreviations and Symbols

List of Abbreviations

ADAM	Adaptive Moment Estimation (optimizer)
AOD	Argon Oxygen Decarburization
ARD	Automatic Relevance Determination
BO	Bayesian Optimization
BOF	Basic Oxygen Furnace
CCD	Central Composite Design
CCUS	Carbon Capture, Utilization and Storage
DAE	Differential-Algebraic Equation
DoE	Design of Experiments
e.g.	for example
EAF	Electric Arc Furnace
EDTA	Ethylenediaminetetraacetic acid
EI	Expected Improvement
FeCr	Ferrochrome
gen.	in general
GIBBON	General-purpose Information-Based Bayesian Optimisation
GP	Gaussian Process
ICC	Intraclass Correlation Coefficient
L-BFGS-B	Limited-memory Broyden-Fletcher-Goldfarb-Shanno-Bounded
LF	Ladle Furnace
LHS	Latin Hypercube Sampling

LOI	Loss-on-Ignition
LOOCV	Leave-One-Out Cross-Validation
MAE	Mean Absolute Error
OFAT	One-Factor-At-a-Time
PID	Proportional-Integral-Derivative
pp	Percentage Points
PSD	Particle Size Distribution
qGIBBON	q-General-purpose Information-Based Bayesian Optimisation
qLogNEI	q-Log Noisy Expected Improvement
qUCB	q-Upper Confidence Bound
RBF	Radial Basis Function
RMSE	Root Mean Squared Error
RSM	Response Surface Methodology
S/L	Solid-to-Liquid Ratio
SCM	Supplementary Cementitious Material
SEM	Scanning Electron Microscopy
spec.	specific
UCB	Upper Confidence Bound
XRD	X-ray Diffraction
XRF	X-ray Fluorescence

List of Indices

0	Initial or reference state
added	Added quantity
Ca	Calcium-related
corrected	Corrected value
dry	Dried state
feedstock	Feedstock material
final	Final state

max	Maximum value
measured	Measured value
Mg	Magnesium-related
observed	Observed value
predicted	Predicted value
residue	Residue after treatment
slag	Slag-related
surrogate	Surrogate measurement
theoretical	Theoretical value
true	True value

List of Symbols

α	Fractional carbonation conversion	[—]
β	Regression coefficient	[—]
ℓ	Lengthscale parameter	[—]
η	Carbonation reaction yield	[%]
μm	Micrometer	[—]
$\pm$	Plus-minus (uncertainty)	[—]
σ	Standard deviation	[—]
σ_f	Signal variance	[—]
σ_n	Observation noise	[—]
ε	Error term	[—]
D_{50}	Median particle diameter	$[\mu\text{m}]$
D_{90}	90th percentile particle diameter	$[\mu\text{m}]$
f^*	Optimal function value	[—]
k_1, k_2, k_3	Kinetic rate constants	$[\text{min}^{-1}]$
K_{sp}	Solubility product constant	[—]
M	Molar mass	$[\text{g/mol}]$
m	Mass	$[\text{g}]$

R^2	Coefficient of determination	[—]
r_i	Standardized residual	[—]
T	Temperature	[C]
t	Reaction time	[min]
w	Weight percentage	[$wt\%$]
bar	Pressure unit (1 bar = 100 kPa)	[—]
CI	Confidence Interval	[%]
g/mL	Grams per milliliter (density)	[—]
g CO ₂ /kg	Grams CO ₂ per kilogram slag	[—]
Henry's law	Gas solubility law	[—]
kPa	Kilopascals	[—]
LOOCV	Leave-One-Out Cross-Validation	[—]
min	Minutes	[—]
mL	Milliliters	[—]
mol/L	Molarity (moles per liter)	[—]
RMSE	Root Mean Squared Error	[%]
rpm	Revolutions per minute	[—]
wt%	Weight percent	[—]
°C	Degrees Celsius	[—]

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A Appendix

A.1 Gaussian Process Fundamentals

A Gaussian Process (GP) is a collection of random variables, any finite subset of which follows a multivariate Gaussian distribution. For regression applications, a GP defines a distribution over functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and is completely specified by its mean function $m(\mathbf{x})$ and covariance function $k(\mathbf{x}, \mathbf{x}')$:

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')) \quad (\text{A.1})$$

The mean function $m(\mathbf{x}) = \mathbb{E}[f(\mathbf{x})]$ represents the expected value of the function at any input point, while the covariance function $k(\mathbf{x}, \mathbf{x}') = \mathbb{E}[(f(\mathbf{x}) - m(\mathbf{x}))(f(\mathbf{x}') - m(\mathbf{x}'))]$ defines the similarity between function values at different input locations.

Given noisy observations $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$, the GP provides a posterior distribution $p(f(\mathbf{x}^*)|\mathcal{D}, \mathbf{x}^*) = \mathcal{N}(\mu(\mathbf{x}^*), \sigma^2(\mathbf{x}^*))$ at new test points $\mathbf{x}^*$, where $\mu(\mathbf{x}^*)$ is the predictive mean and $\sigma^2(\mathbf{x}^*)$ quantifies model uncertainty.

A.1.1 Kernel Functions

Kernel functions encode prior assumptions about the smoothness and structural properties of the underlying function.

Squared Exponential (RBF) Kernel

The Squared Exponential kernel generates infinitely differentiable functions:

$$k_{\text{SE}}(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp \left(-\frac{1}{2\ell^2} \|\mathbf{x} - \mathbf{x}'\|^2 \right) \quad (\text{A.2})$$

where σ_f^2 is the output variance and ℓ is the characteristic lengthscale.

Automatic Relevance Determination (ARD)

ARD kernels assign individual lengthscales to each input dimension:

$$k_{\text{ARD}}(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp \left(-\frac{1}{2} \sum_{d=1}^D \frac{(x_d - x'_d)^2}{\ell_d^2} \right) \quad (\text{A.3})$$

Shorter lengthscales ℓ_d indicate higher sensitivity to parameter d , enabling automatic feature selection.

Matérn Kernel Family

Matérn kernels provide flexible smoothness control:

$$k_{\text{Matérn}}(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu}r}{\ell} \right)^\nu K_\nu \left(\frac{\sqrt{2\nu}r}{\ell} \right) \quad (\text{A.4})$$

where $r = \|\mathbf{x} - \mathbf{x}'\|$ and $\nu \in \{0.5, 1.5, 2.5\}$ controls function smoothness. For experimental data, $\nu = 1.5$ and $\nu = 2.5$ often provide better fits than the infinitely smooth SE kernel.

Scale Kernel

The Scale Kernel wraps base kernels to improve numerical stability and provide explicit output scaling:

$$k_{\text{Scale}}(\mathbf{x}, \mathbf{x}') = \sigma_{\text{scale}}^2 \cdot k_{\text{base}}(\mathbf{x}, \mathbf{x}') \quad (\text{A.5})$$

where σ_{scale}^2 is a learnable output scale parameter and k_{base} is the underlying kernel (e.g., RBF, ARD, or Matérn). This formulation separates amplitude scaling from the base kernel's shape parameters, facilitating more stable hyperparameter optimization and preventing numerical issues during training.

Composite Kernels

The complete kernel structure combines signal and noise components:

$$k(\mathbf{x}, \mathbf{x}') = k_{\text{signal}}(\mathbf{x}, \mathbf{x}') + \sigma_n^2 \delta(\mathbf{x}, \mathbf{x}') \quad (\text{A.6})$$

where $\delta(\mathbf{x}, \mathbf{x}')$ is the Kronecker delta function modeling observation noise.

A.1.2 Acquisition Functions

Acquisition functions guide experiment selection by balancing exploration (sampling in high-uncertainty regions) and exploitation (sampling near predicted optima) based on the GP posterior. They operate on the posterior mean $\mu(\mathbf{x})$ and variance $\sigma^2(\mathbf{x})$ to identify the most promising next evaluation point.

Expected Improvement

Expected Improvement (EI) quantifies the expected gain from evaluating a point $\mathbf{x}$ relative to the current best observation $f^+ = \max_{i=1, \dots, n} y_i$. It balances exploitation by favoring regions with high predicted values and exploration by considering prediction uncertainty [77]:

$$\alpha_{\text{EI}}(\mathbf{x}) = \mathbb{E}[\max(0, f(\mathbf{x}) - f^+)] \quad (\text{A.7})$$

For a GP posterior $f(\mathbf{x}) \sim \mathcal{N}(\mu(\mathbf{x}), \sigma^2(\mathbf{x}))$, this expectation can be computed analytically:

$$\alpha_{\text{EI}}(\mathbf{x}) = \sigma(\mathbf{x})[Z\Phi(Z) + \phi(Z)] \quad (\text{A.8})$$

where $Z = (\mu(\mathbf{x}) - f^+)/\sigma(\mathbf{x})$ is the standardized improvement and $\Phi(\cdot)$, $\phi(\cdot)$ are the standard normal CDF and PDF, respectively.

Upper Confidence Bound

The Upper Confidence Bound (UCB) acquisition function provides a simpler yet effective exploration-exploitation strategy by directly combining the posterior mean and uncertainty [18]:

$$\alpha_{\text{UCB}}(\mathbf{x}) = \mu(\mathbf{x}) + \kappa\sigma(\mathbf{x}) \quad (\text{A.9})$$

where $\kappa > 0$ is a hyperparameter controlling the exploration-exploitation trade-off. Larger values of κ favor exploration in uncertain regions, while smaller values emphasize exploitation of promising areas. The UCB acquisition has theoretical convergence guarantees and is particularly robust for optimization problems with limited computational budgets [78].

qLogNoisyExpectedImprovement

For parallel experimental designs requiring batch optimization in noisy environments, qLogNoisyExpectedImprovement (qLogNEI) extends the EI framework to handle multiple candidate points simultaneously [20]:

$$\alpha_{\text{qLogNEI}}(\mathbf{X}) = \mathbb{E} \left[\max \left(0, \max_{i=1, \dots, q} \log f(\mathbf{x}_i) - \log f^+ \right) \right] \quad (\text{A.10})$$

where $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_q\}$ represents the candidate batch of size q . The logarithmic transformation serves multiple purposes: it provides numerical stability when dealing with small improvement values, handles noisy observations more robustly, and naturally incorporates relative improvements rather than absolute differences. This acquisition function accounts

for the mutual information between batch points, preventing redundant sampling in similar regions.

GIBBON

GIBBON (General-purpose Information-Based Bayesian Optimisation) employs an information-theoretic approach that quantifies the expected reduction in uncertainty about the global optimum location [86]:

$$\alpha_{\text{GIBBON}}(\mathbf{x}) = I[y(\mathbf{x}); \mathbf{x}^*] \quad (\text{A.11})$$

where $I[y(\mathbf{x}); \mathbf{x}^*]$ represents the mutual information between a function evaluation at $\mathbf{x}$ and the global optimum location $\mathbf{x}^*$. In practice, this is approximated using entropy calculations:

$$\alpha_{\text{GIBBON}}(\mathbf{x}) \approx H[p(\mathbf{x}^*)] - \mathbb{E}_{y(\mathbf{x})} [H(p(\mathbf{x}^* | y(\mathbf{x})))] \quad (\text{A.12})$$

Here, $H[\cdot]$ denotes the entropy, $p(\mathbf{x}^*)$ is the current belief about the optimum location, and the expectation is taken over the posterior predictive distribution. GIBBON is particularly effective for multimodal optimization problems, where traditional acquisition functions may struggle to maintain sufficient global exploration.

A.2 Materials and Characterization

A.2.1 Ferrochrome Slag Characterization

The phase composition and particle size distribution of the as-received ferrochrome slag were analyzed to assess its suitability for aqueous carbonation. Crystalline phases were identified by X-ray diffraction (XRD) combined with Rietveld refinement. The particle size distribution was determined via ultrasonic scattering analysis.

Phase Composition

Figure A.1 shows the X-ray diffraction pattern of the slag sample along with the corresponding reference patterns for the identified phases. Table A.1 provides the quantitative results based on Rietveld refinement.

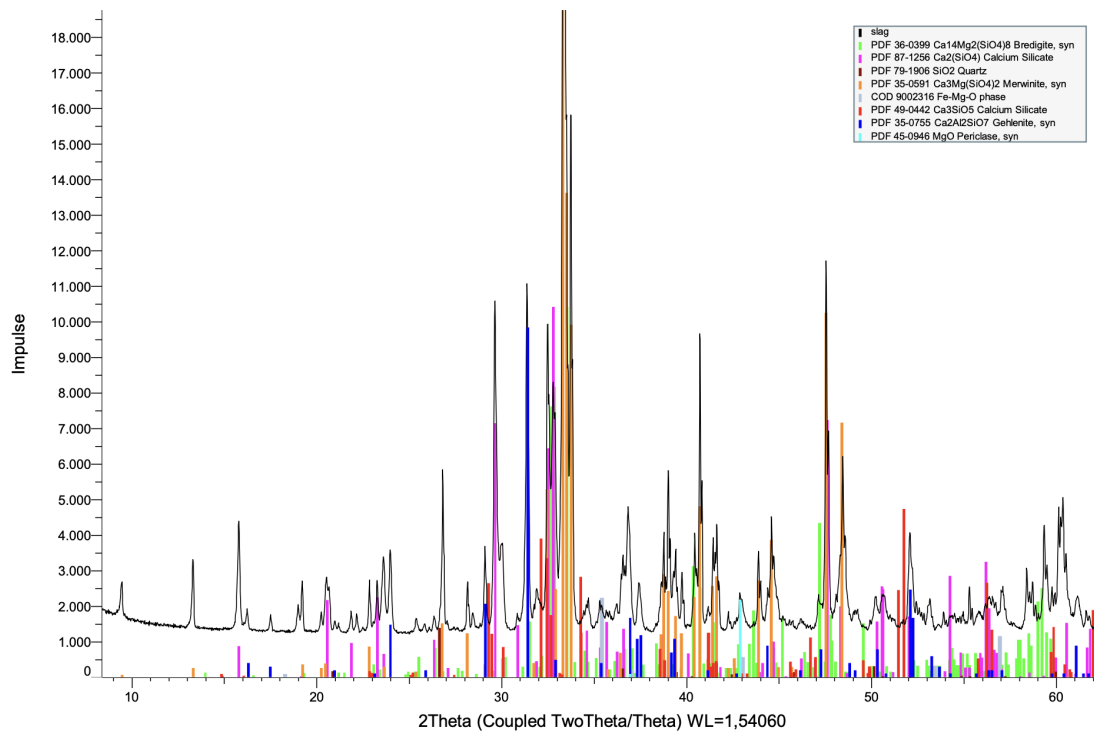


Figure A.1: X-ray diffraction (XRD) pattern of the ferrochrome slag sample. Colored ticks indicate reference peak positions of the identified crystalline phases (e.g., Merwinite, Gehlenite, Periclase). Measurement was performed using Cu-K α radiation ($\lambda = 1.54060 \text{ \AA}$).

Table A.1: Phase composition of the as-received slag sample based on XRD and Rietveld refinement.

Phase (Chemical Formula)	Measured value	Unit
Merwinite ($\text{Ca}_3\text{Mg}(\text{SiO}_4)_2$)	approx. 64	%
$\gamma\text{-C}_2\text{S}$ ($2\text{CaO}\cdot\text{SiO}_2$)	approx. 13	%
Gehlenite ($\text{Ca}_2\text{Al}(\text{AlSiO}_7)$)	approx. 15	%
Bredigite ($\text{Ca}_7\text{Mg}(\text{SiO}_4)_4$)	2–4	%
C_3S ($3\text{CaO}\cdot\text{SiO}_2$)	<1	%
Periclase (MgO)	1	%
Quartz (SiO_2)	1	%
Fe-Mg-O phase (e.g., MgFe_2O_4)	<1	%

Particle Size Distribution

The particle size distribution (PSD) of the ferrochrome slag was measured using the CILAS 1190 Liquid system based on ultrasonic scattering. The resulting size spectrum is shown in Figure A.2, indicating both cumulative and differential volume distributions. Table A.2 summarizes the key percentiles of the volume-based distribution.

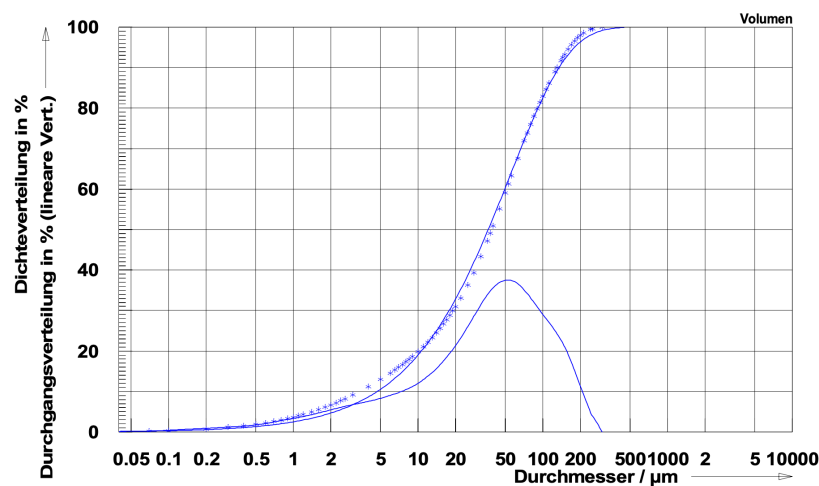


Figure A.2: Particle size distribution (PSD) of the used ferrochrome slag determined by ultrasonic particle size analysis (CILAS 1190 Liquid). The plot shows both the cumulative distribution and the density function of particle sizes in volume percent.

The feedstock was stored in an airtight container.

Table A.2: Key quantiles of the ferrochrome slag particle size distribution (volume-based).

Quantile	Diameter [μm]	Description
d_{10}	3.37	10 % of particles are smaller than this
d_{50}	39.11	Median diameter (50 % threshold)
d_{90}	130.79	90 % of particles are smaller than this

A.2.2 Chemical Additives and Reagents

Table A.3: Specifications of chemical additives used in carbonation experiments.

Additive	Supplier	Purity	Form	CAS-No.
CaO	Thermo (Acros)	96%	Powder	1305-78-8
Ca(OH) ₂	Thermo (Acros)	98%	Powder	1305-62-0
CaCl ₂	Thermo	96%	Granules	10043-52-4
CaCO ₃	Merck / Sigma	$\geq 99\%$	Powder	471-34-1
CaSO ₄ ·2H ₂ O	VWR	99.7%	Powder	10101-41-4
NaCl	Merck / Sigma	$\geq 99\%$	Crystalline	7647-14-5
NaHCO ₃	VWR (BDH)	101.1%	Powder	144-55-8

All additives were stored in airtight containers. Calcium-bearing additives were mixed with the dry feedstock. Sodium salts were prepared with the solution.

A.3 Data Validation and Reproducibility Assessment

To evaluate the overall reproducibility and reliability of the experimental methodology, systematic validation was conducted using historical data from previous experimental campaigns with olivine feedstock. Nine experimental conditions were replicated simultaneously across all three autoclaves under identical reaction parameters, providing a robust dataset for statistical analysis.

The reproducibility was quantified using the intraclass correlation coefficient (ICC) based on a one-way random-effects model. The ICC represents the proportion of total variance attributable to systematic differences between experimental conditions, as opposed to random measurement error:

$$\text{ICC} = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2} \quad (\text{A.13})$$

where σ_S^2 represents the between-group variance (systematic experimental effects) and σ_ε^2 represents the within-group error variance across replicates.

The analysis yielded an ICC value of 0.973, indicating excellent reliability according to established statistical benchmarks. This high reproducibility validates the experimental methodology and supports the use of single-reactor experiments in the optimization campaign, as inter-reactor variability is demonstrably negligible compared to the experimental effects under investigation.

Table A.4: Summary of statistical metrics for reproducibility assessment based on intraclass correlation analysis.

Metric	Value
Total standard deviation	11.72%
Mean of standard deviations	1.93%
Standardized coefficient of variation	0.041
Intraclass correlation coefficient (ICC)	0.973

A.4 Sampling Method Comparison

To justify the selection of orthogonal sampling for the initial dataset, six sampling strategies were evaluated using design space coverage metrics. The comparison was conducted for 15 sampling points across the four-dimensional parameter space.

Table A.5: Comparison of sampling methods for initial experimental design.

Sampling Method	Maximin Distance	Mean Correlation	Discrepancy
Orthogonal LHS	21.27	0.003	0.003
Optimized (normal) LHS	13.39	0.155	0.011
Sobol	8.88	0.080	0.009
(normal) LHS	8.36	0.211	0.012
Random	3.61	0.178	0.031

Orthogonal LHS demonstrated superior performance across all metrics, with the highest maximin distance (optimal space-filling), lowest parameter correlation (balanced coverage), and minimal discrepancy (uniform distribution). These properties ensure robust Gaussian Process training under limited data conditions.

A.5 Summary of Model Assumptions and Design Decisions

Table A.6: Core modeling assumptions and architectural decisions [77, 78, 20, 18].

Decision / Assumption	Rationale / Purpose
RBF kernel with isotropic lengthscales (initial)	Robust behavior under limited data and good generalization performance.
Transition to ARD kernel after 30 experiments	Enables learning of individual input sensitivities when data suffices.
Zero-mean prior	No prior assumption on output level; consistent with exploratory modeling.
Homoskedastic noise assumption	Constant noise variance across all observations; reflects controlled laboratory conditions as we have no reference campaign yet.
Gaussian likelihood	Captures experimental reproducibility noise.
Log-normal priors on σ_n and ℓ	Supports stable hyperparameter learning under limited data.
Gamma priors on ARD lengthscales	Prevents extreme scale values; promotes realistic dependencies.

Table A.7: Implementation decisions and data handling strategy [77, 78, 20, 18].

Decision / Assumption	Rationale / Purpose
Input standardization (z-score)	Improves numerical conditioning and kernel interpretability.
Prediction discretization (1°C, 1 min, mg scale)	Aligns model output with experimental resolution and feasibility.
Use of dried mass gain as surrogate yield	Enables rapid model updates within tight campaign timeframe.
Inclusion of pseudo-experiments at $t = 0$ min	Prevents unphysical extrapolation; enforces chemical realism.
Batch optimization with parallel autoclaves	Maximizes throughput while respecting model-driven guidance.
Hybrid acquisition function (qLogNEI + qGIBBON)	Balances exploitation and targeted exploration in uncertain subspaces.
Greedy batch construction for qGIBBON	Custom extension to enable parallel batch BO in entropy-based frameworks.

Table A.8: Validation and model monitoring procedures [17, 88].

Decision / Assumption	Rationale / Purpose
Independent validation experiments (6 total)	Assess generalization in both high-yield and high-uncertainty regions.
Monitoring of standardized residuals ($\pm 2.5\sigma$)	Enables detection and filtering of outliers and artifacts.
Empirical noise monitoring ($\sigma_{\text{pred}} \approx 1.5\text{--}5\%$)	Ensures realistic uncertainty levels and calibration.
Sequential model retraining after each batch	Maintains up-to-date posterior and guides next acquisition step.

Table A.9: Experimental assumptions [80, 13].

Decision / Assumption	Rationale / Purpose
Calcium-bearing additives undergo complete conversion ($\eta \approx 100\%$)	Approximate upper-bound reactivity to assess model performance in high-yield and high-uncertainty regions.
Restricted chloride content for SCM purposes ($\text{Cl} \leq 0.1\text{ wt\%}$ in final product)	Ensure compliance with material standards for supplementary cementitious materials (SCMs) and avoid durability issues related to chloride-induced corrosion [80].
Retention of 28 % of dissolved salts in the solid product	Empirical estimate to account for salt entrapment despite filtration and drying; relevant for Cl^- carryover and SCM compatibility assessment.

A.6 Final Gaussian Process Implementation

A.6.1 Final Model settings

The final Gaussian Process (GP) surrogate model used for adaptive Bayesian optimization was based on a Matérn 2.5 kernel with Automatic Relevance Determination (ARD). This kernel choice reflects moderate smoothness assumptions and enables directional sensitivity across the input space.

- **Covariance Kernel:** Matérn 2.5 with ARD, implemented as a scaled product kernel with individual lengthscales ℓ_i for each input dimension.
- **Noise Model:** Homoskedastic observation noise was modeled via a GaussianLikelihood, assuming a constant noise variance σ_n^2 across all training points. This reflects the assumption of uniformly distributed experimental uncertainty under controlled laboratory conditions.
- **Noise Prior:** A log-normal prior was placed on the noise standard deviation σ_n , defined as

$$\log \sigma_n \sim \mathcal{N}\left(\frac{1}{2} \log p(\mathcal{D}), \sigma_0^2\right)$$

where $p(\mathcal{D})$ denotes the marginal likelihood and σ_0 controls the prior spread. This prior anchors noise learning toward empirically plausible values without imposing strong regularization.

- **Lengthscale Prior:** A Gamma prior with parameters $\alpha = 10$ and $\beta = 6$ was used for each lengthscale ℓ_i :

$$\ell_i \sim \text{Gamma}(10, 6) \quad \text{with} \quad \mathbb{E}[\ell_i] = \frac{10}{6} = 1.67$$

The weakly informative prior favors moderate smoothness and prevents overfitting in low-data regimes.

- **Output Scale:** The signal variance σ_f^2 was optimized jointly with the kernel hyperparameters as part of the scaled kernel structure.

This model configuration provided a balance between expressiveness and regularization, ensuring robust surrogate predictions over the full design space. Model quality was validated

through cross-validation, LOOCV residual diagnostics, and independent test experiments.

A.7 Kinetic Model Fit for Selected Configurations

To validate the mechanistic interpretation of carbonation kinetics, the experimental data were fitted to three classical rate-limiting models representing gas–liquid mass transfer, diffusion through a product layer, and chemical surface reaction. The coefficient of determination (R^2) for each fit was used to infer the most probable controlling mechanism under various process conditions. Table A.10 summarizes the results for six representative configurations. These include the experimentally determined optimum, as well as systematically varied scenarios: (i) removal or substitution of additives, (ii) reduced reaction time, and (iii) elevated temperature. The dominant kinetic regime was determined by the model with the highest R^2 value. Overall, the analysis confirms a shift in controlling mechanisms depending on process conditions. In particular, high CaO dosages and extended durations favor mass transfer control, whereas additive-free or high-temperature conditions promote diffusion limitation. See Section 4.4.3 for interpretation.

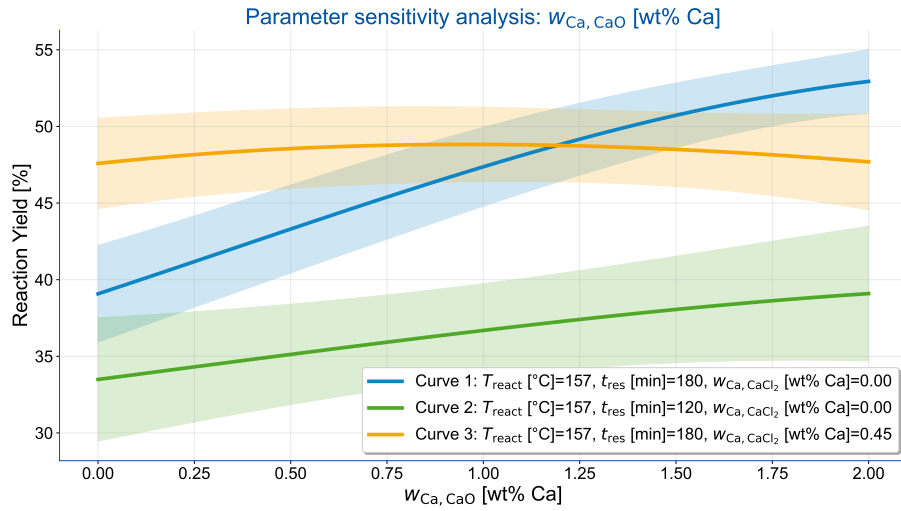
Table A.10: Dominant kinetic mechanisms for key process configurations.

Configuration (T, t, Ca as CaO, Ca as CaCl ₂)	MT R^2	D R^2	Ch R^2	Dominant
157°C, 180 min, 2.0 wt%, 0.0 wt%	0.990	0.966	0.989	MT
157°C, 180 min, 0.0 wt%, 0.0 wt%	0.861	0.938	0.880	D
157°C, 180 min, 0.0 wt%, 0.45 wt%	0.888	0.889	0.891	Ch
157°C, 40 min, 2.0 wt%, 0.0 wt%	0.998	0.997	0.999	Ch
200°C, 180 min, 2.0 wt%, 0.0 wt%	0.956	0.988	0.967	D
120°C, 180 min, 2.0 wt%, 0.0 wt%	0.948	0.907	0.942	MT

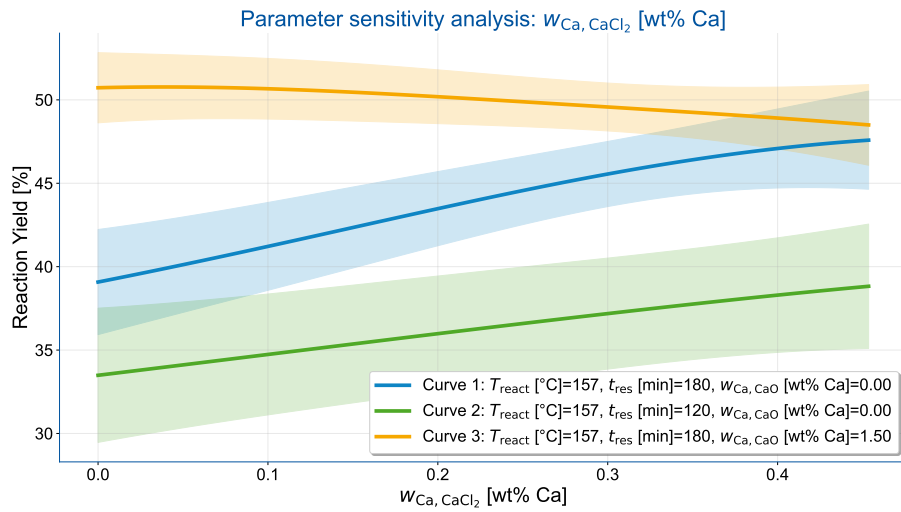
MT = Mass Transfer, D = Diffusion, Ch = Chemical Reaction.

R^2 values indicate model fit quality; dominant mechanism: highest R^2 .

A.8 Supplementary Process Visualizations



- (a) Ca as CaO at temperature of 157°C for reaction time of 180 min / 120 min with and without CaCl_2 (0.45 wt%) addition. Effect increases with time and may be modulated by CaCl_2 .



- (b) Ca as CaCl_2 at temperature of 157°C for reaction time of 180 min / 120 min with and without CaO addition (1.50 wt%). Time amplifies effect; CaO may modulate response negatively.

Figure A.3: Effect of Ca addition route (CaO vs. CaCl_2), varying process conditions at optimal temperature (157°C). In the upper plot, Curve 3, from 1.50 wt% Ca as CaO, exceeds the Ca constraint.

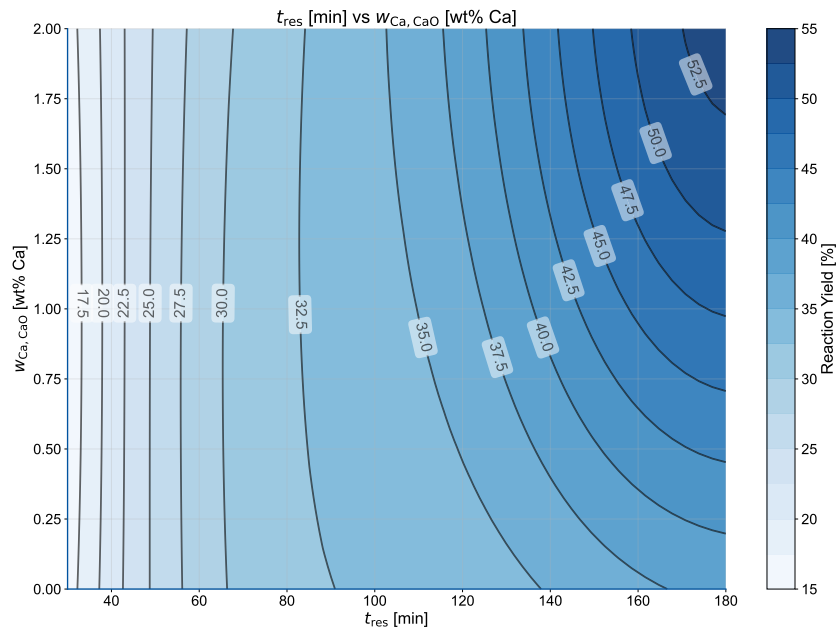


Figure A.4: Response surface contour plot showing carbonation yield (%) as a function of reaction time and CaO addition under optimal temperature conditions (157°C). For higher reaction times the positive influence of CaO increases.

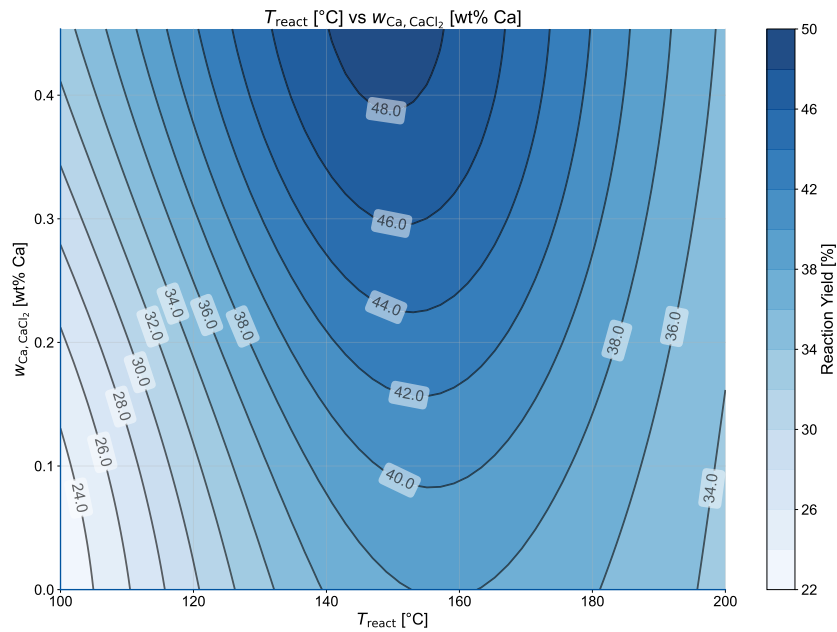


Figure A.5: Response surface contour plot showing carbonation yield (%) as a function of temperature and CaCl_2 addition under optimal time and zero CaO conditions (t : 180 min, Ca as CaCl_2 : 0 wt %). The plot reveals temperature-additive interactions and confirms whether the temperature optimum (157°C) moves to lower values across CaCl_2 concentrations.

A.9 Retrospective Benchmarking

A.9.1 Polynomial Function for Retrospective Benchmarking

To establish a high-quality ground truth function for Bayesian optimization benchmarking, a comprehensive polynomial feature engineering analysis up to degree 3 was conducted. The objective was to create a realistic yet interpretable response surface that captures the complex dependencies observed in chemical processes while maintaining analytical tractability.

The feature selection process followed a multi-criteria approach, combining statistical significance testing (F-test), mutual information analysis, and forward selection algorithms. Starting from the full set of polynomial feature combinations—including all base variables (reaction temperature T_{react} [100–200 °C], residence time t_{res} [30–180 min], CaO concentration c_{CaO} [0–2 wt%], and CaCl₂ concentration c_{CaCl_2} [0–0.45 wt%])—as well as their quadratic and three-way interaction terms, a total of 11 statistically significant and performance-relevant features were systematically identified. Feature inclusion was based on their incremental contribution to predictive accuracy and robustness.

Optimized Linear Model

The optimized 11-feature polynomial model demonstrated substantial improvement over conventional approaches:

- **Coefficient of determination:** $R^2 = 0.830$
- **Adjusted R^2 :** 0.81 (accounting for model complexity)
- **Prediction accuracy:** RMSE = 0.056, MAE = 0.044
- **Model complexity:** 11 features from 53 data points (appropriate complexity ratio)

This performance improvement validates the systematic feature engineering approach and establishes a robust ground truth function for subsequent Bayesian optimization studies.

- **Explicit mathematical representation:** The model takes the transparent form:

$$\begin{aligned}
 f(x) = & 0.2645 \cdot c_{\text{CaCl}_2} - 2.721 \times 10^{-2} \cdot c_{\text{CaO}} + 1.763 \times 10^{-3} \cdot t_{\text{res}} \\
 & + 8.502 \times 10^{-4} \cdot T_{\text{react}} - 3.603 \times 10^{-3} \cdot (c_{\text{CaCl}_2} \times c_{\text{CaO}} \times T_{\text{react}}) \\
 & + 1.420 \times 10^{-3} \cdot (c_{\text{CaCl}_2}^2 \times T_{\text{react}}) - 1.160 \times 10^{-3} \cdot (c_{\text{CaCl}_2} \times c_{\text{CaO}} \times t_{\text{res}}) \\
 & + 5.586 \times 10^{-4} \cdot (t_{\text{res}} \times c_{\text{CaO}}) + 3.081 \times 10^{-5} \cdot (T_{\text{react}} \times t_{\text{res}}) \\
 & - 3.367 \times 10^{-7} \cdot (t_{\text{res}} \times T_{\text{react}}^2) - 1.986 \times 10^{-7} \cdot (t_{\text{res}}^2 \times T_{\text{react}}) + \varepsilon
 \end{aligned} \tag{A.14}$$

where each coefficient has direct physical interpretation within the specified operating ranges. The global optimum was found at $T_{\text{react}} = 157.9^\circ\text{C}$, $t_{\text{res}} = 180\text{ min}$, $c_{\text{CaO}} = 2.0\text{ wt\%}$, $c_{\text{CaCl}_2} = 0.0\text{ wt\%}$, yielding a predicted response of 55.08 %.

A.9.2 Classical Design of Experiments Implementation

Three classical DOE methods were implemented as baseline comparisons for the Bayesian Optimization approach, each adapted to utilize a fixed experimental budget of 47 evaluations.

Three-Level Factorial Design The factorial design was implemented as a 3^k full factorial with levels at 0, 0.5, 1, systematically exploring all combinations of low, medium, and high parameter values. For four dimensions, this generates up to $3^4 = 81$ possible design points. When constrained by the experimental budget, the design uses the first 47 points in lexicographic order, ensuring systematic coverage of the parameter space with emphasis on main effects and two-factor interactions. This approach allows for the estimation of quadratic effects through the inclusion of intermediate levels while maintaining the structured nature of classical factorial designs.

Latin Hypercube Sampling LHS was implemented using stratified random sampling with permuted intervals. Each dimension is divided into n equal intervals, where n corresponds to the number of sample points. For each dimension, a random permutation of interval indices is generated, and within each interval, a point is sampled uniformly at random. This ensures that each interval in every dimension is sampled exactly once, providing excellent space-filling properties and avoiding clustering of design points. LHS naturally accommodates any sample size, making it particularly suitable for budget-constrained experimental designs.

Experimental Evaluation All DOE methods were evaluated using the same 47-point experimental budget to ensure fair comparison with the Bayesian Optimization approach. Each design point was evaluated using the ground truth function with observational noise ($\sigma = 0.04$), and a Gaussian Process surrogate model was fitted to the resulting data. The predicted optimum from each DOE method was identified through optimization of the fitted GP model and compared against both the true function optimum and the BO performance metrics.

Implementation of model quality metrics

The $\text{RMSE}_{\text{high}}$ and R^2 metrics were computed by evaluating the trained Gaussian Process models on a standardized test grid of 512 points generated via Sobol sampling in the normalized $[0,1]^4$ input space. For each test point, the GP posterior mean $\mu_{\text{GP}}(x)$ was compared against the true function value $f_{\text{true}}(x)$ from the synthetic linear ground truth. The global R^2 coefficient was calculated as $R^2 = 1 - \frac{\sum_i (f_{\text{true}}(x_i) - \mu_{\text{GP}}(x_i))^2}{\sum_i (f_{\text{true}}(x_i) - f_{\text{true}})^2}$ across all test points. For $\text{RMSE}_{\text{high}}$, a binary mask was applied to identify the top 20% of true function values ($f_{\text{true}}(x_i) > 0.9 \cdot \max(f_{\text{true}})$), and the root mean square error was computed only for this subset: $\text{RMSE}_{\text{high}} = \sqrt{\frac{1}{|S|} \sum_{i \in S} (f_{\text{true}}(x_i) - \mu_{\text{GP}}(x_i))^2}$, where S represents the high-performing region.

A.9.3 Results of Benchmarking

Table A.11: Relative performance ranges for all optimization strategies based on theoretical best (BS = batch size).

Strategy	Min	Max	Mean	Range
<i>Bayesian Optimization – qGIBBON</i>				
Init = 5	0.9284	0.9992	0.9750	0.0708
Init = 10	0.9366	0.9998	0.9813	0.0632
Init = 20	0.9417	0.9984	0.9804	0.0567
Init = 20; BS = 3	0.9170	0.9985	0.9170	0.0815
<i>Bayesian Optimization – qLogNEI</i>				
Init = 5	0.9201	0.9989	0.9684	0.0788
Init = 10	0.9215	0.9964	0.9659	0.0749
Init = 20	0.9102	0.9985	0.9645	0.0883
<i>Traditional DoE Methods</i>				
Factorial	0.8702	0.9508	0.9280	0.0806
LHS	0.8878	0.9488	0.9117	0.0610
Random	0.7684	0.9248	0.8862	0.1564

Values are relative to the global maximum of the synthetic ground truth. Bold entries indicate best results.

Table A.12: Performance comparison of different acquisition functions averaged over multiple independent runs (4 % observation noise).

Strategy	Init. Points	Avg. Regret [%]	R ²	RMSE _{high} [%]
qLogNEI	5	1.74	0.891	2.37
qLogNEI	10	1.88	0.900	2.39
qLogNEI	20	2.16	0.917	2.85
UCB	5	3.75	0.684	7.54
UCB	10	4.24	0.799	6.14
UCB	20	3.26	0.885	4.04
qGIBBON	5	1.38	0.858	3.06
qGIBBON	10	1.03	0.893	2.92
qGIBBON	20	1.08	0.925	2.37

RMSE_{high} refers to the Root Mean Square Error in the top 20 % of function values. Bold values indicate best performance.

A.10 List of Experiments

These experimental inputs do not exactly match the conditions proposed by the BO framework but rather reflect the actual process settings applied during execution.

Table A.13: Pseudo Experiments (Reaction Time = 0 min). CaO and CaCl₂ dosages must be understood as wt% Ca from CaO/CaCl₂.

Temperature [°C]	Reaction Time [min]	CaO [w%]	CaCl ₂ [w%]	Yield [-]
163	0	0.2720	0.2060	0.00
129	0	1.2710	0.3110	0.00
171	0	1.2360	0.1240	0.00
174	0	1.6500	0.1500	0.00
164	0	0.2050	0.1960	0.00

Table A.14: Experimental Data; first 5 experiments are applicable experiments from the screening phase. Next 15 experiments are from initial sampling via LHS. The remaining are from adaptive BO. CaO and CaCl₂ dosages must be understood as wt% Ca from CaO/CaCl₂.

Temperature [°C]	Reaction Time [min]	CaO [w%]	CaCl ₂ [w%]	Yield [-]
180	180	0.0000	0.0000	0.2695
180	180	1.0006	0.0000	0.4644
180	180	0.0000	0.0000	0.3107
180	180	1.0034	0.2008	0.4211
180	180	0.0000	0.0000	0.3827
107	163	1.8253	0.0563	0.2959
103	54	1.2221	0.3127	0.2226
167	41	0.3030	0.3373	0.3143
136	138	0.2344	0.4377	0.3727
195	111	0.8762	0.2506	0.3652
129	89	0.5603	0.3914	0.2446
129	49	0.3602	0.2369	0.2521
106	94	0.5961	0.3091	0.1949
170	38	1.4022	0.4218	0.3084
186	159	1.1978	0.3936	0.3610
143	177	0.8147	0.3012	0.4668
115	143	1.6881	0.1873	0.2837
159	74	1.8368	0.0744	0.3240
153	127	0.9463	0.0404	0.3724

Temperature [°C]	Reaction Time [min]	CaO [w%]	CaCl ₂ [w%]	Yield [-]
195	105	0.1172	0.1416	0.3773
186	180	0.0000	0.0000	0.3703
200	62	0.0000	0.0000	0.3063
117	99	0.9148	0.4146	0.2702
153	180	0.1715	0.2838	0.4825
153	180	0.0000	0.3171	0.4522
152	180	1.5609	0.4398	0.5046
146	180	1.6381	0.3691	0.4800
146	180	0.0000	0.3893	0.5043
147	85	0.0000	0.3647	0.3605
149	180	1.6066	0.2586	0.5223
157	180	1.5637	0.3004	0.4605
156	160	1.5409	0.3077	0.5337
180	180	0.0000	0.0000	0.3759
169	110	1.1106	0.3344	0.3595
124	45	1.0091	0.1314	0.2358
157	172	2.0040	0.2427	0.4686
182	38	0.0858	0.4536	0.3814
190	72	1.7853	0.0116	0.3130
148	180	2.0026	0.0000	0.5101
197	40	1.7910	0.0289	0.2654
135	151	1.3679	0.3907	0.4641
156	170	1.5423	0.3069	0.4889
148	180	2.0211	0.0000	0.5578
180	180	0.0000	0.0000	0.3937
148	180	2.0369	0.0000	0.5431
180	180	0.0000	0.0000	0.3807
180	180	0.0000	0.0000	0.3803

Table A.15: Independent Validation Experiments. CaO and CaCl₂ dosages must be understood as wt% Ca from CaO/CaCl₂.

Temperature [°C]	Reaction Time [min]	CaO [w%]	CaCl ₂ [w%]	Yield [-]
149	180	1.9997	0.0000	46.98
143	180	1.8668	0.0000	42.70
157	180	1.5380	0.3120	50.18
106	177	0.0972	0.1141	24.79
199	169	0.1901	0.3965	35.86
118	111	0.2358	0.0000	23.49